



Marine communities and ecosystem functioning

Housekeeping

01

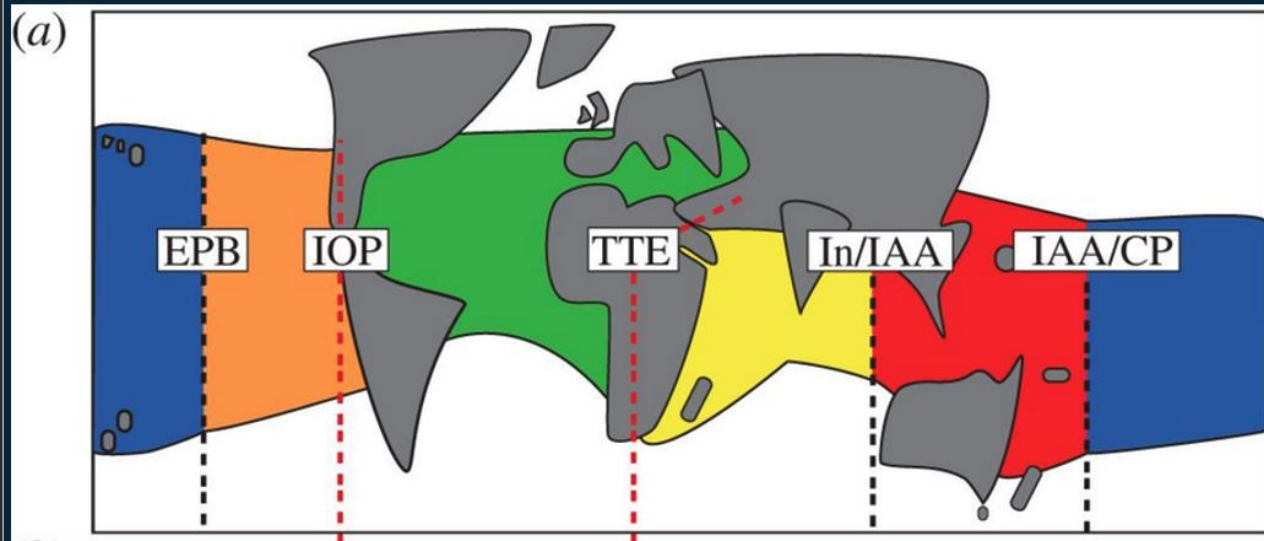
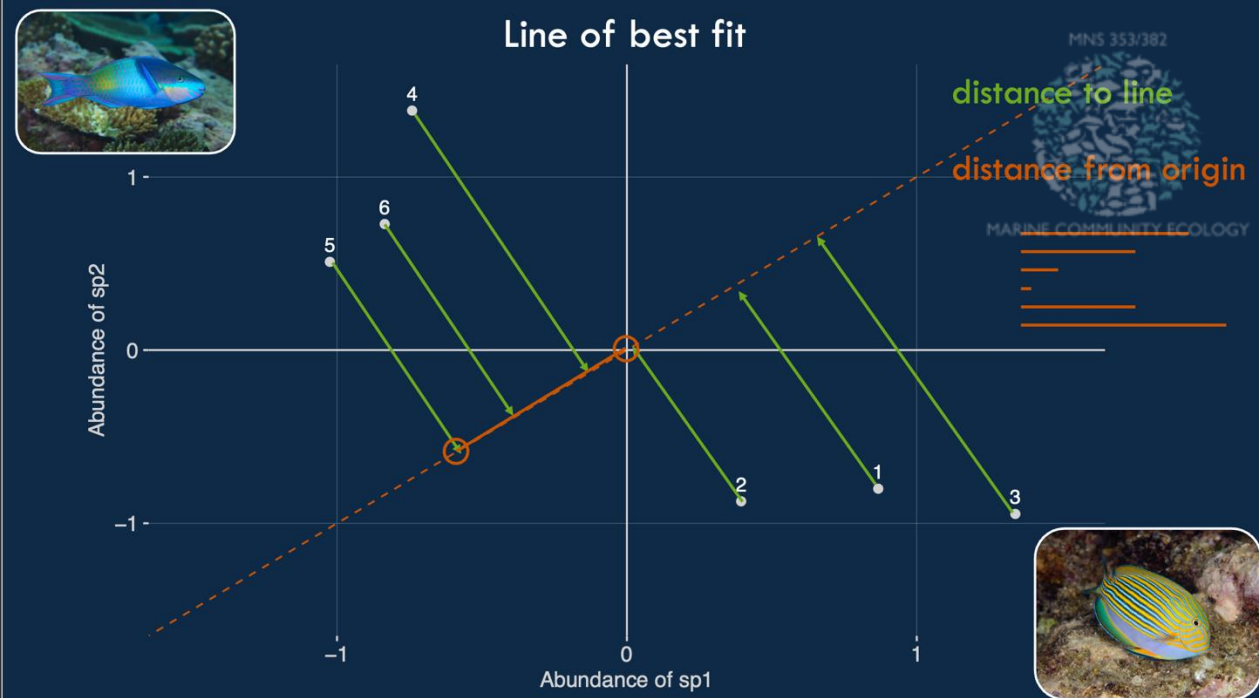
Assignment 2 still in progress 📌📌😞

02

Assignment 3: on Canvas

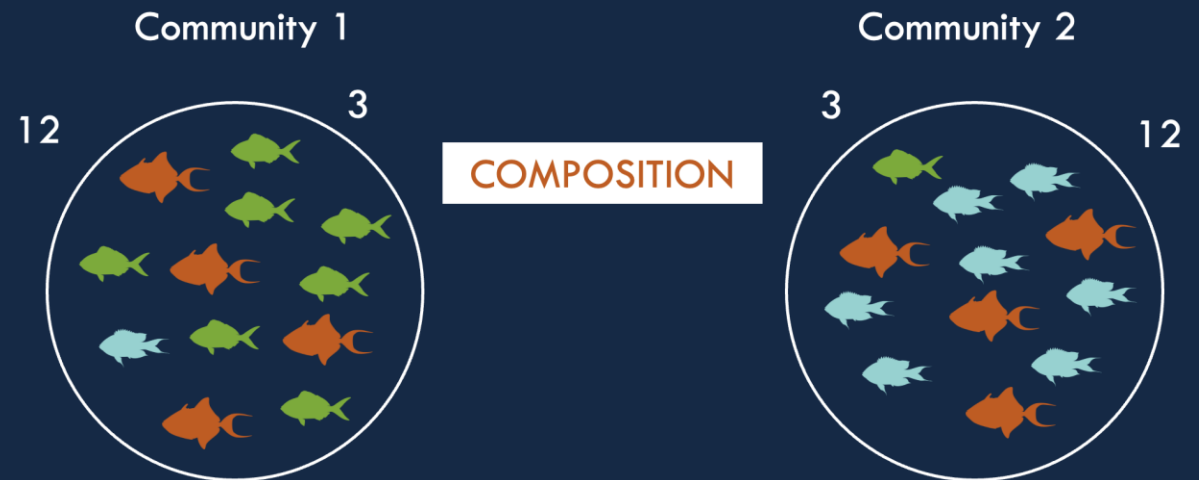
03

Thursday: functional trait based analyses (yay!)



Cowman & Bellwood 2013

First order properties: multiple communities

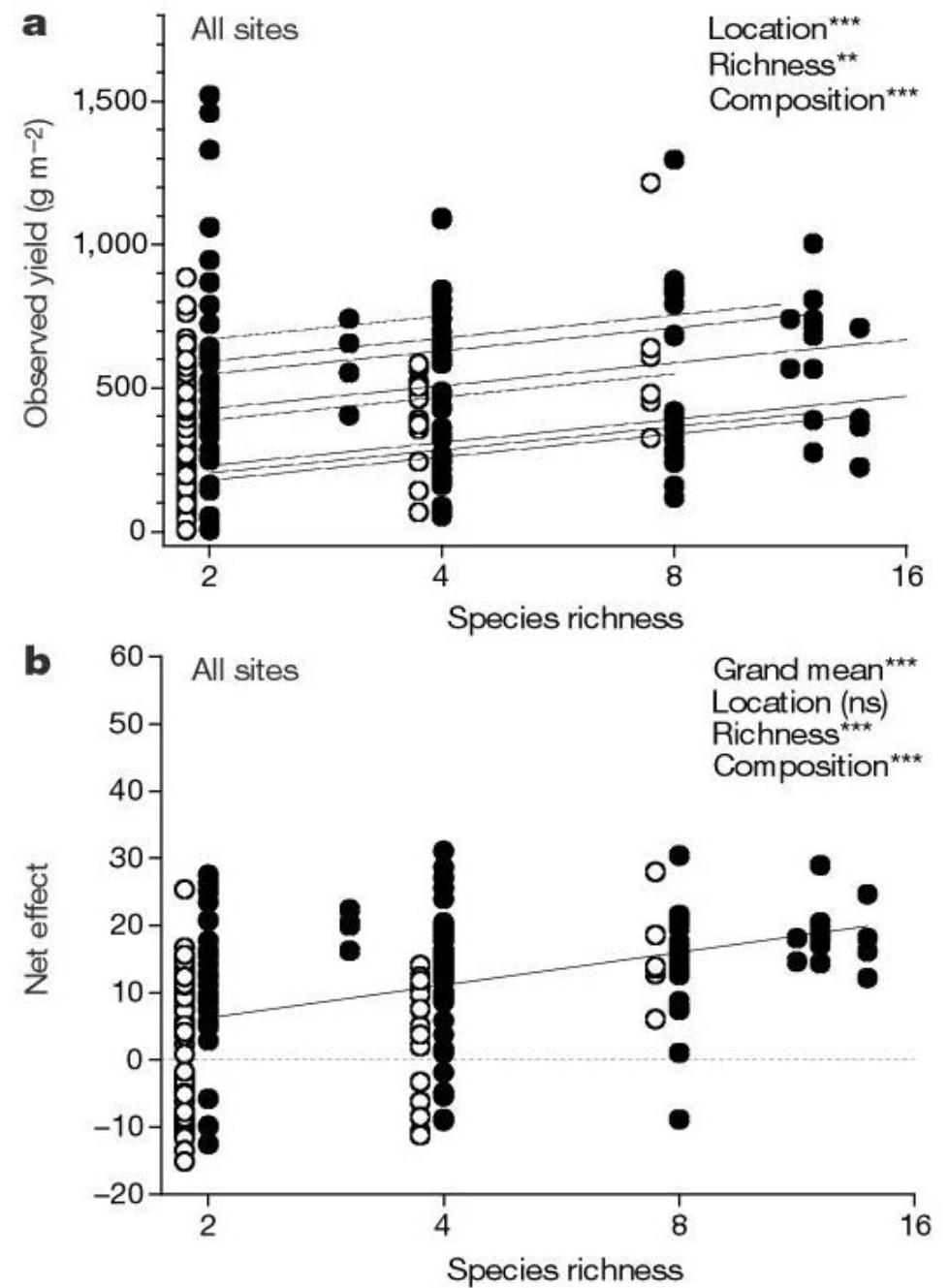


LOOK AT ALL THE SHITS



**I DON'T GIVE ABOUT
ECOLOGICAL COMMUNITIES**







Community structure

production

decomposition

trophic transfer

nutrient cycling

stability

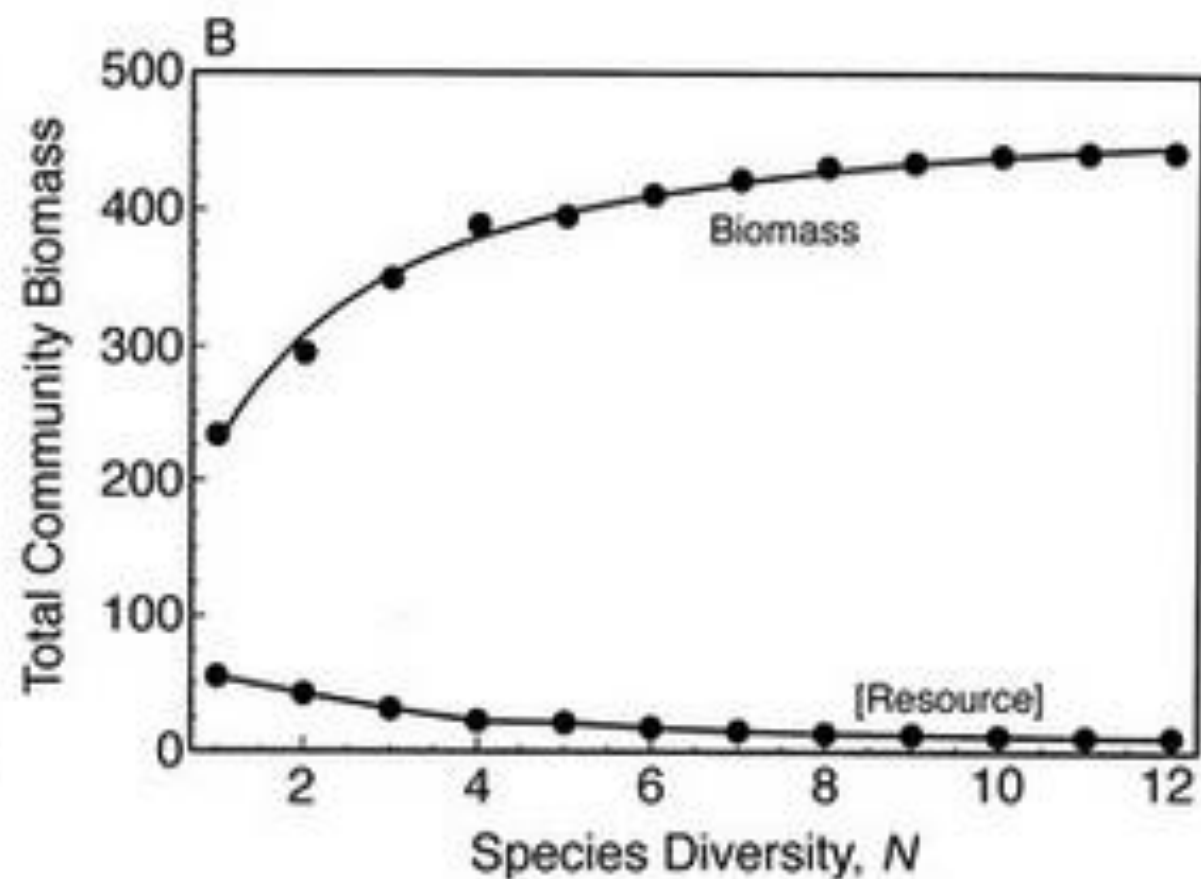
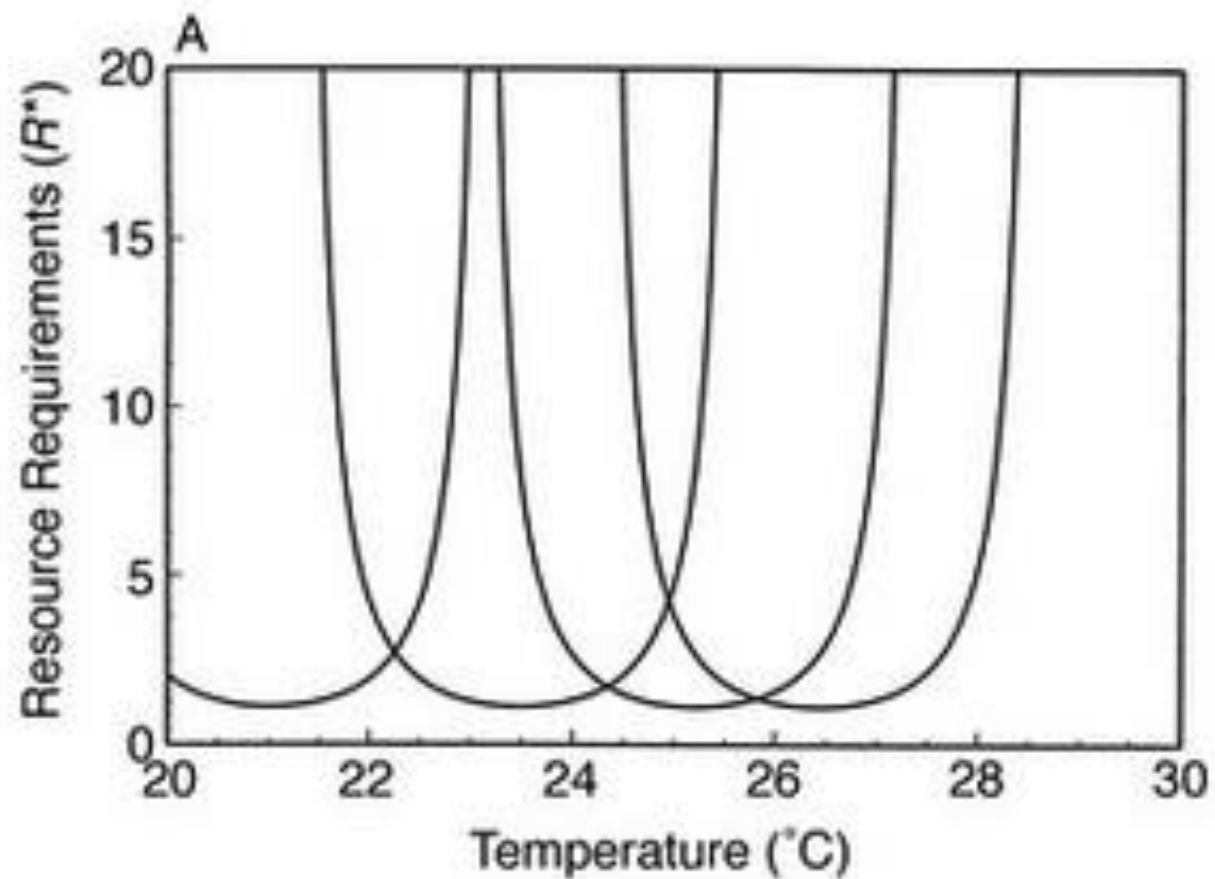
Ecosystem functioning





Theory: coexistence requires differences in resource use





More species = more traits = more function:
the biodiversity-ecosystem functioning
relationship (BEF)



Complementarity vs. redundancy



$\neq$



$=$



Complementarity vs. redundancy



$\neq$



$=$



Complementarity vs. redundancy



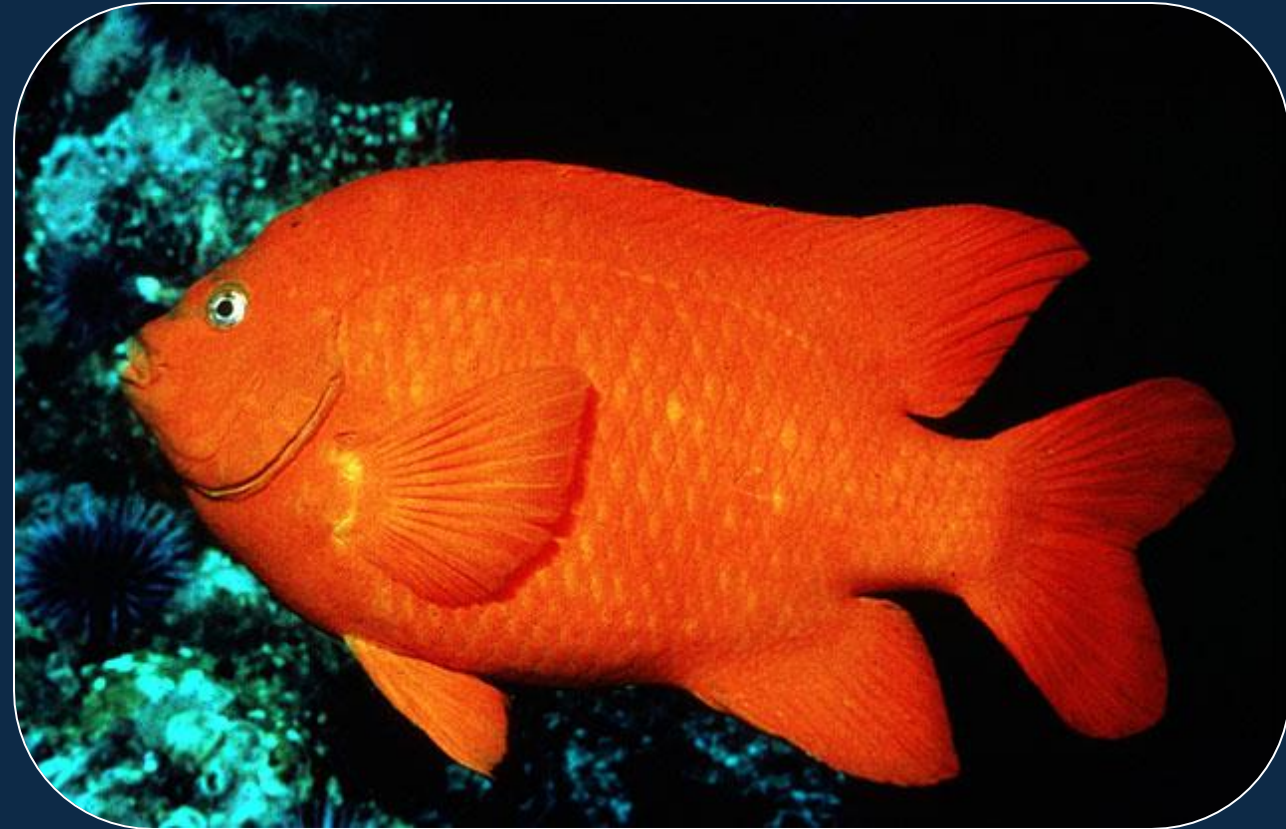
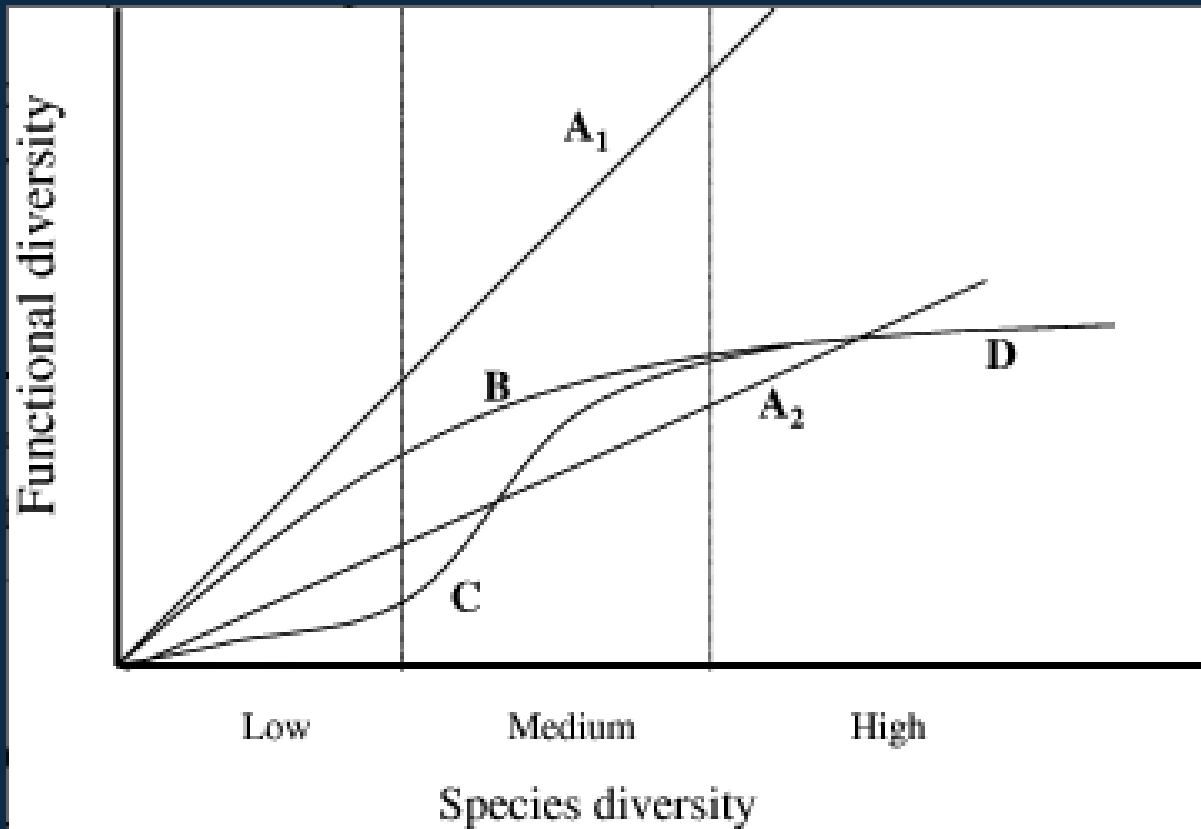
$\neq$



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Early explorations



Fishes

Herbivorous fish

Girella nigricans

Planktivorous fish

Chromis punctipinnis, *Sebastes mystinus*

Mobile small-invertebrate-eating fish

Embiotoca jacksoni, *Embiotoca lateralis*, *Oxyjulis californica*, *Damalichthys vacca*, *Hypsypops rubicundus*, *Alloclinus holderi*,

Territorial small-invertebrate-eating fish

Rhinogobiops nicholsii, *Lythrypnus dalli*

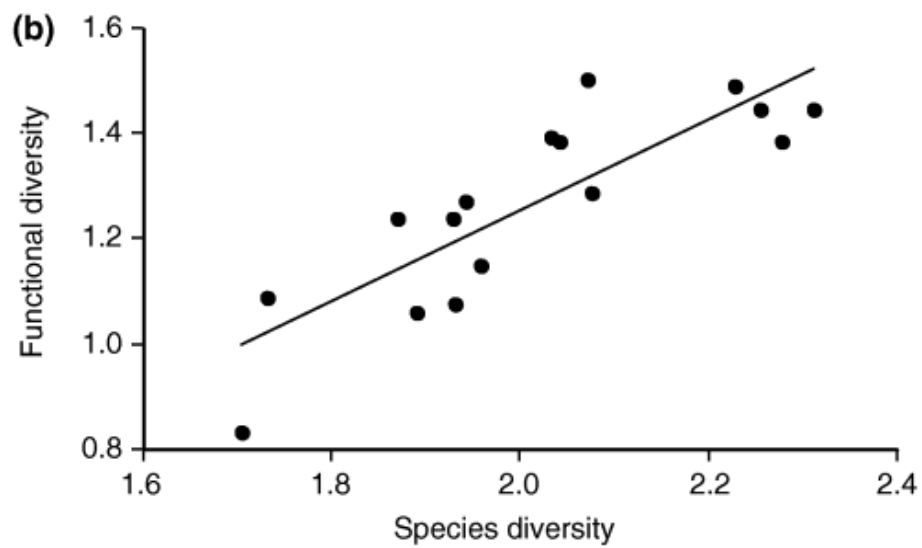
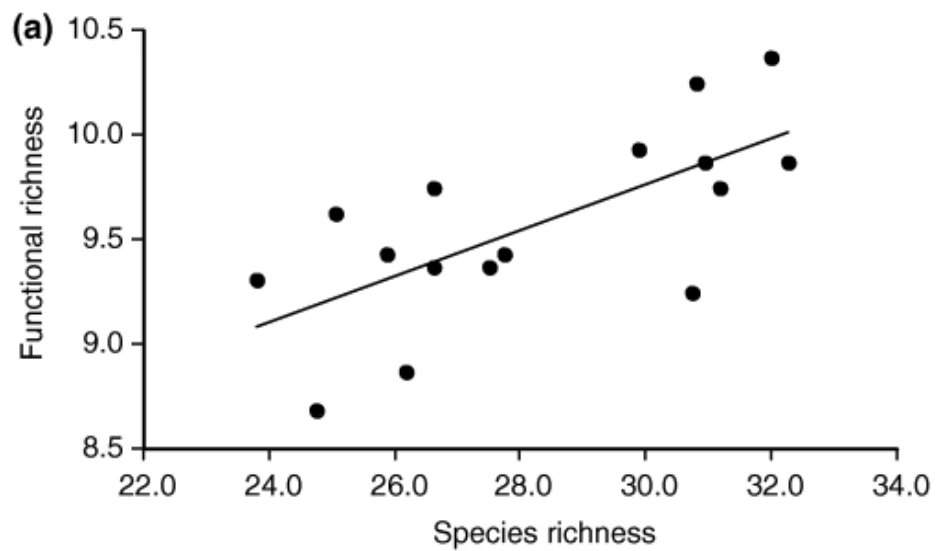
Large-invertebrate-eating fish

Semicossyphus pulcher

Invertebrate-eating fish/piscivores

Sebastes serranoides, *Sebastes atrovirens*, *Paralabrax clathratus*







Primary production, predation,
herbivory, nutrient cycling

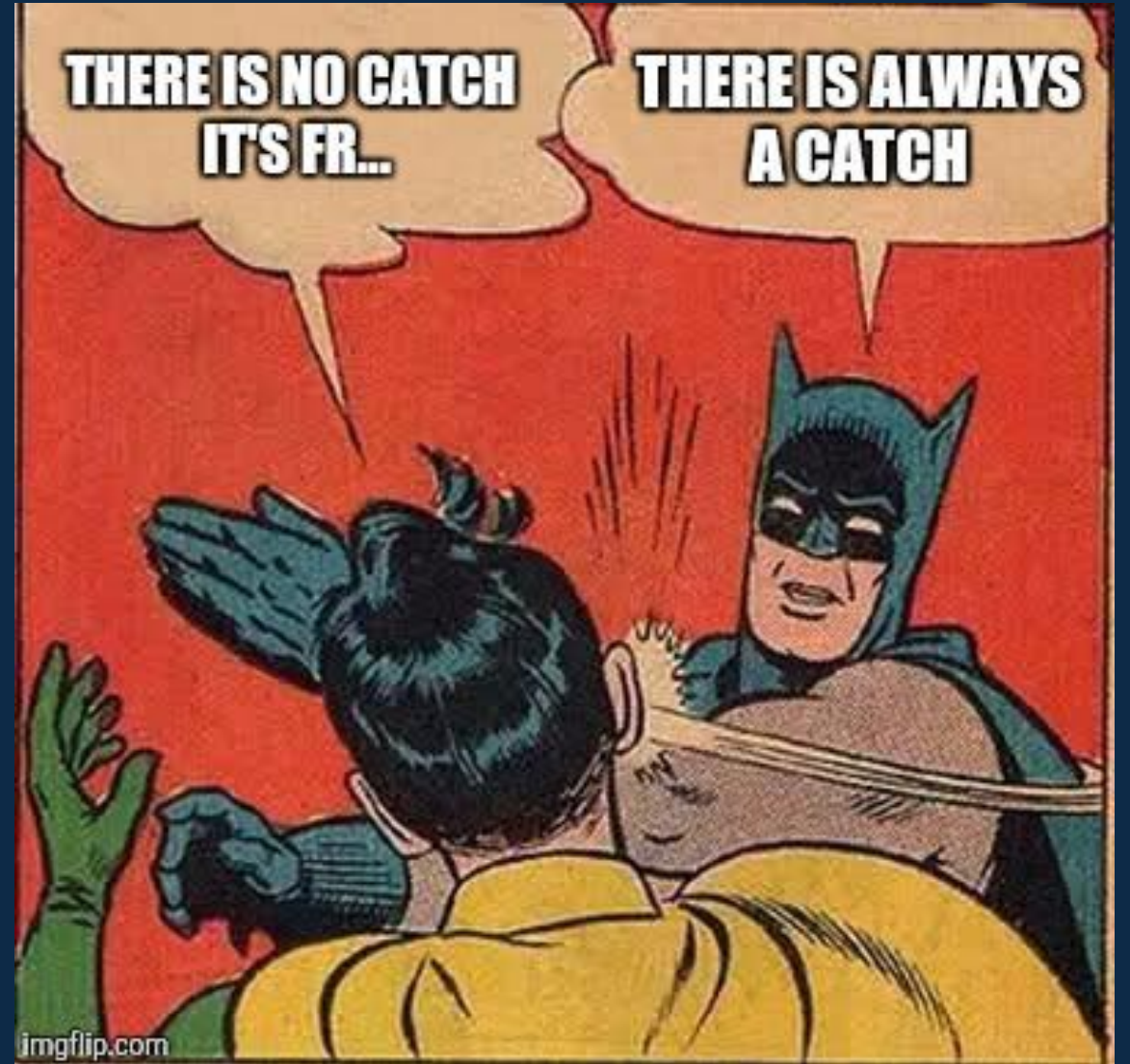
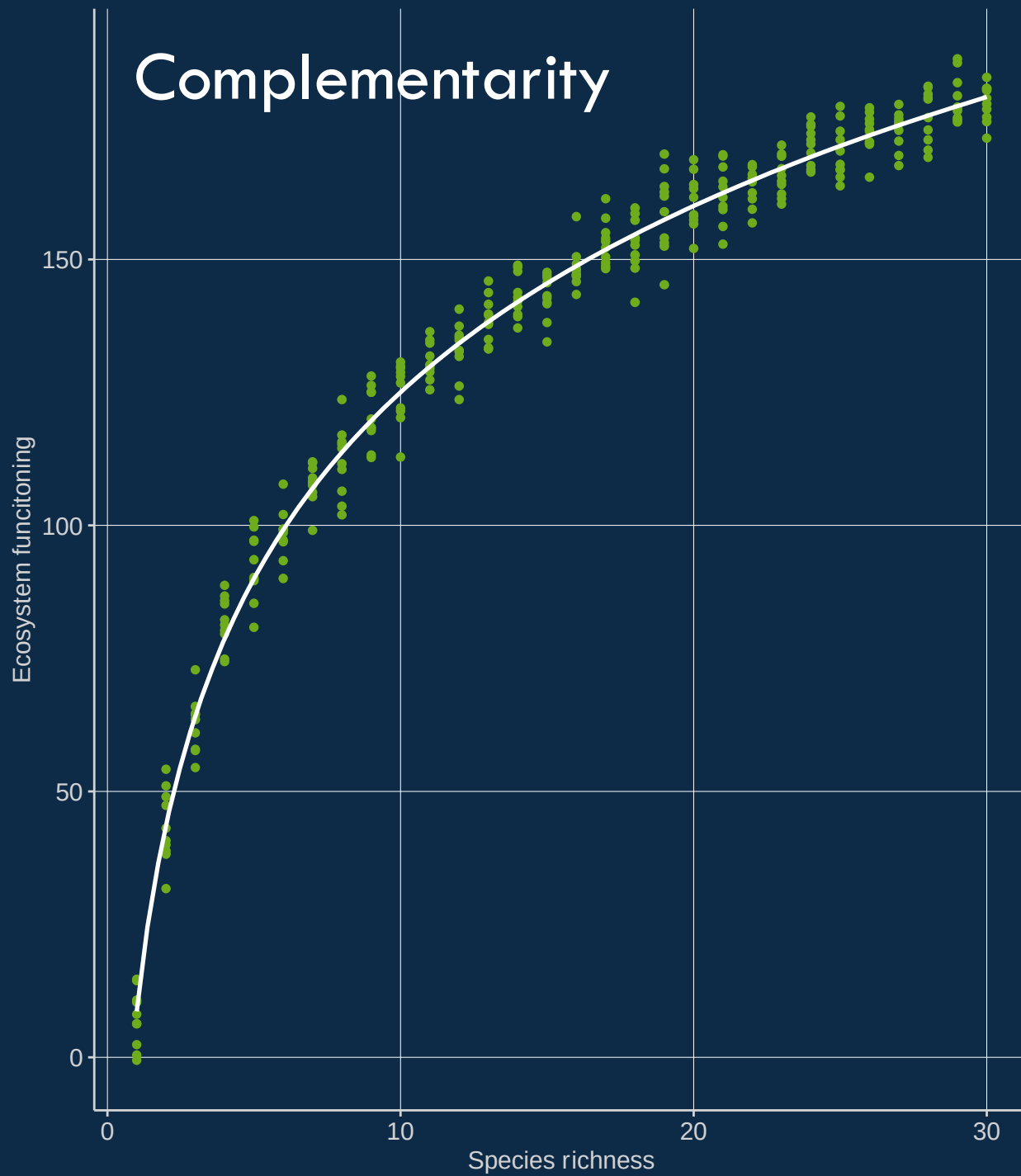


LET'S PUT THE FUN

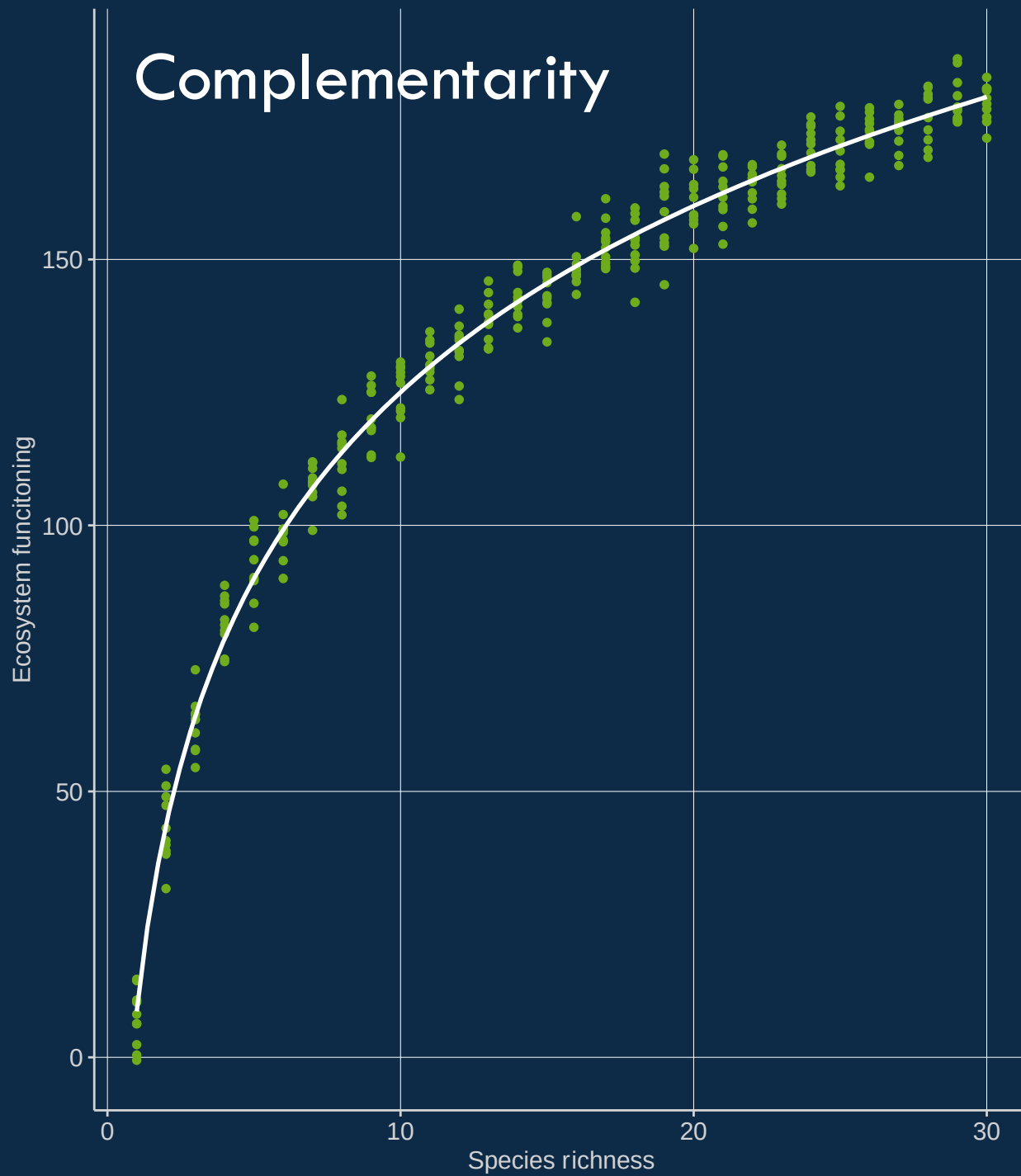


IN FUNCTION

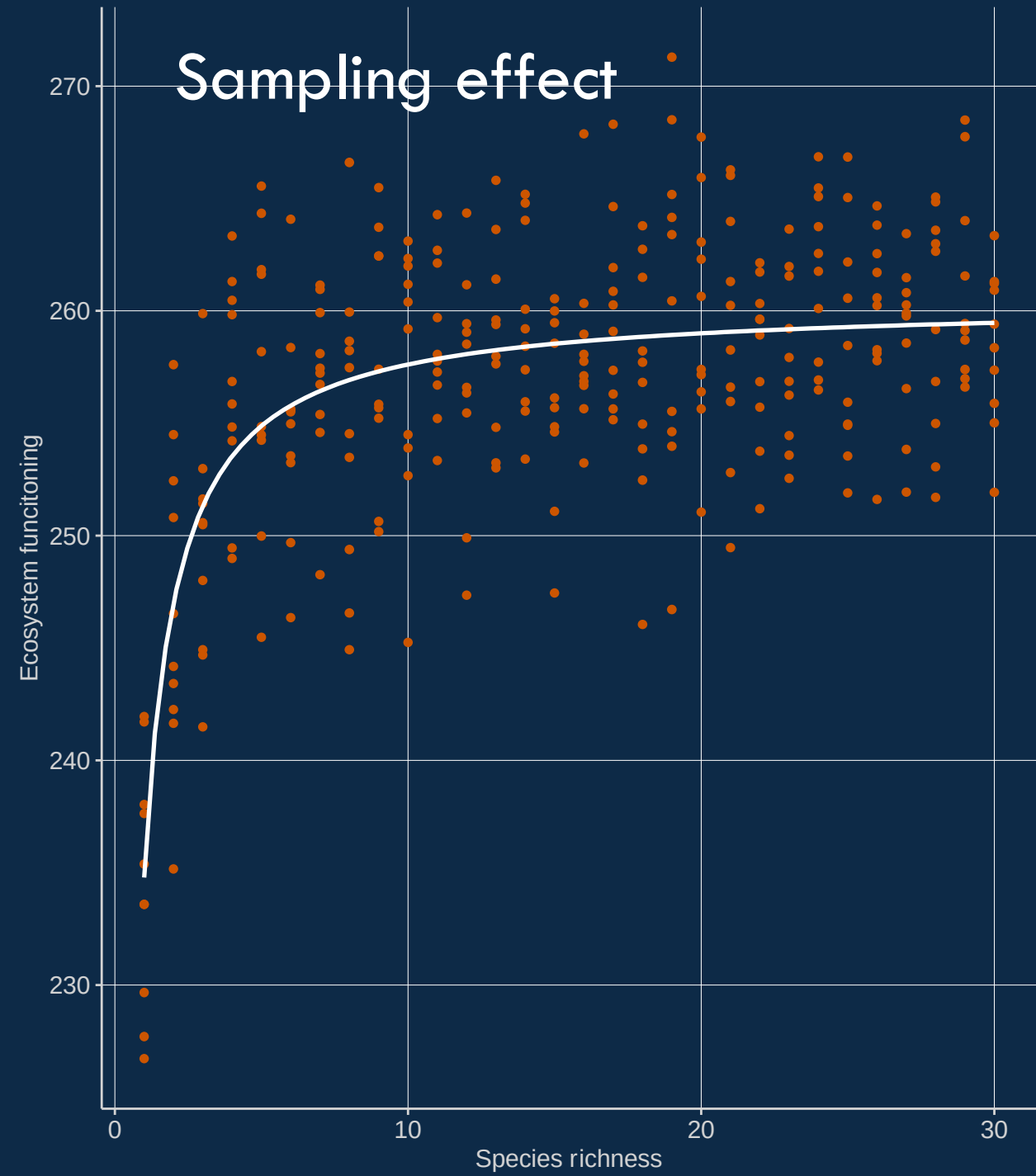
Complementarity



Complementarity

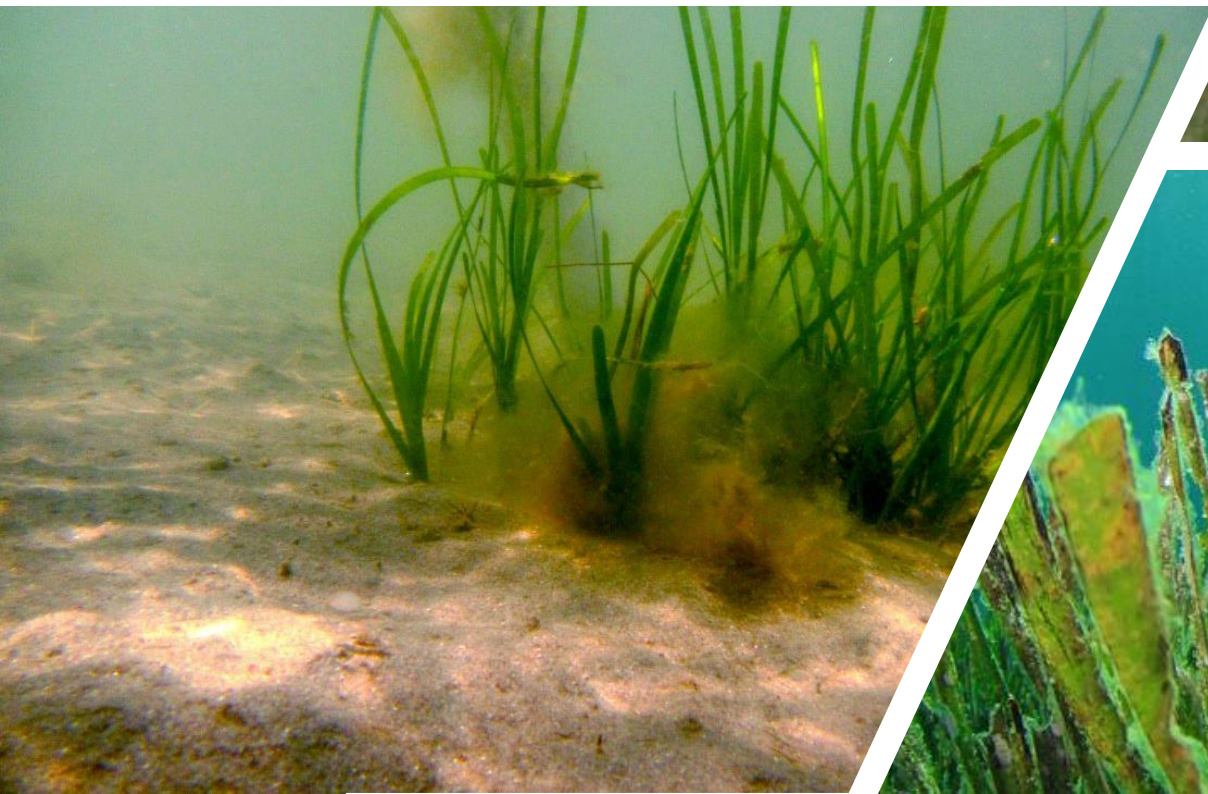


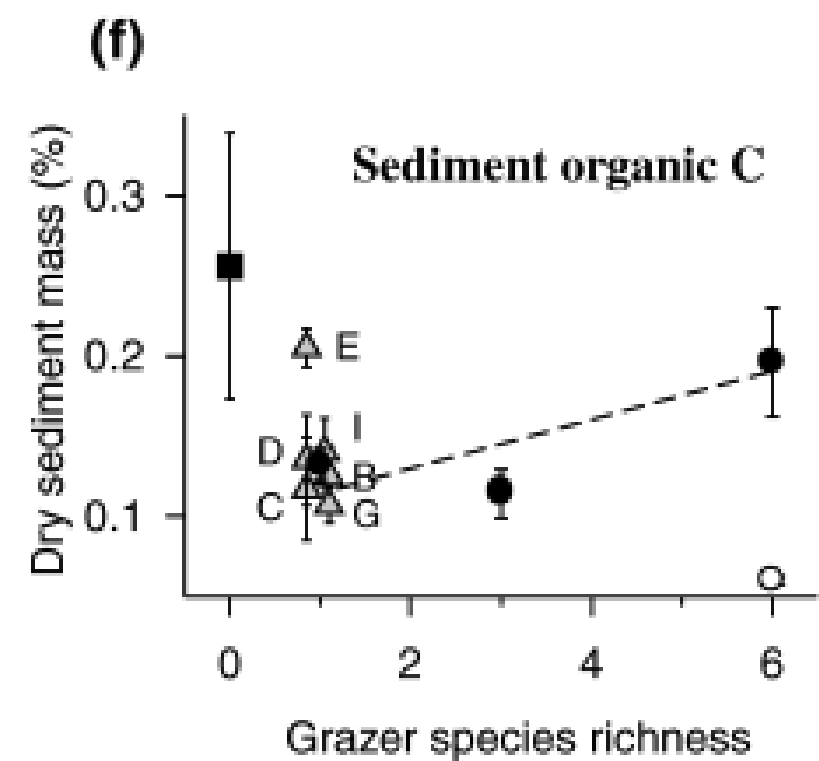
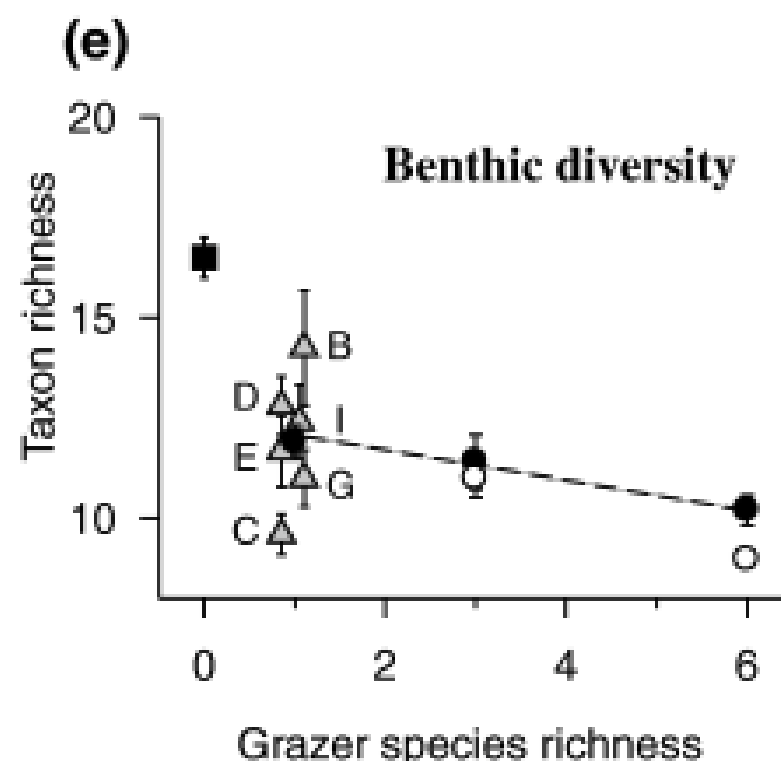
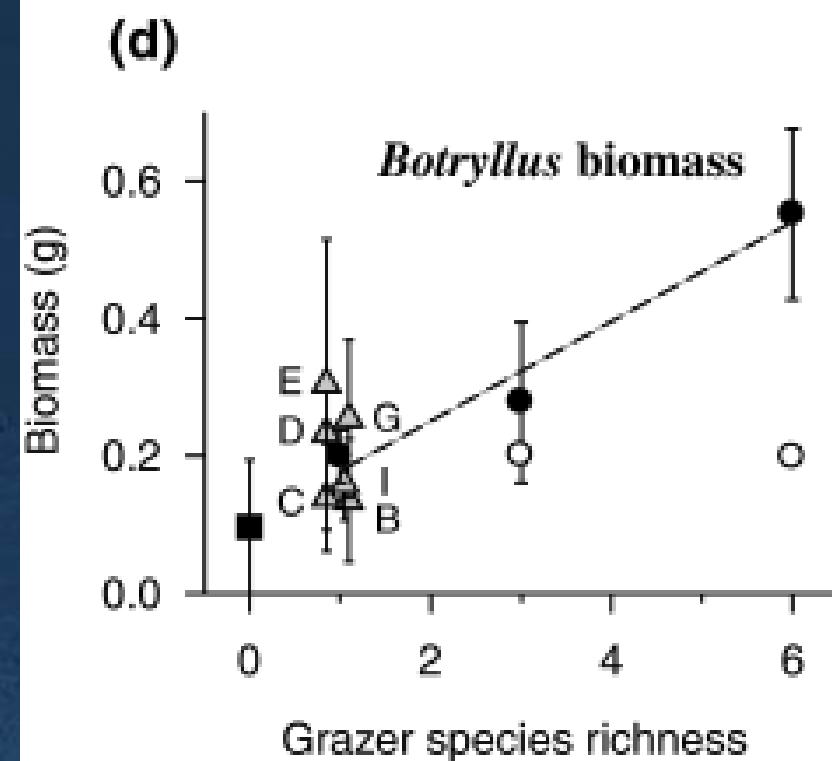
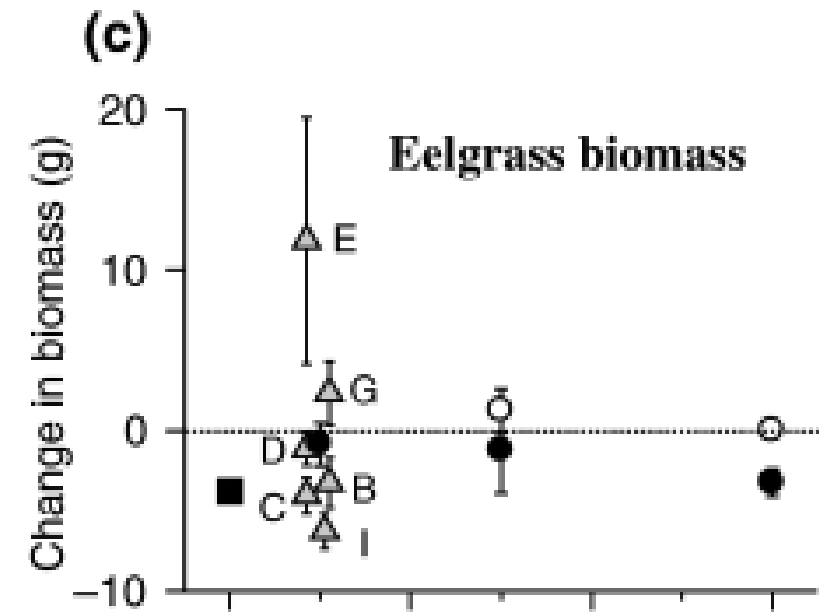
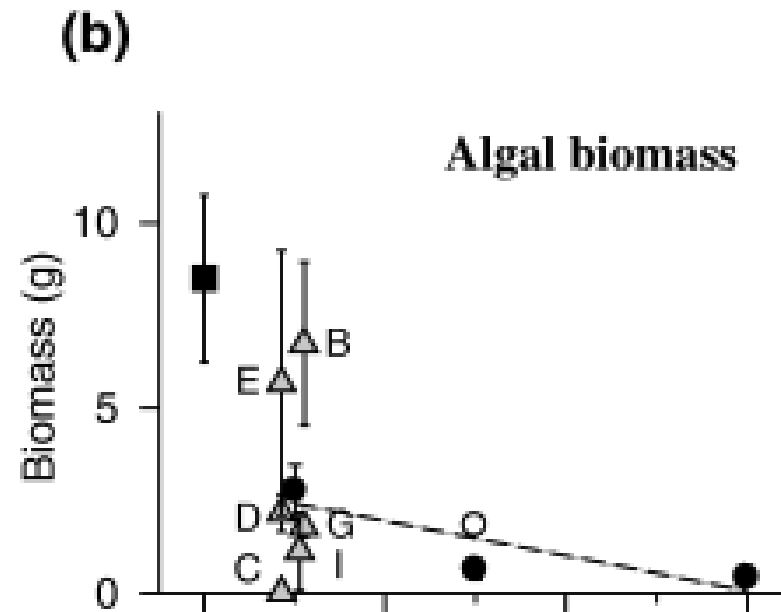
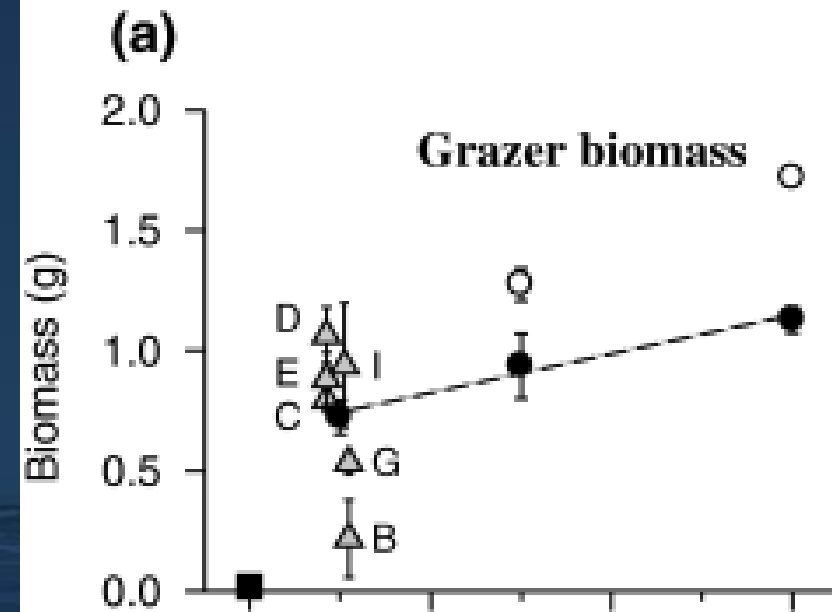
Sampling effect







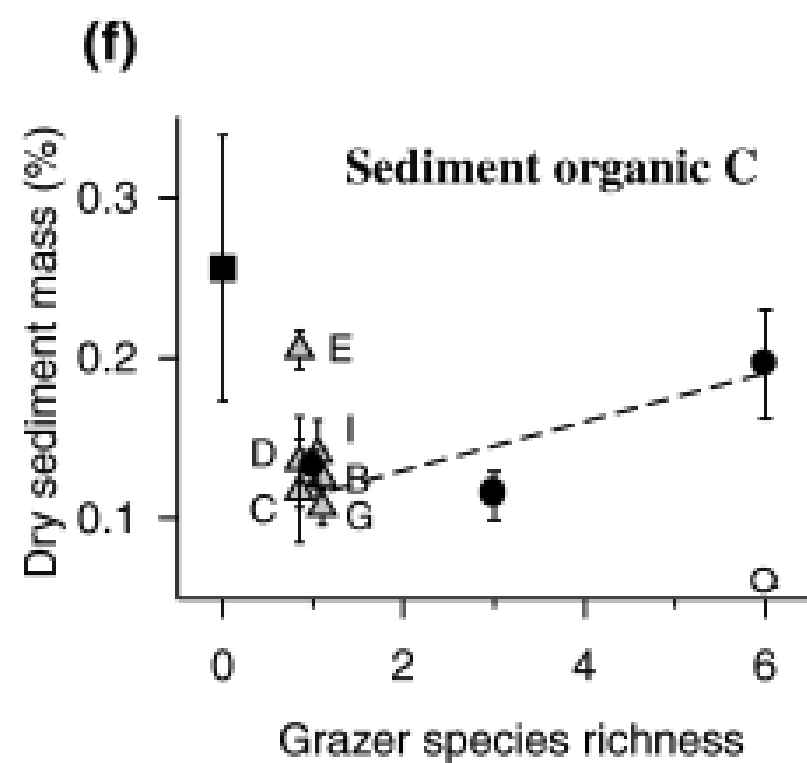
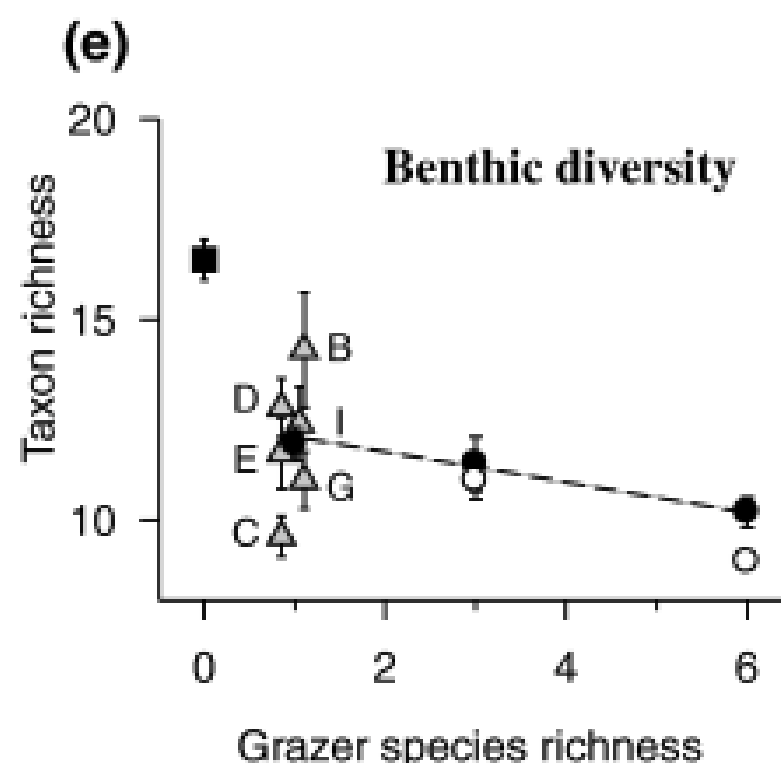
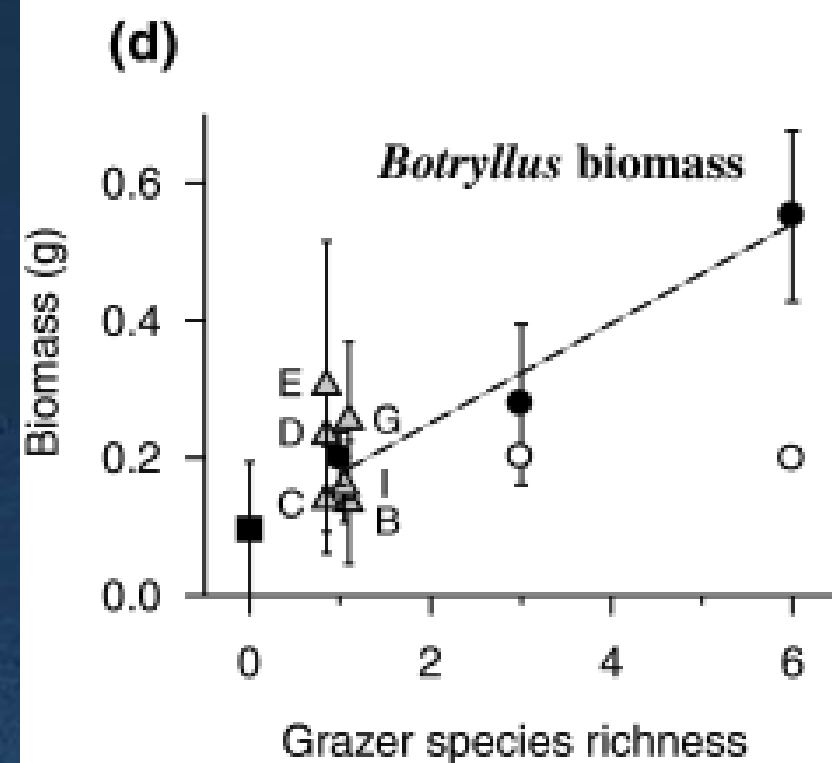
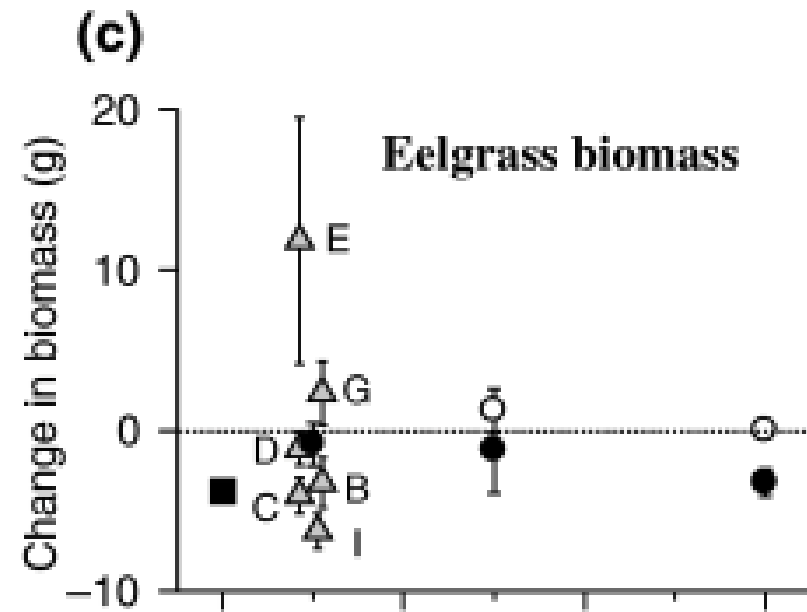
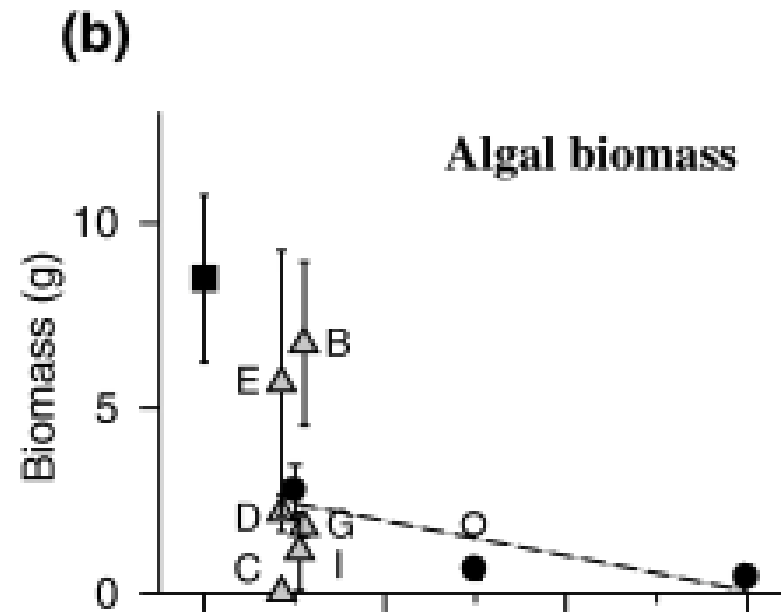
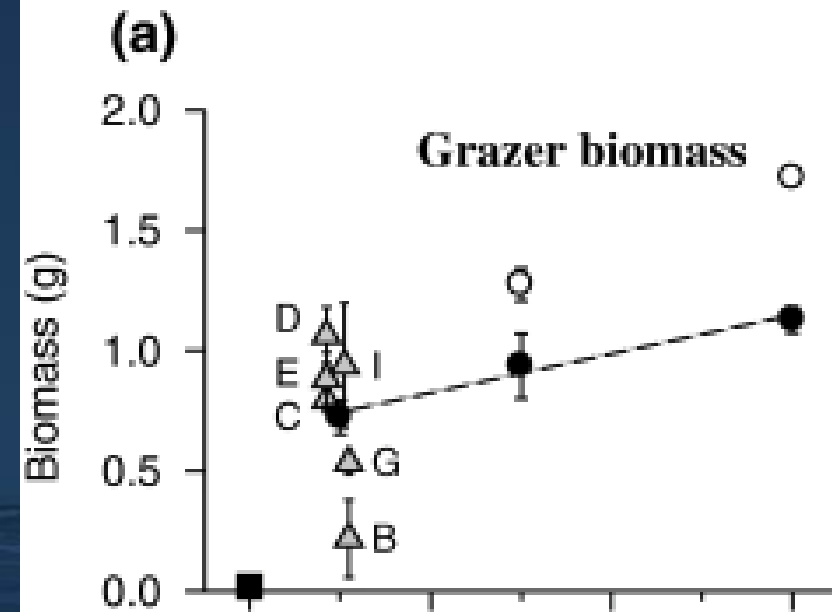


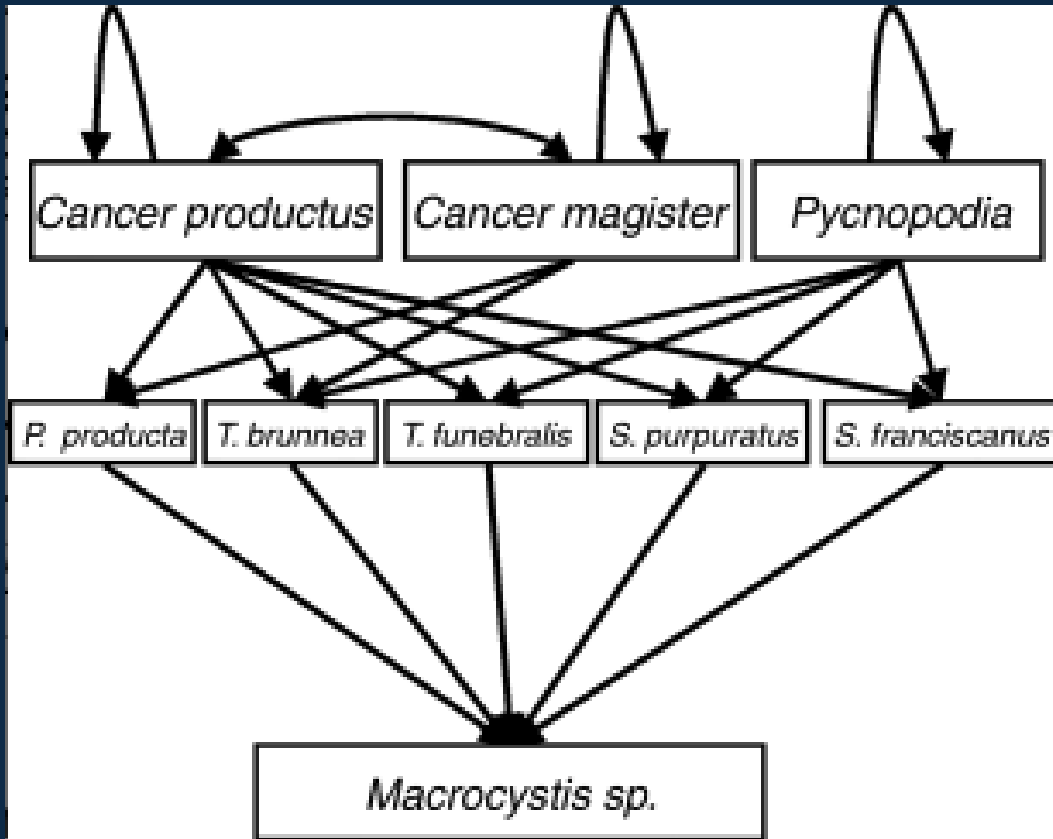


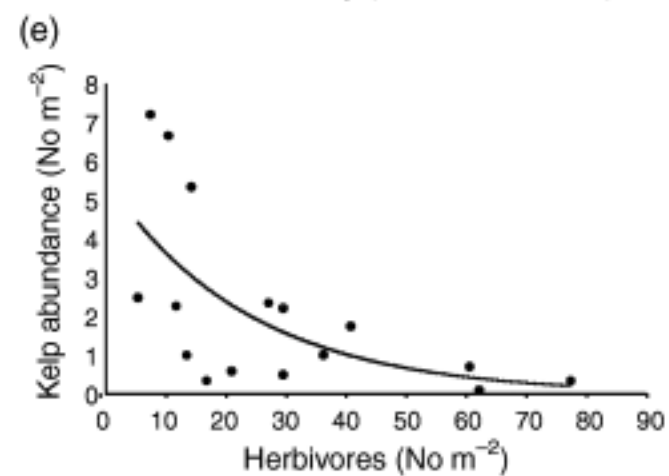
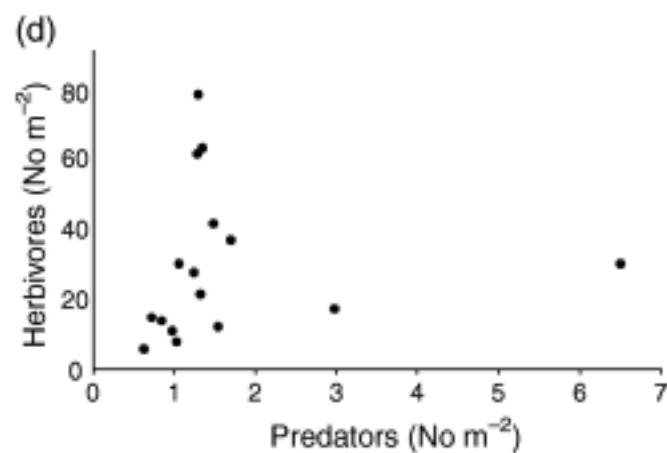
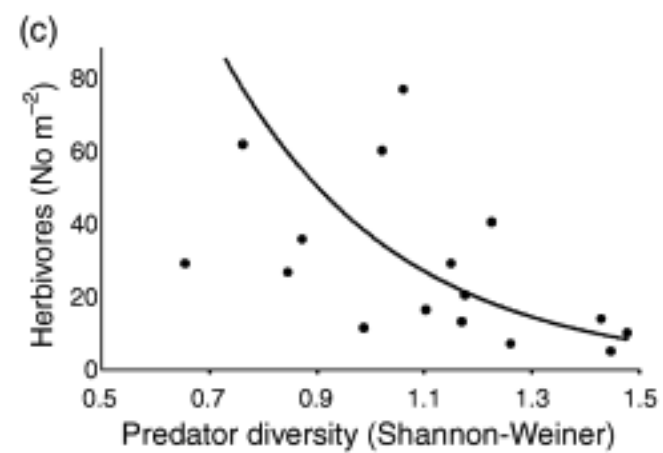
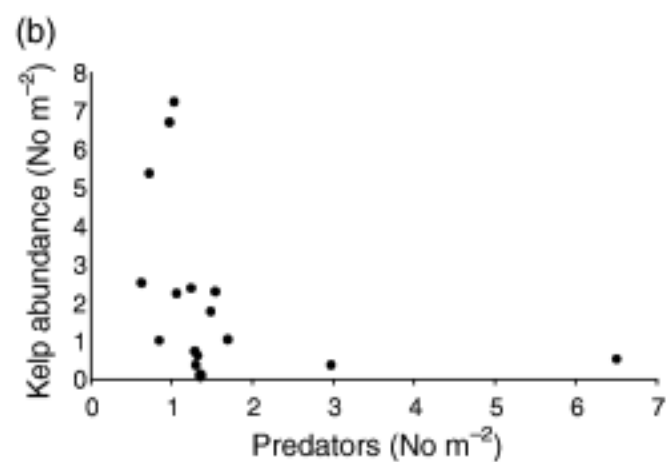
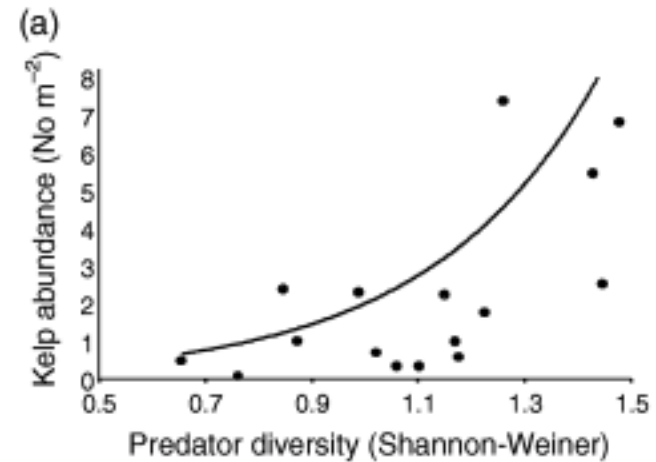
Take 5 minutes to mullet 🐟 over:

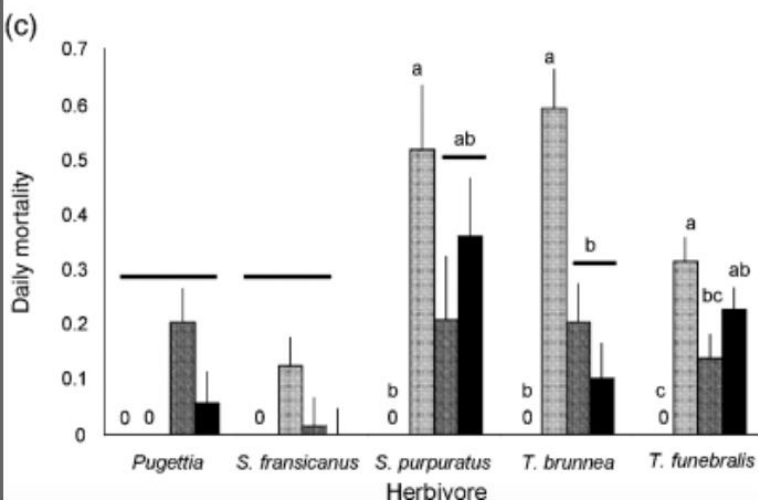
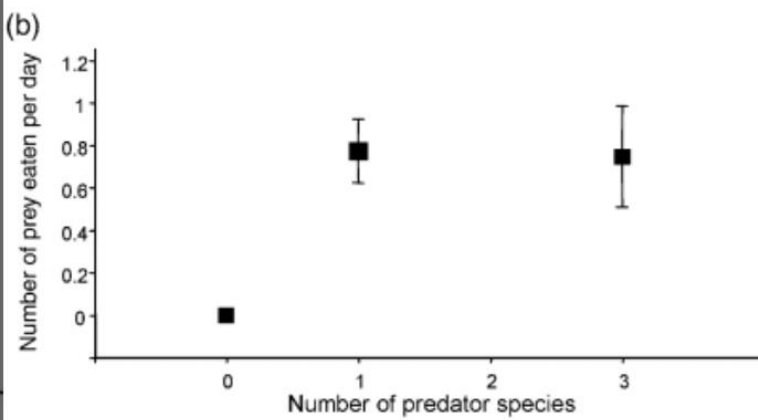
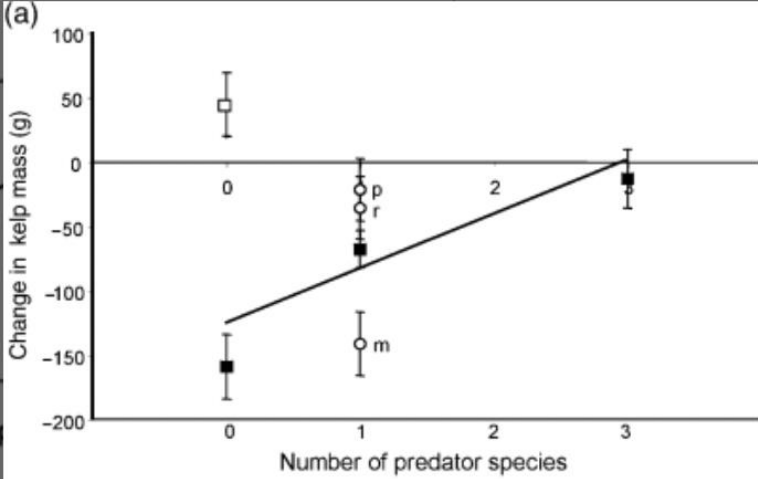
- How does the diversity of grazers affect seagrass ecosystems?
- Higher grazer diversity = higher functioning?



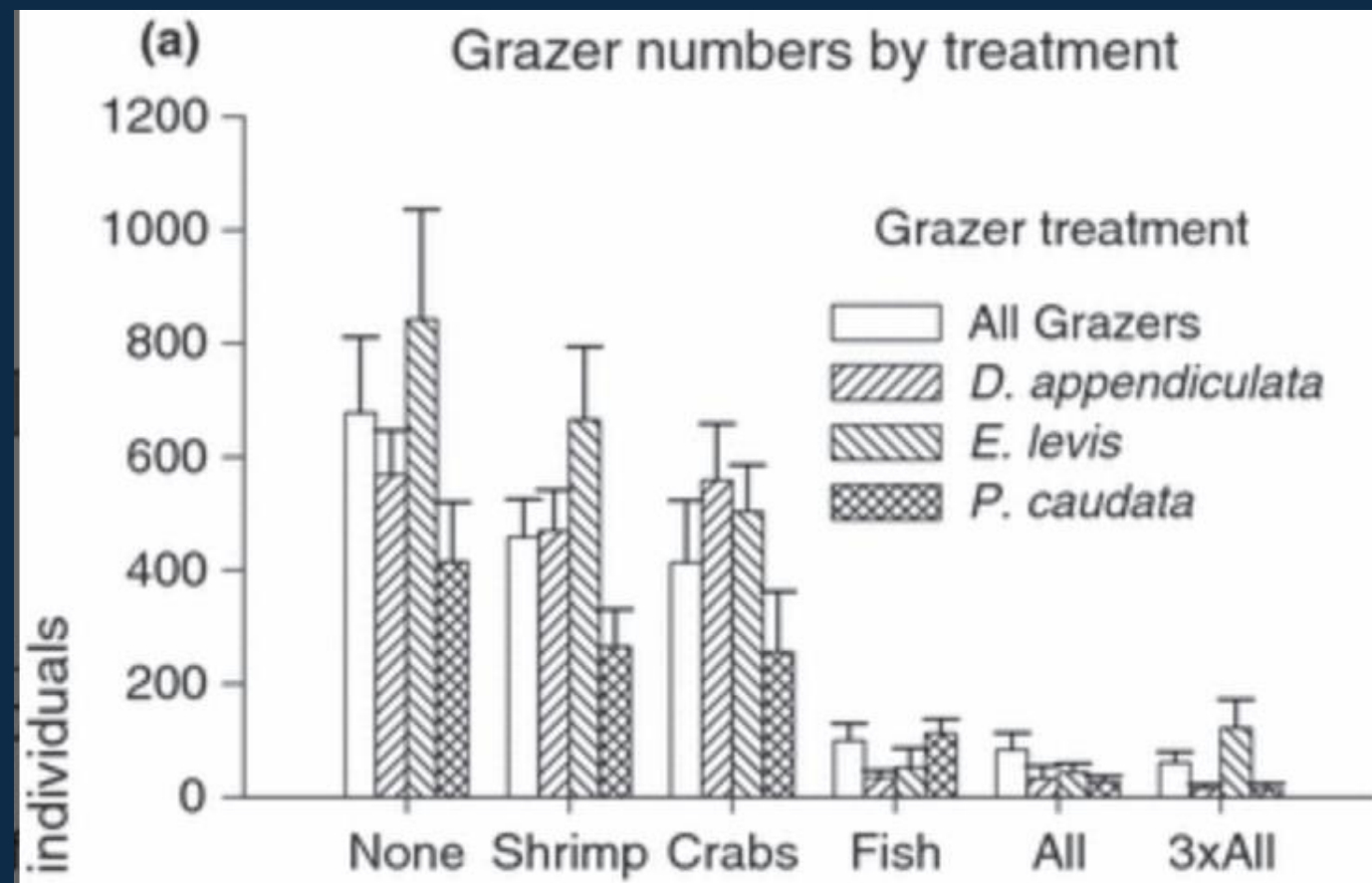






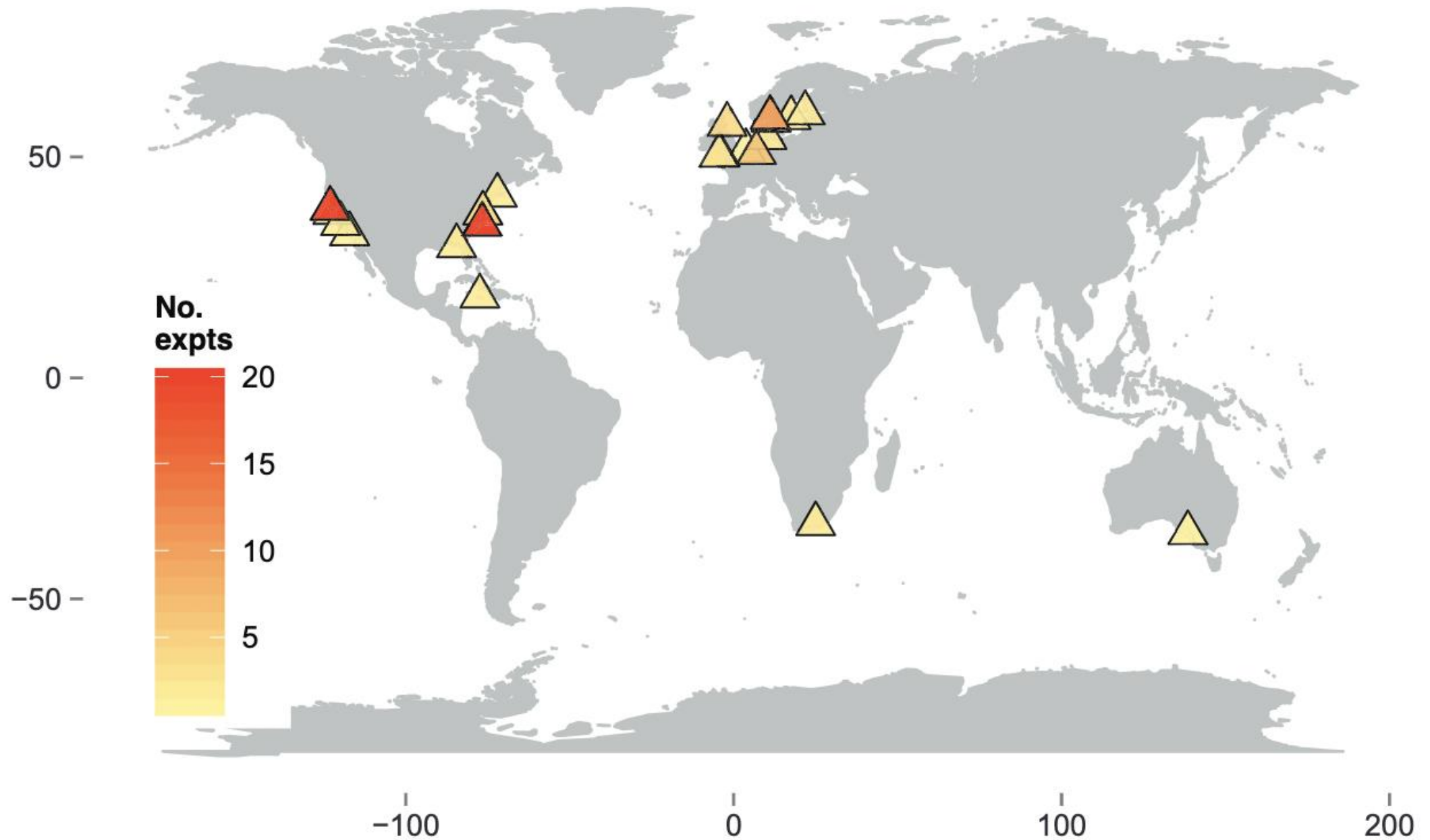


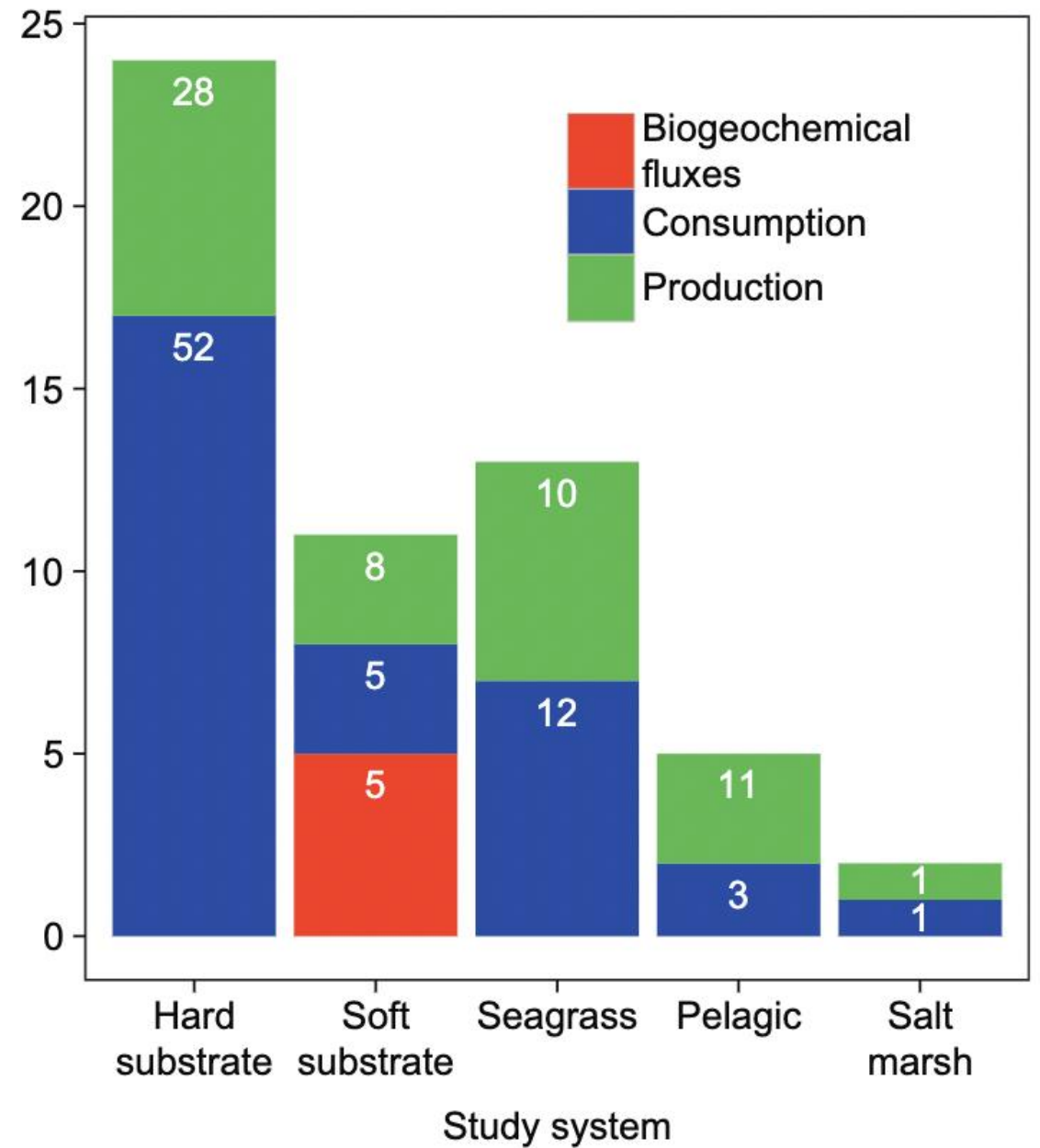
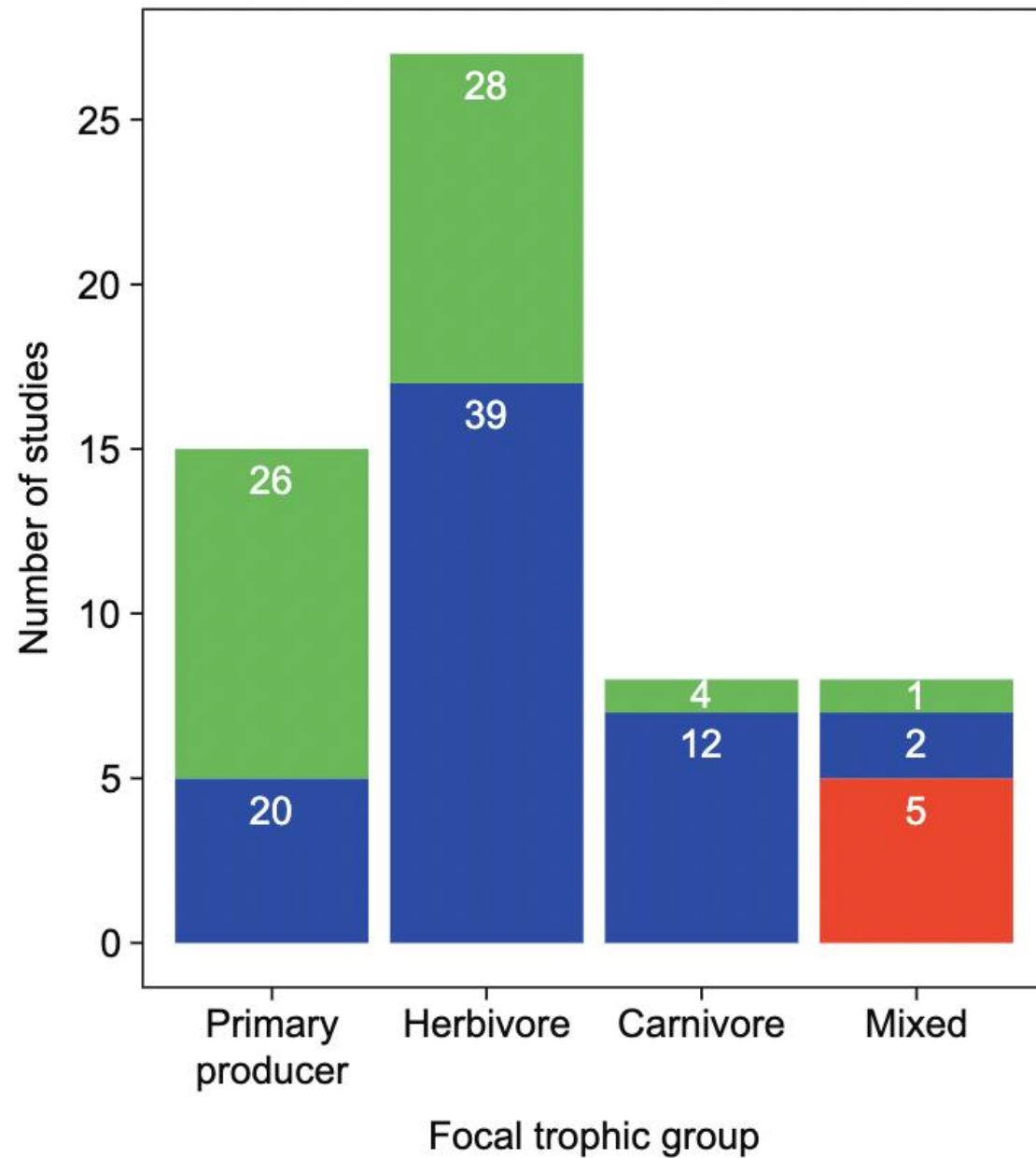


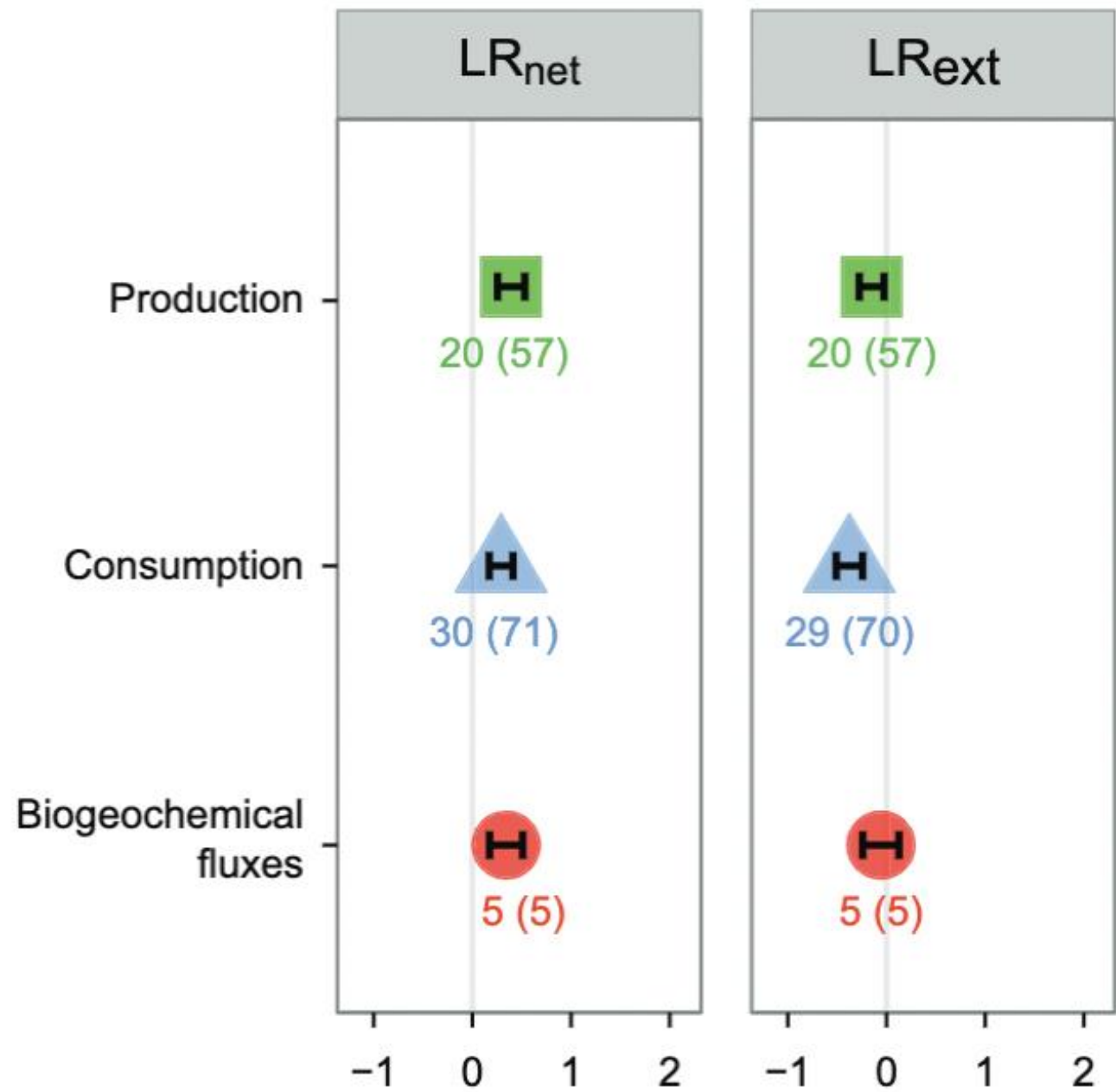


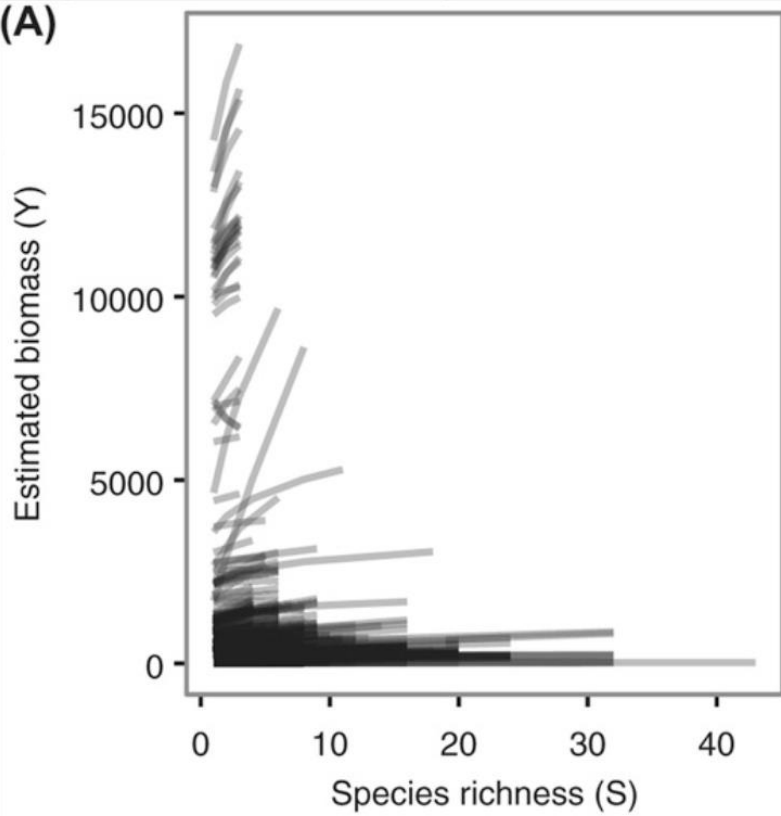
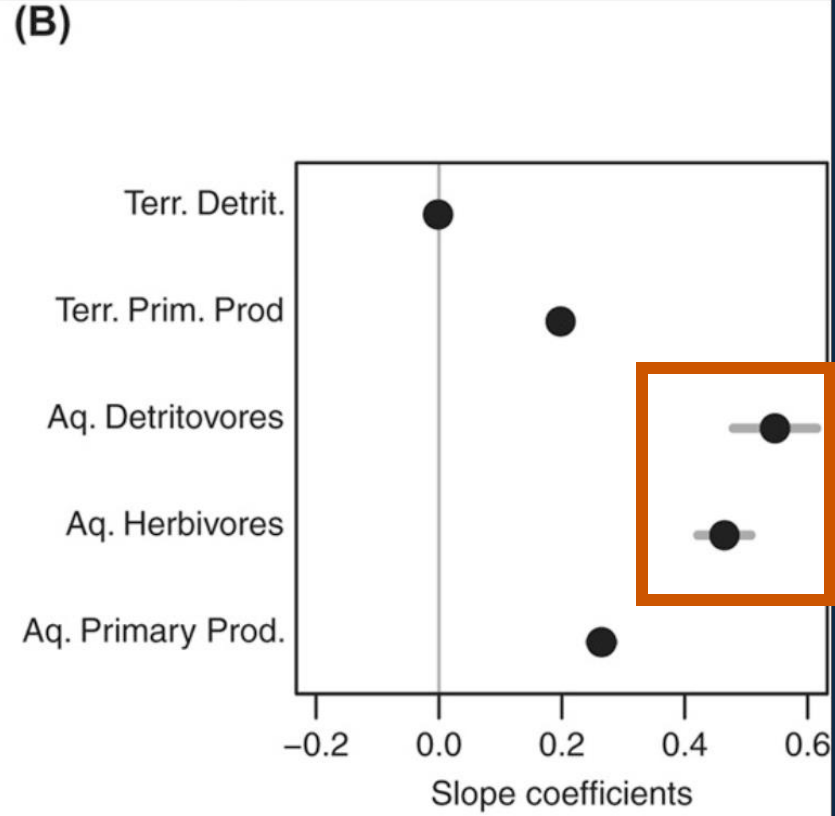
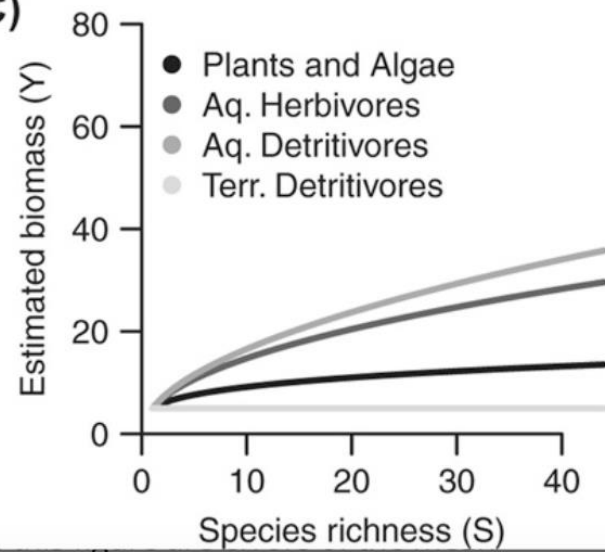


Steep increase in papers
examining the BEF since the
2000s

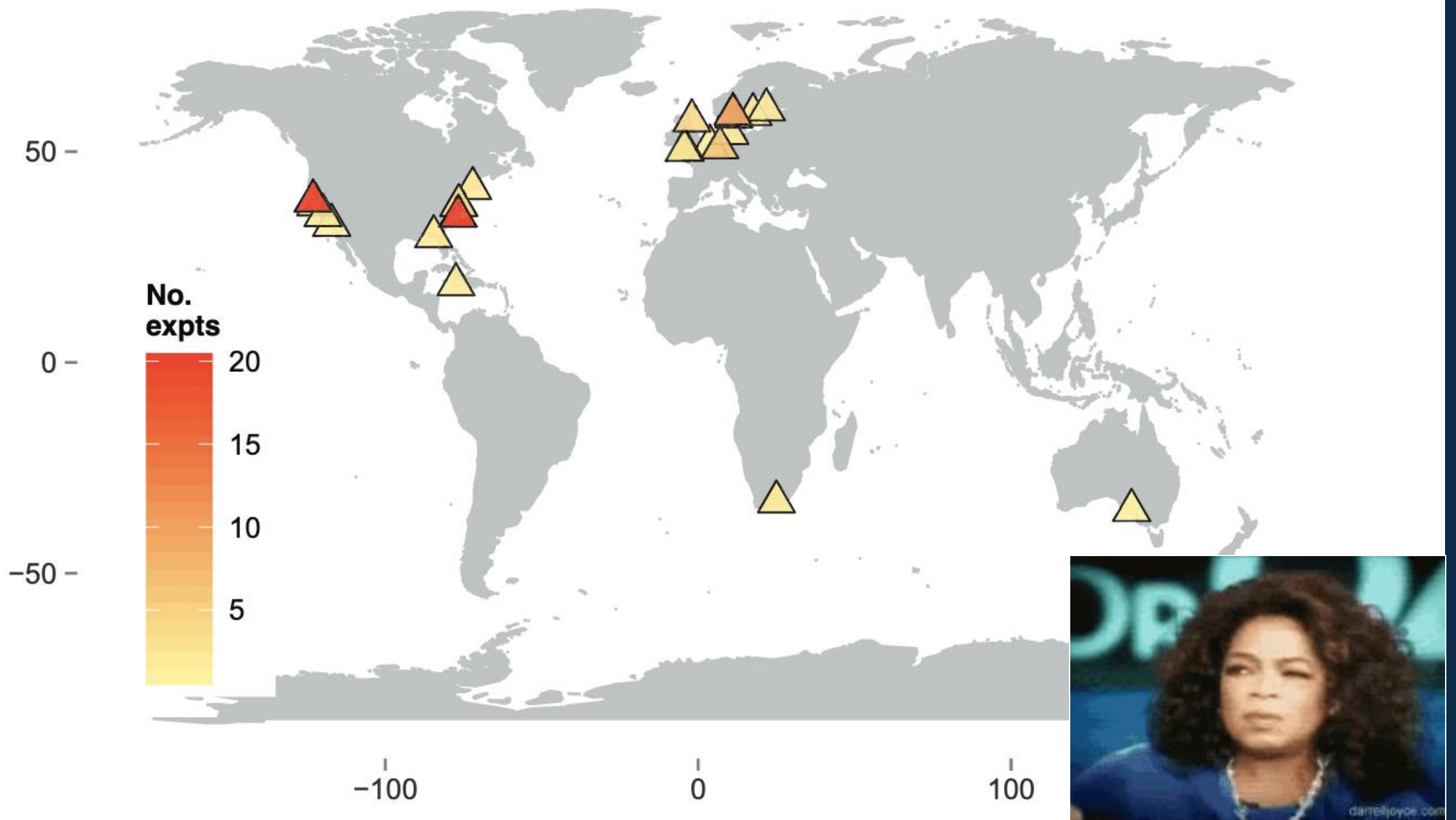






(A)**(B)****(C)**

O'Connor et al. 2017, Oikos



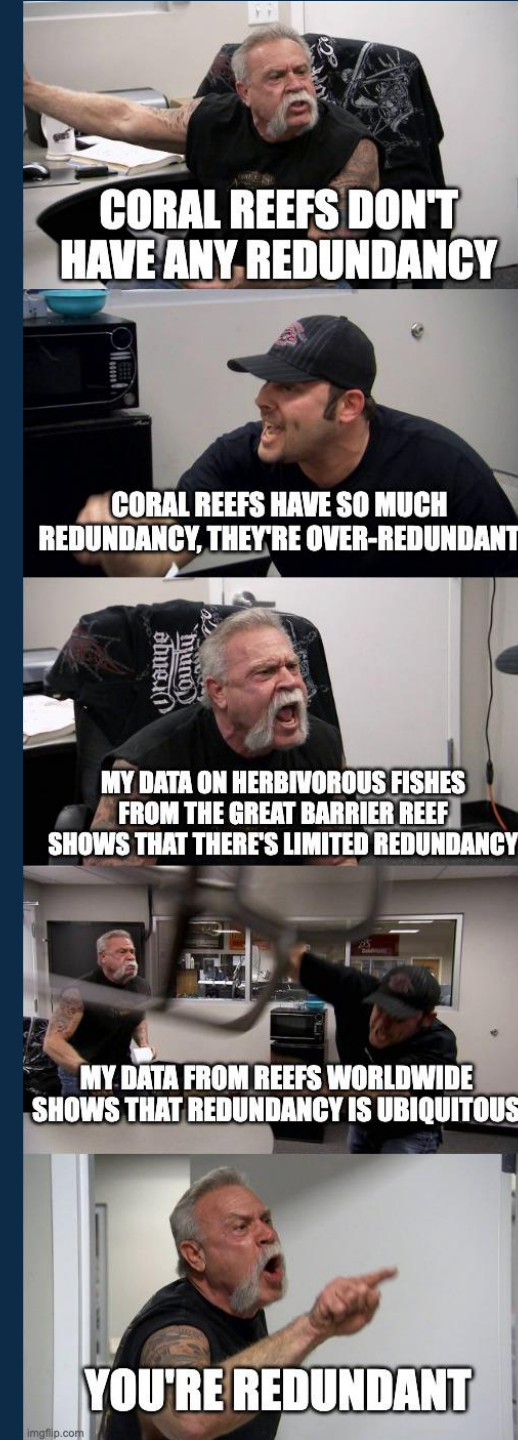
Gamfeldt et al. 2014, Oikos



REPORT

Key herbivores reveal limited functional redundancy on inshore coral reefs

C. L. Johansson · I. A. van de Leemput ·
M. Depczynski · A. S. Hoey · **D. R. Bellwood**



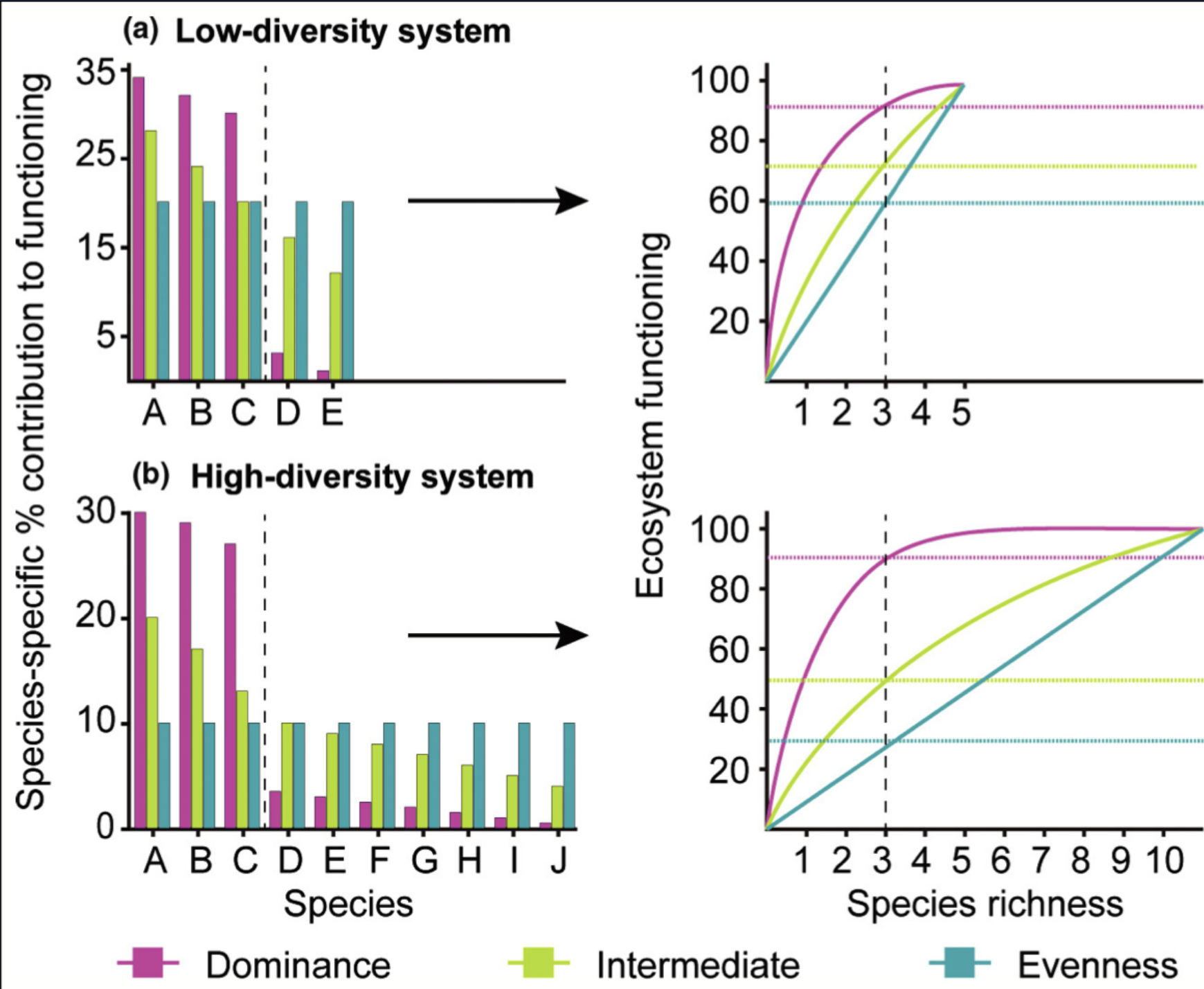
Functional over-redundancy and high functional vulnerability in global fish faunas on tropical reefs

David Mouillot^{a,b,1,2}, Sébastien Villéger^{a,1}, Valeriano Parravicini^{c,d,1}, Michel Kulbicki^c, Jesus Ernesto Arias-González^e, Mariana Bender^{a,f}, Pascale Chabanet^g, Sergio R. Floeter^f, Alan Friedlander^h, Laurent Vigliolaⁱ, and **David R. Bellwood^{b,j}**

^aEcologie des Systèmes Marins Côtiers, Unité Mixte de Recherche 5119, Centre National de la Recherche Scientifique-Institut de Recherche pour le Développement-Université Montpellier 2-Institut Français de Recherche pour l'Exploitation de la Mer, Université Montpellier 2, 34095 Montpellier, France; ^bAustralian Research Council Centre of Excellence for Coral Reef Studies, James Cook University, Townsville, QLD 4811, Australia; ^cInstitut de Recherche pour le Développement, Unité de Recherche 227 Biocomplexité des écosystèmes coralliens de l'Indo-Pacifique, LABEX Corail, Laboratoire Arago, BP 44, 66651 Banyuls sur Mer, France; ^dCenter for Synthesis and Analysis of Biodiversity-Fondation pour la Recherche sur la Biodiversité Immeuble Henri Poincaré, Domaine du Petit Arbois, 13857 Aix-en-Provence Cedex 3, France; ^eLaboratorio de Ecología de Ecosistemas de Arrecifes Coralinos, Departamento de Recursos del Mar,



SPECIAL FEATURE

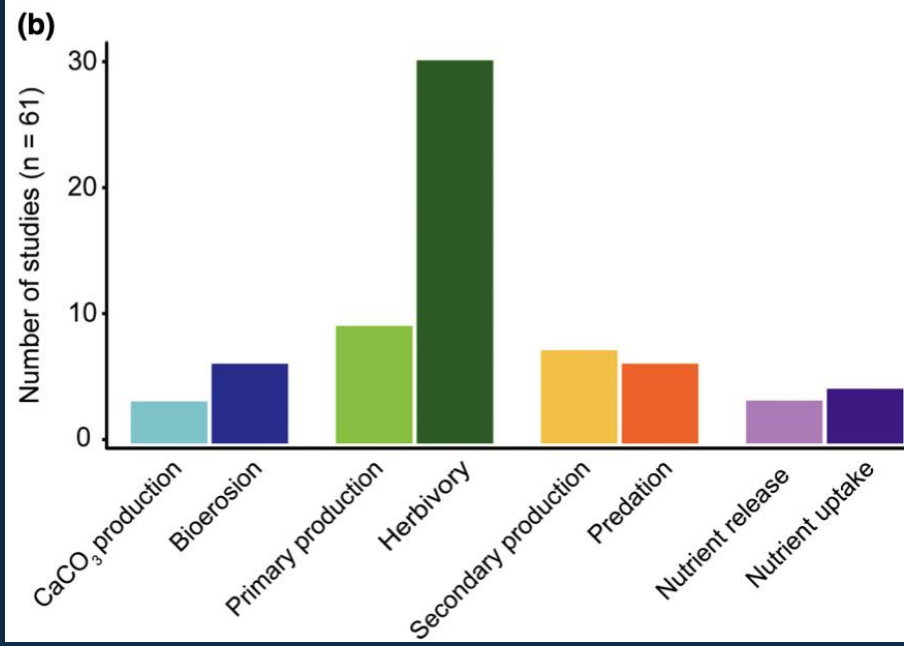
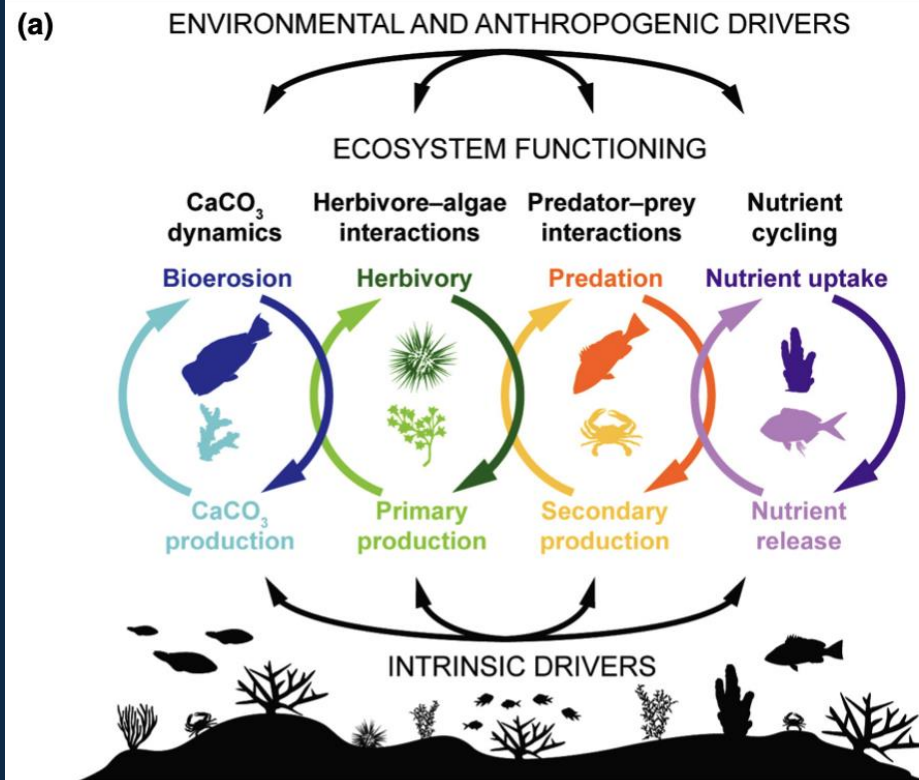
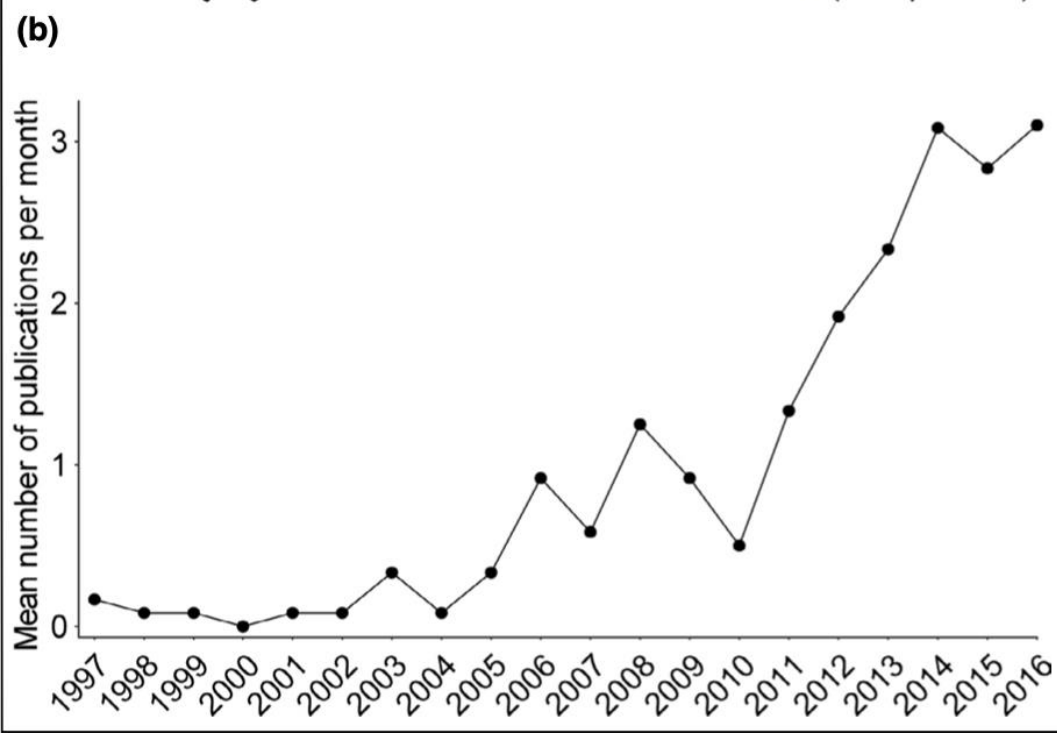
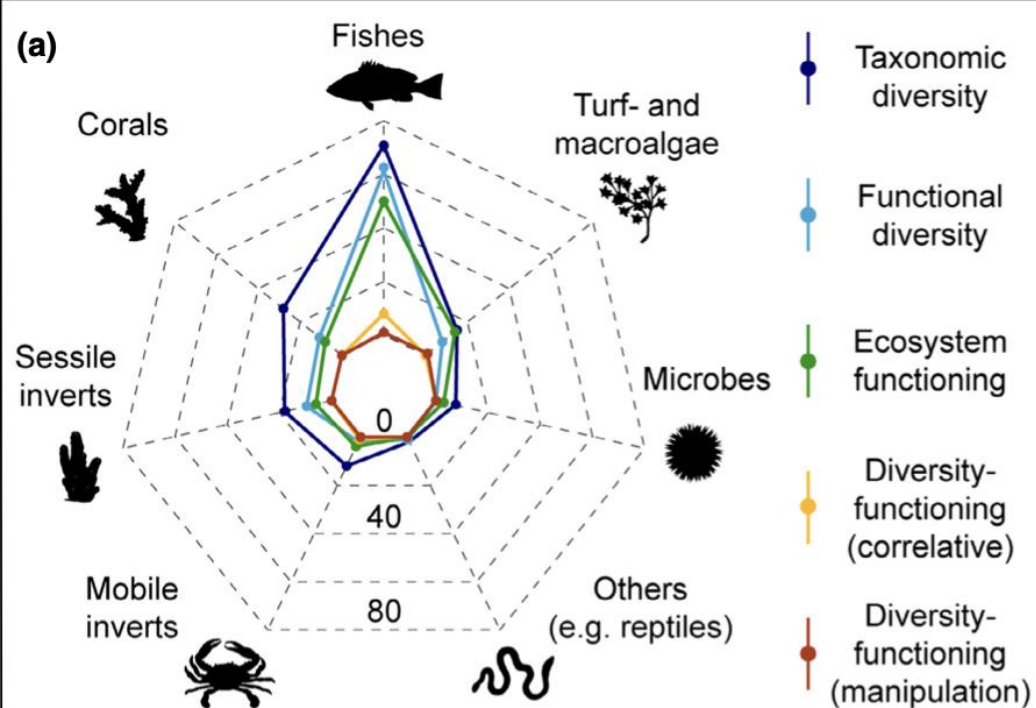


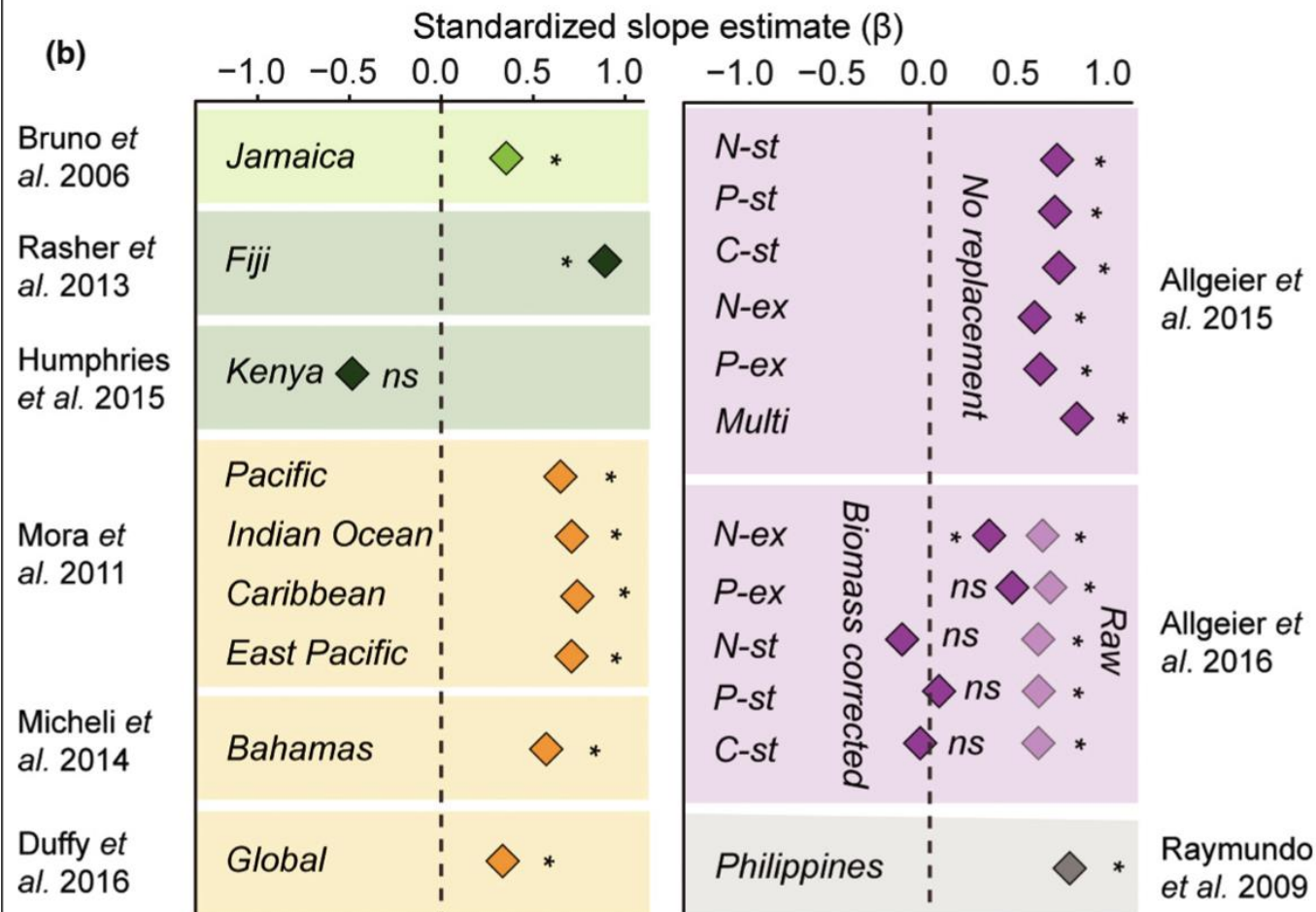
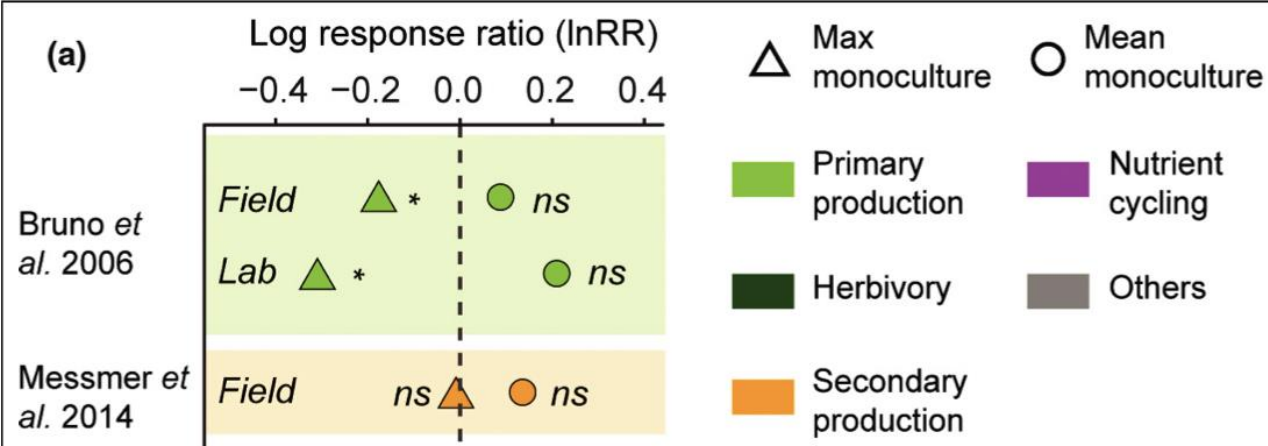


MMS 353 582

MAKING COMMUNITY ECOLOGY

Tane Sinclair-Taylor
Expedition Photographer



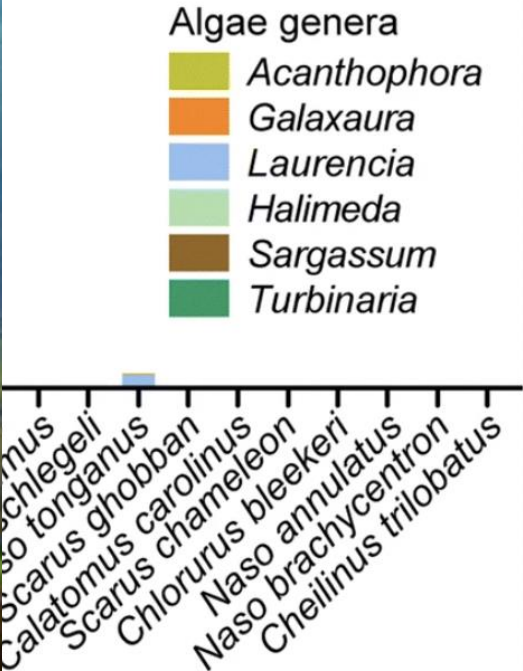
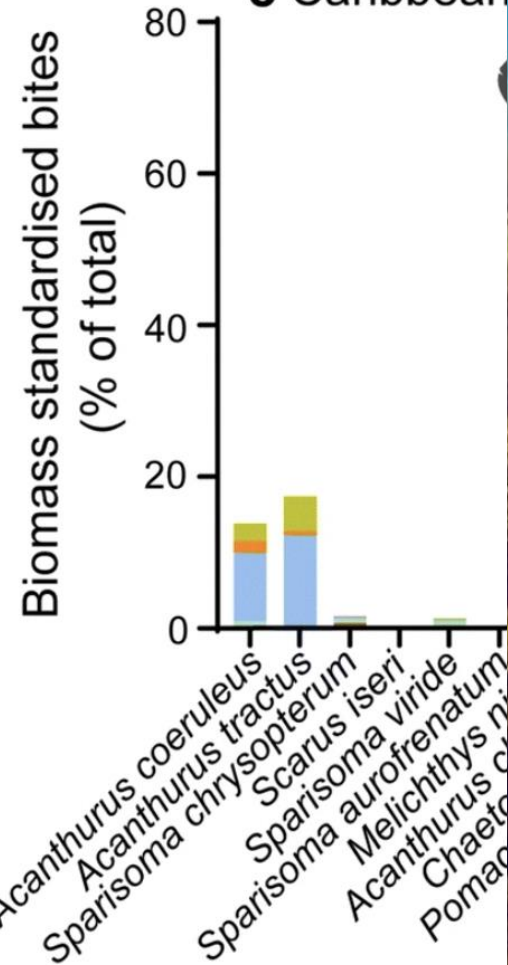


Group discussion

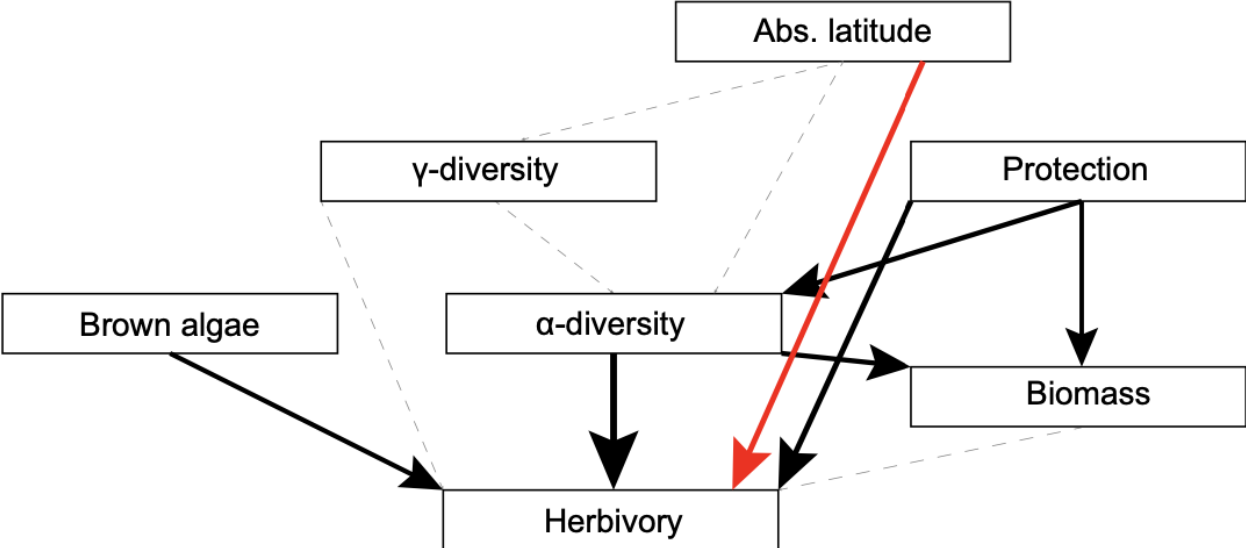
What do these results suggest?

Estimated function

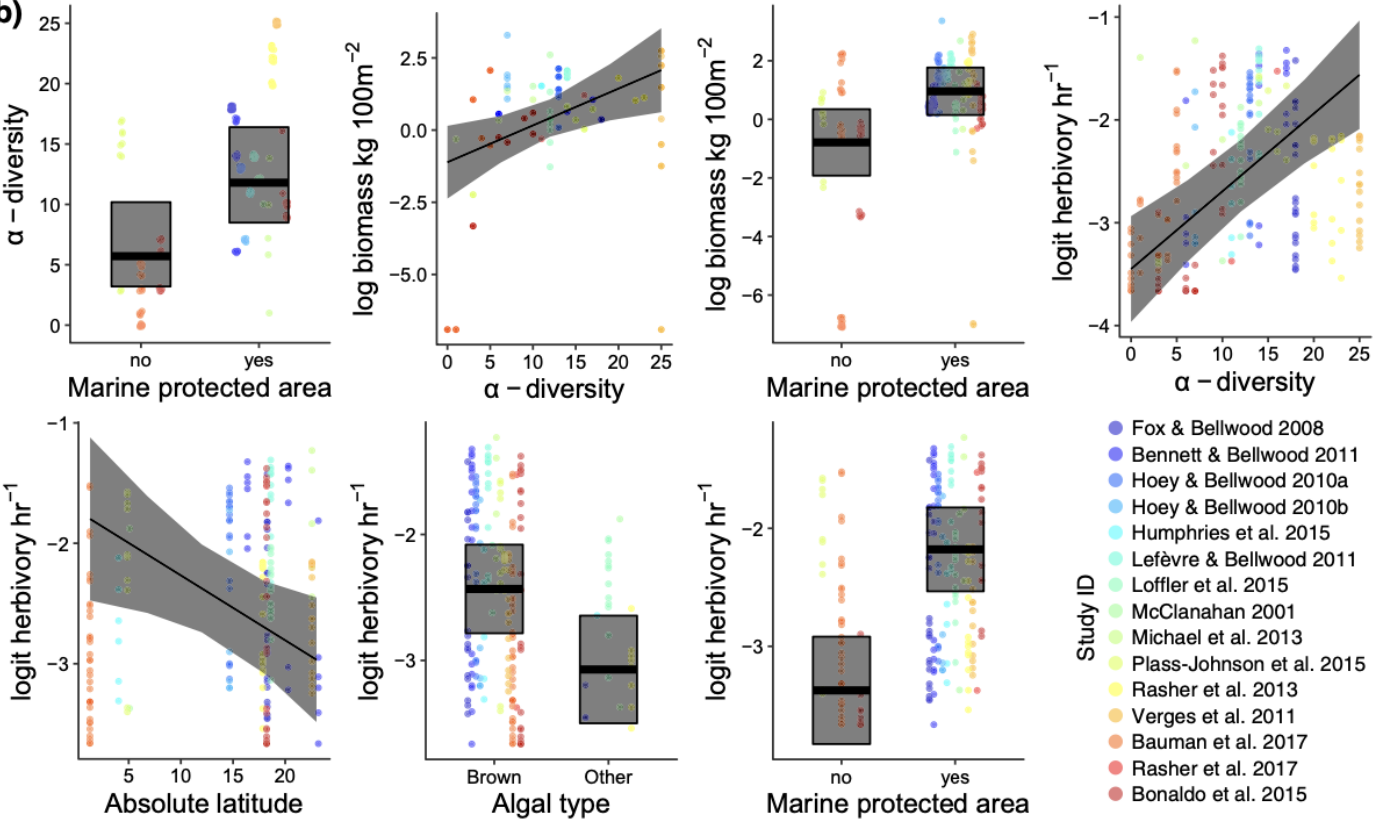
e Caribbean



a)

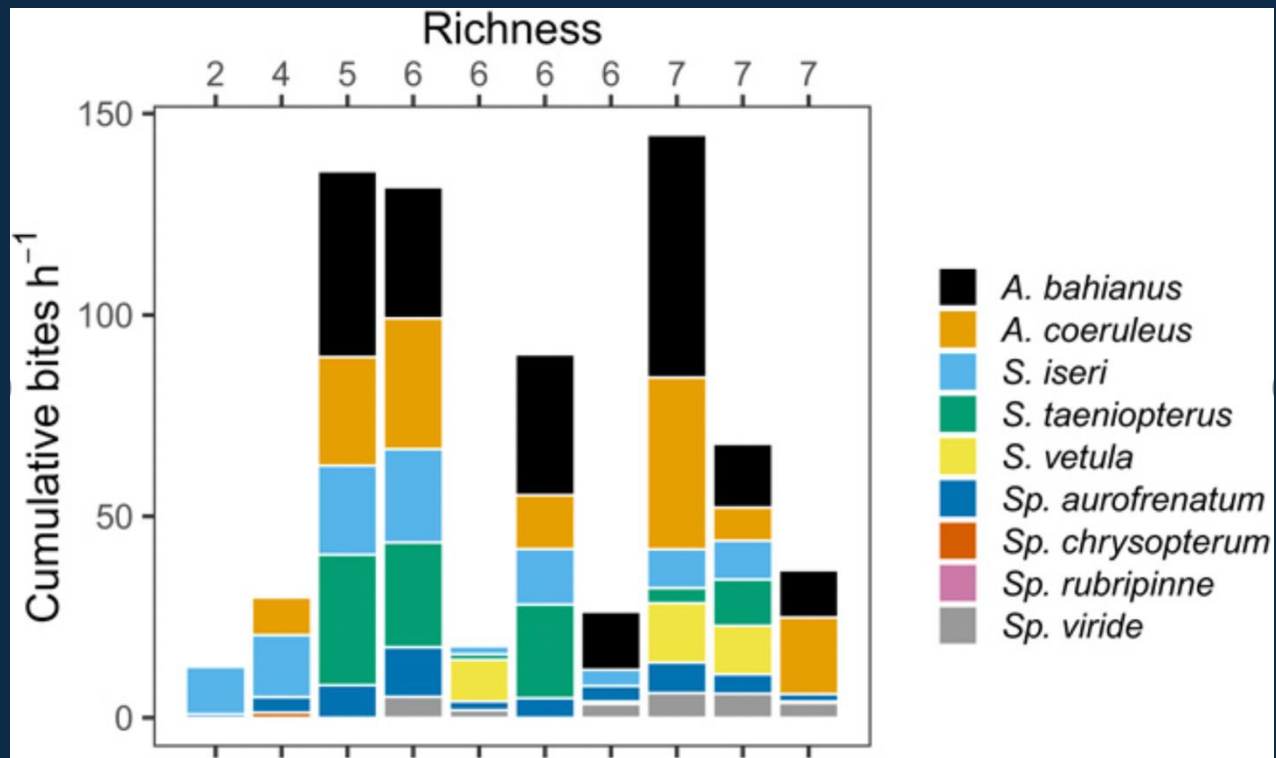
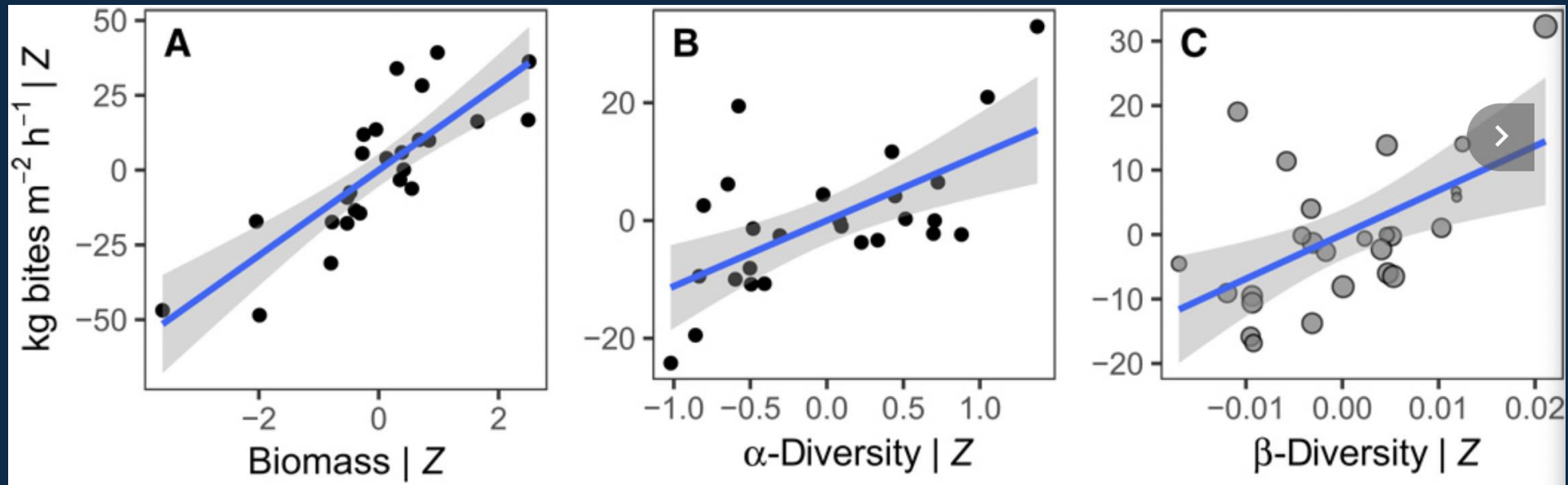


b)



Topor et al. 2019, Cons. Lett.

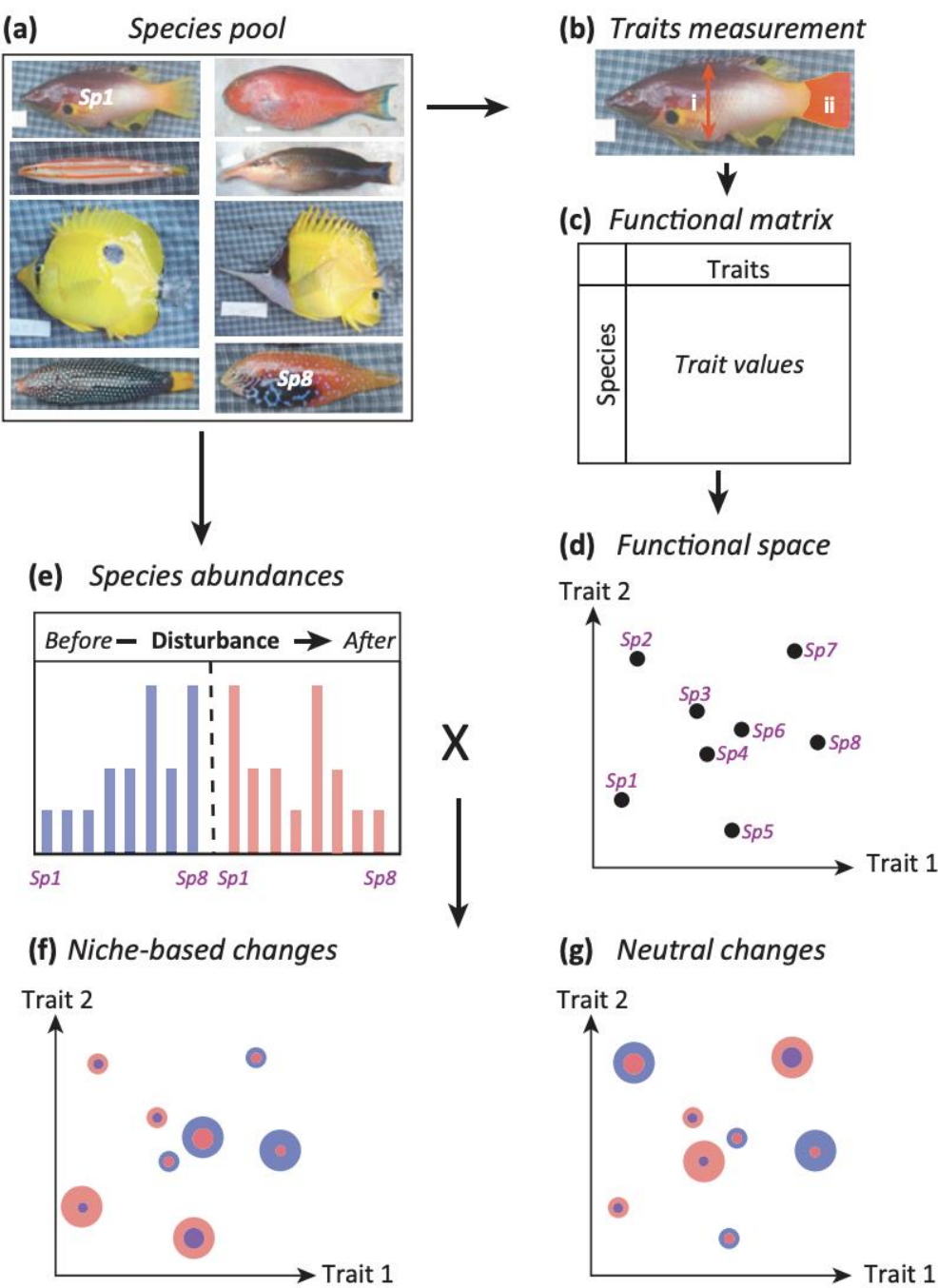
Lefcheck et al.
2019, Sci. Adv.











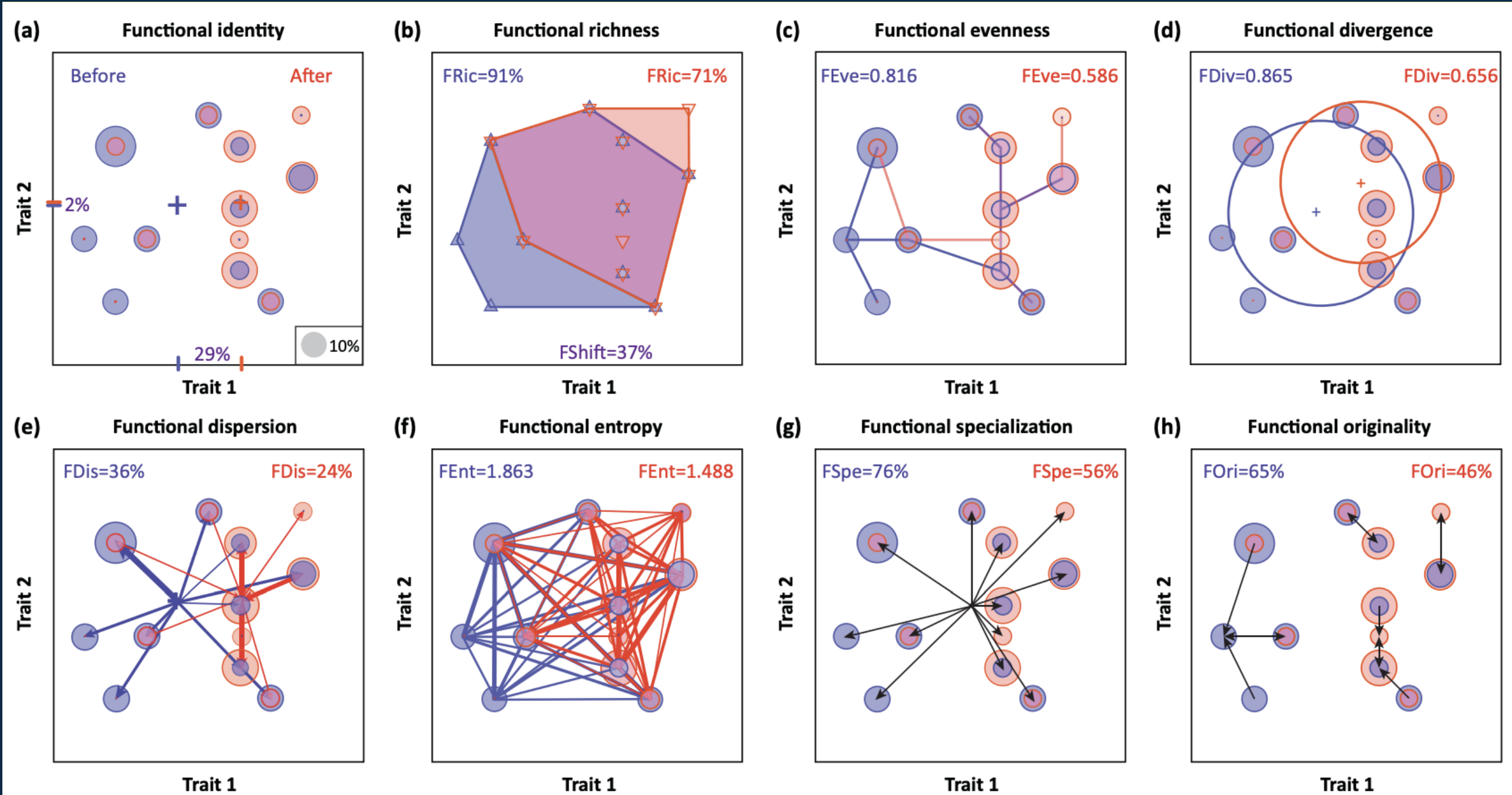


Table I. List of functions

Main goal

Building the functional trait space

Calculating indices of functional community structure

Assessing change in community structure over time



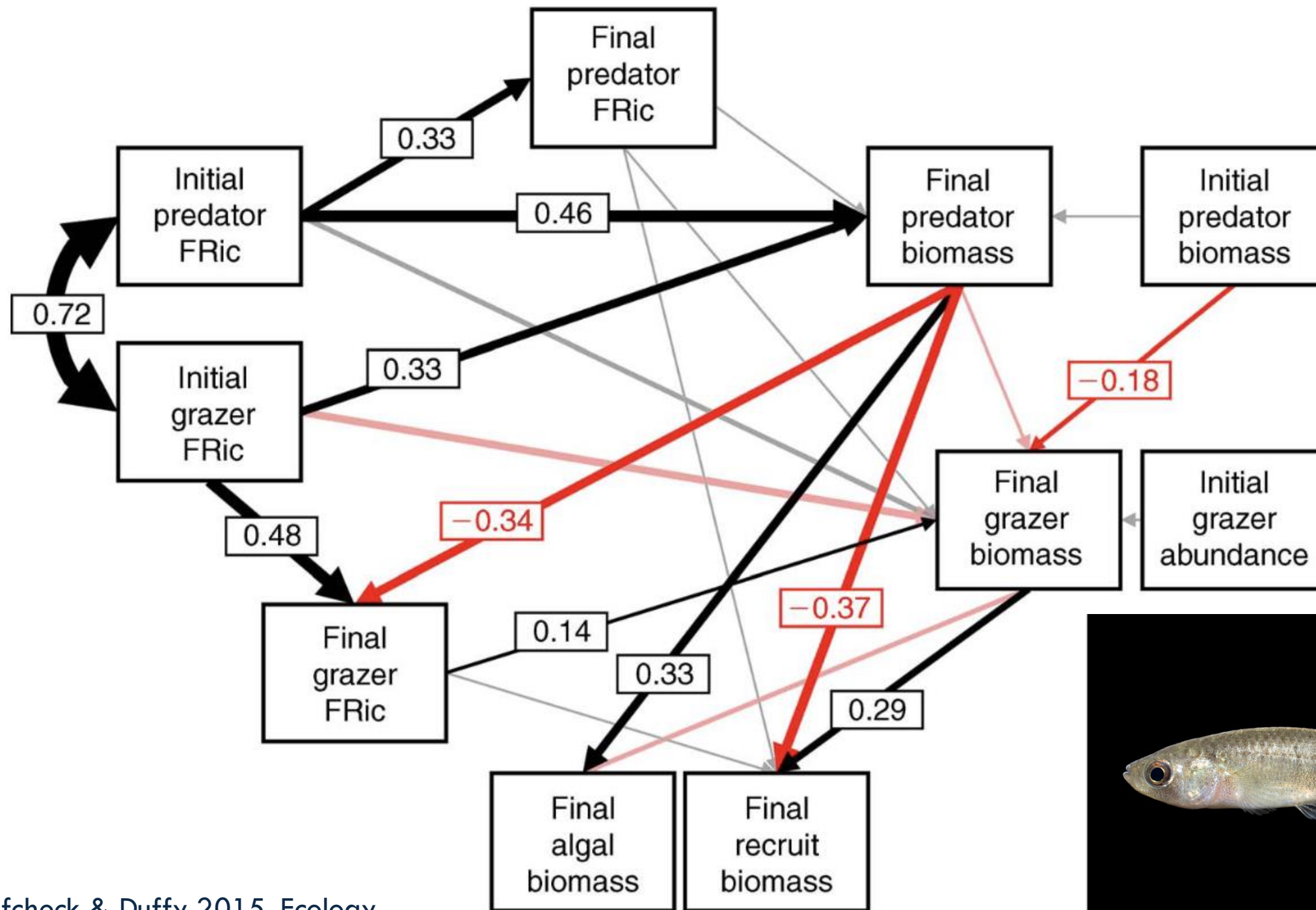
Use functions from 9 different packages to build a functional trait space

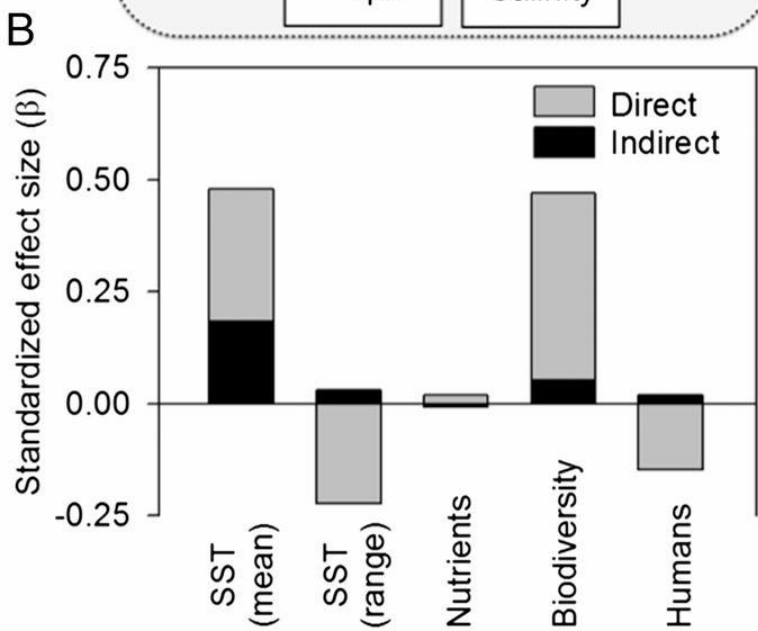
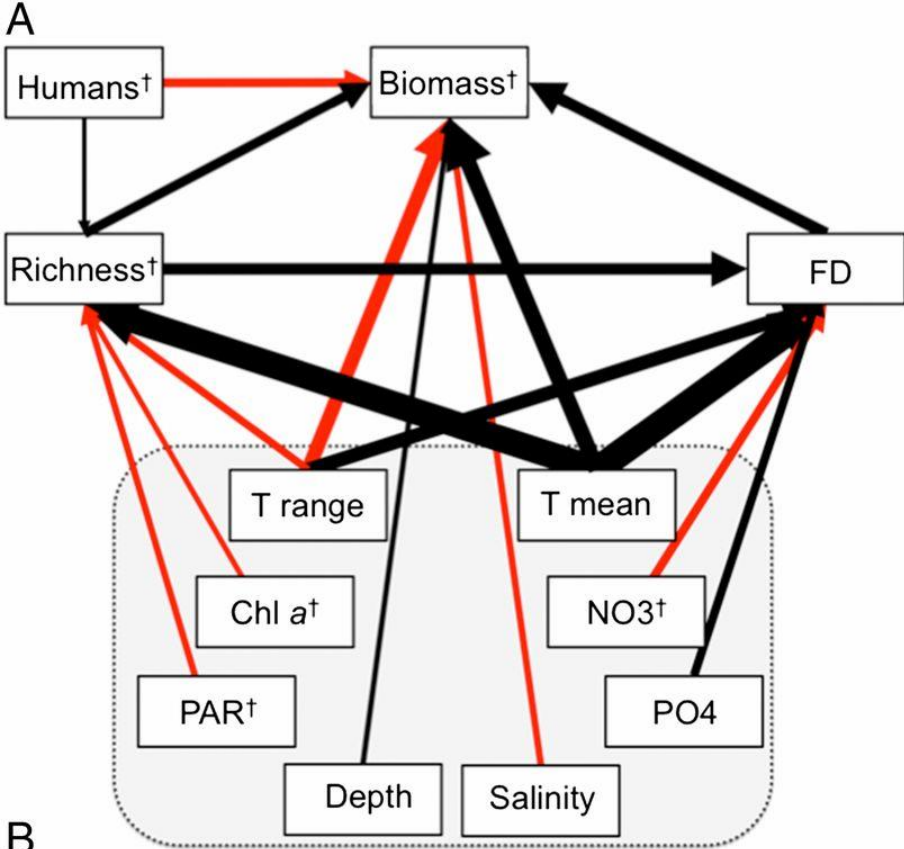


Use the FD package, 12 years later

Table II. List of R libraries

	R library
	base
	ade4
	cluster
	ape
	FD
	FD
t	rccd
	base
	ade4
	FD



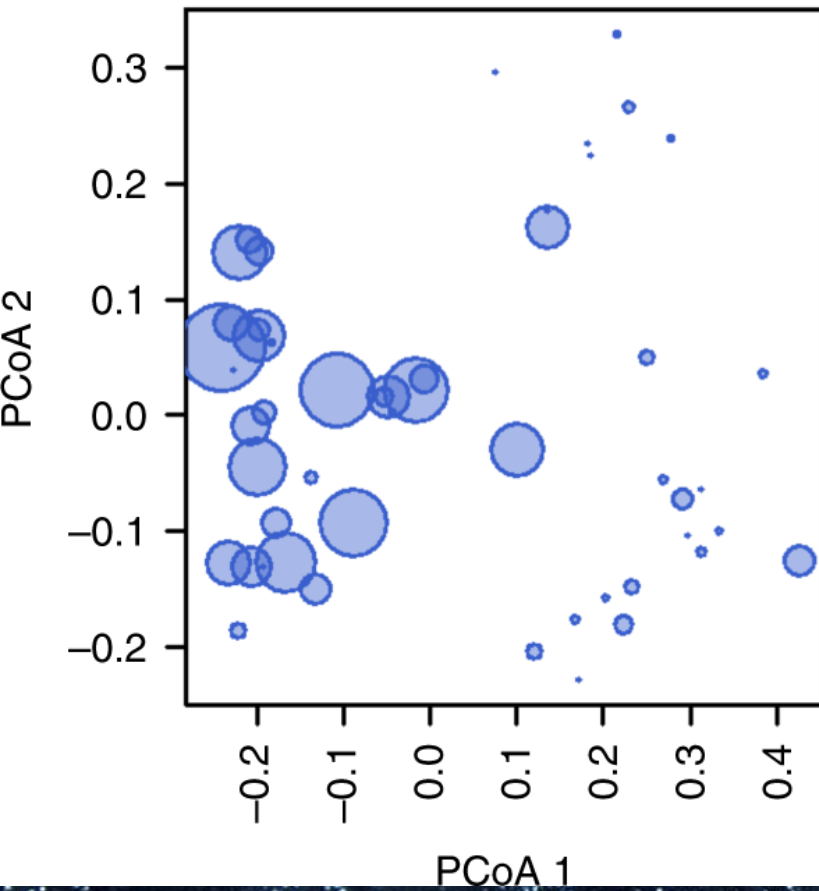


Duffy et al. 2016, PNAS

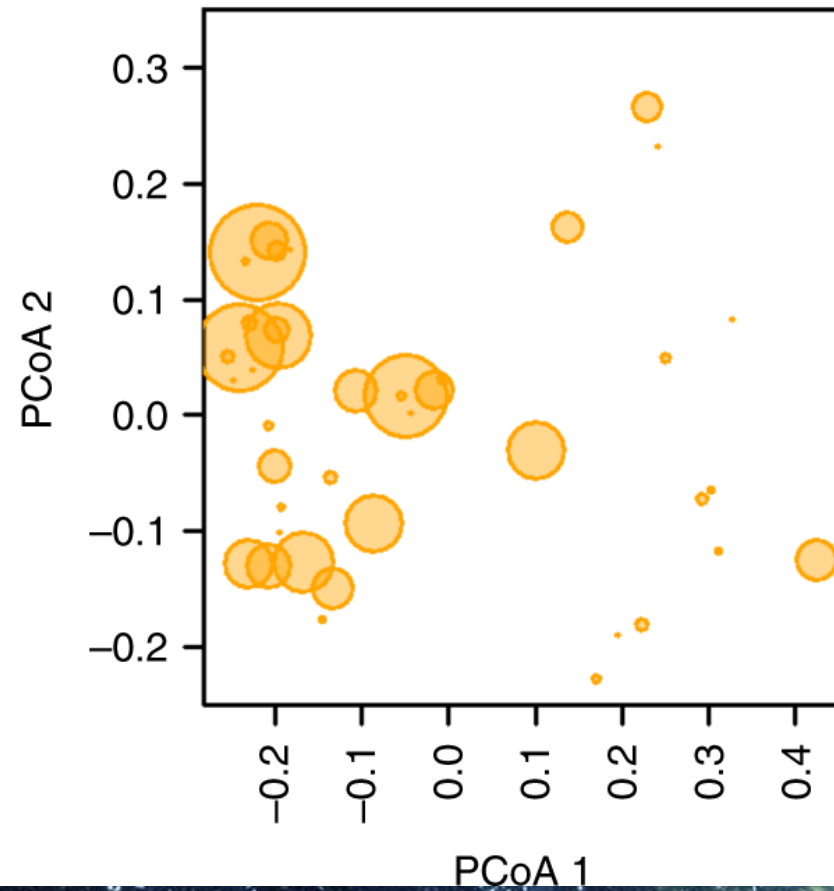




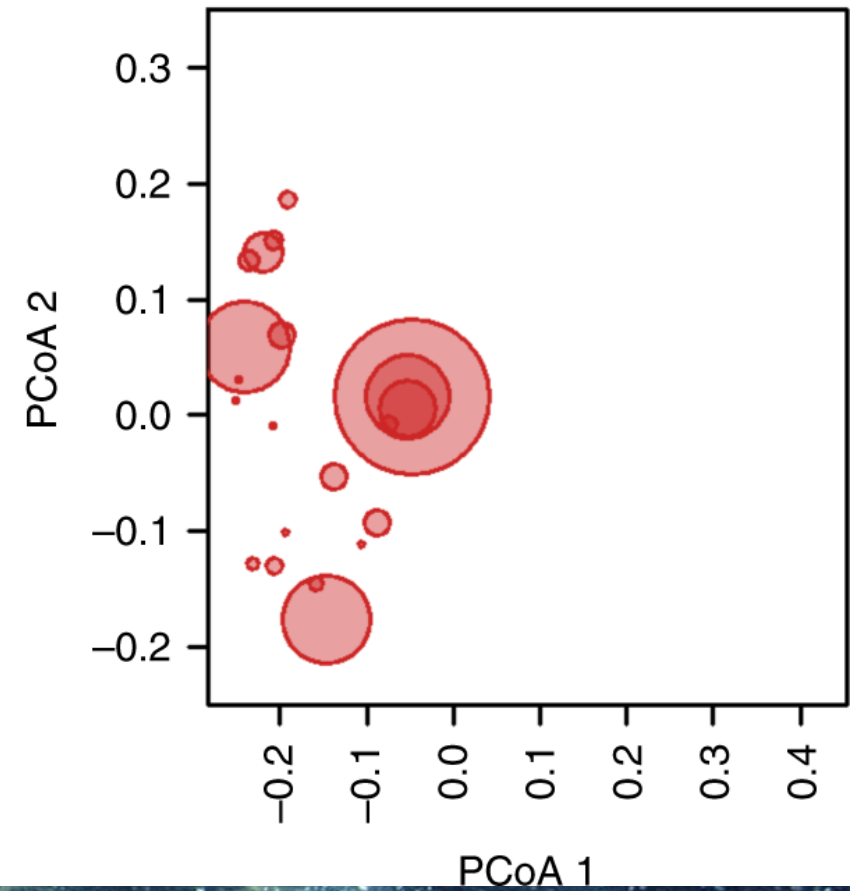
Ambient pH

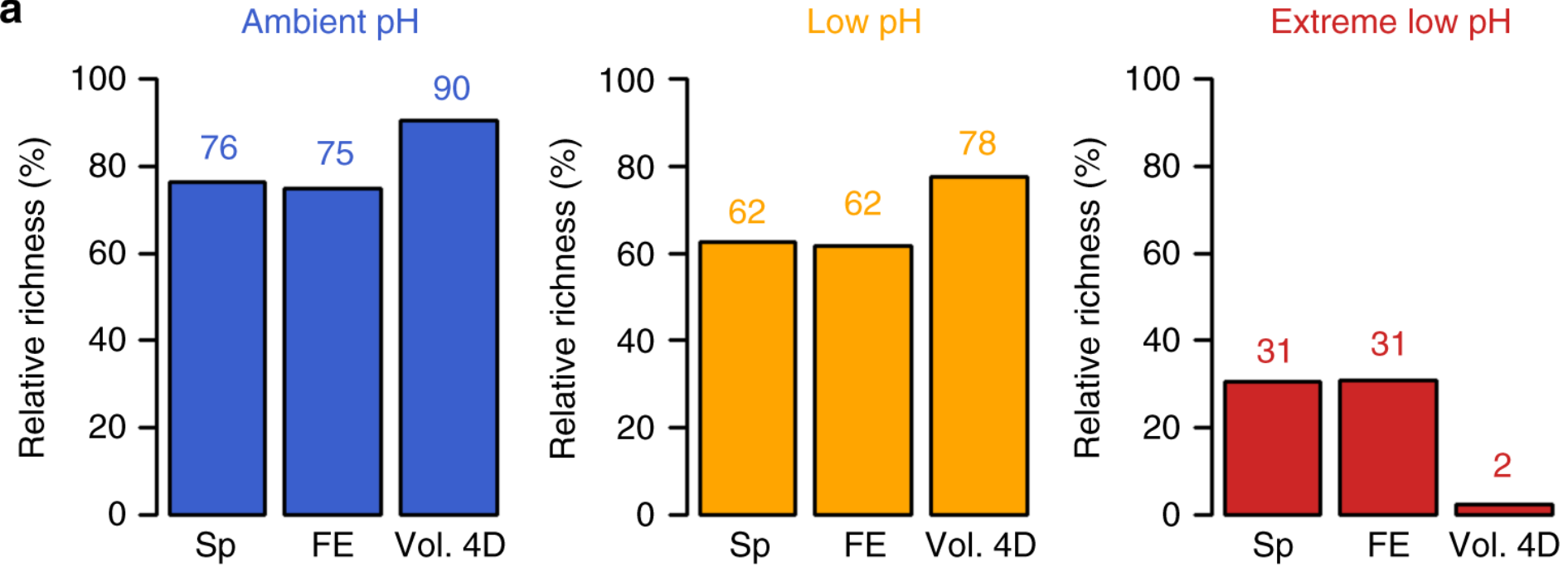
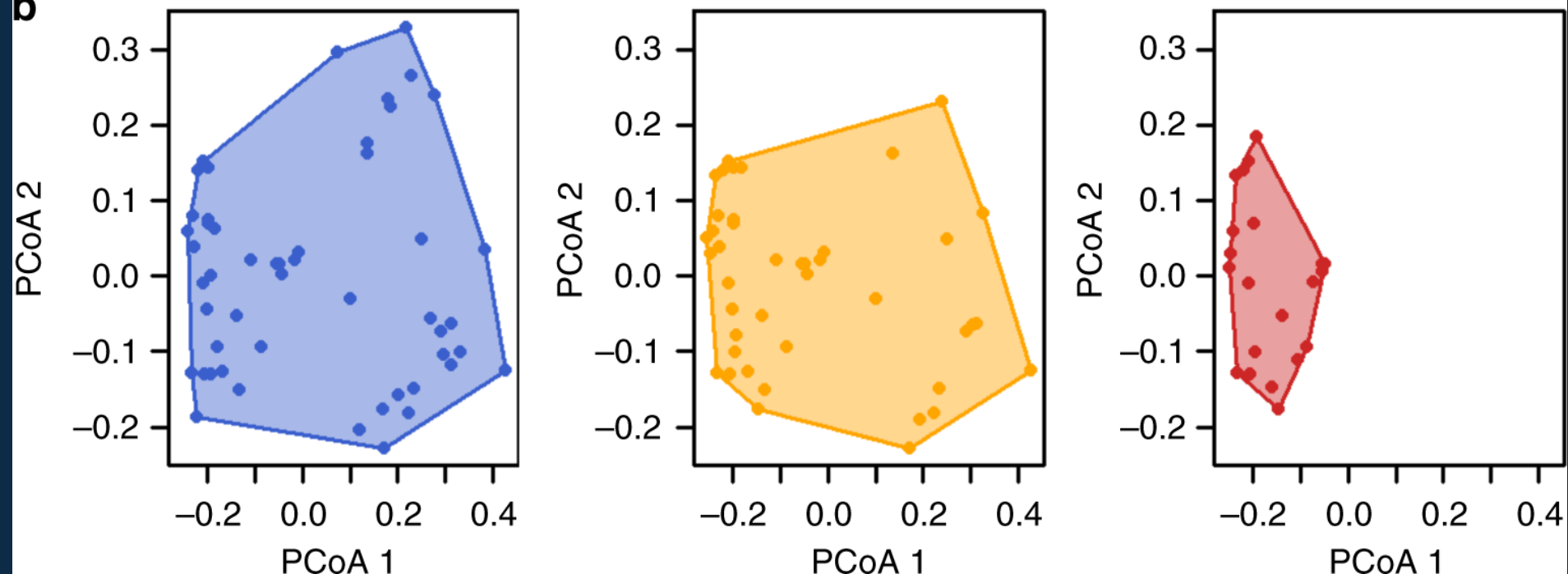


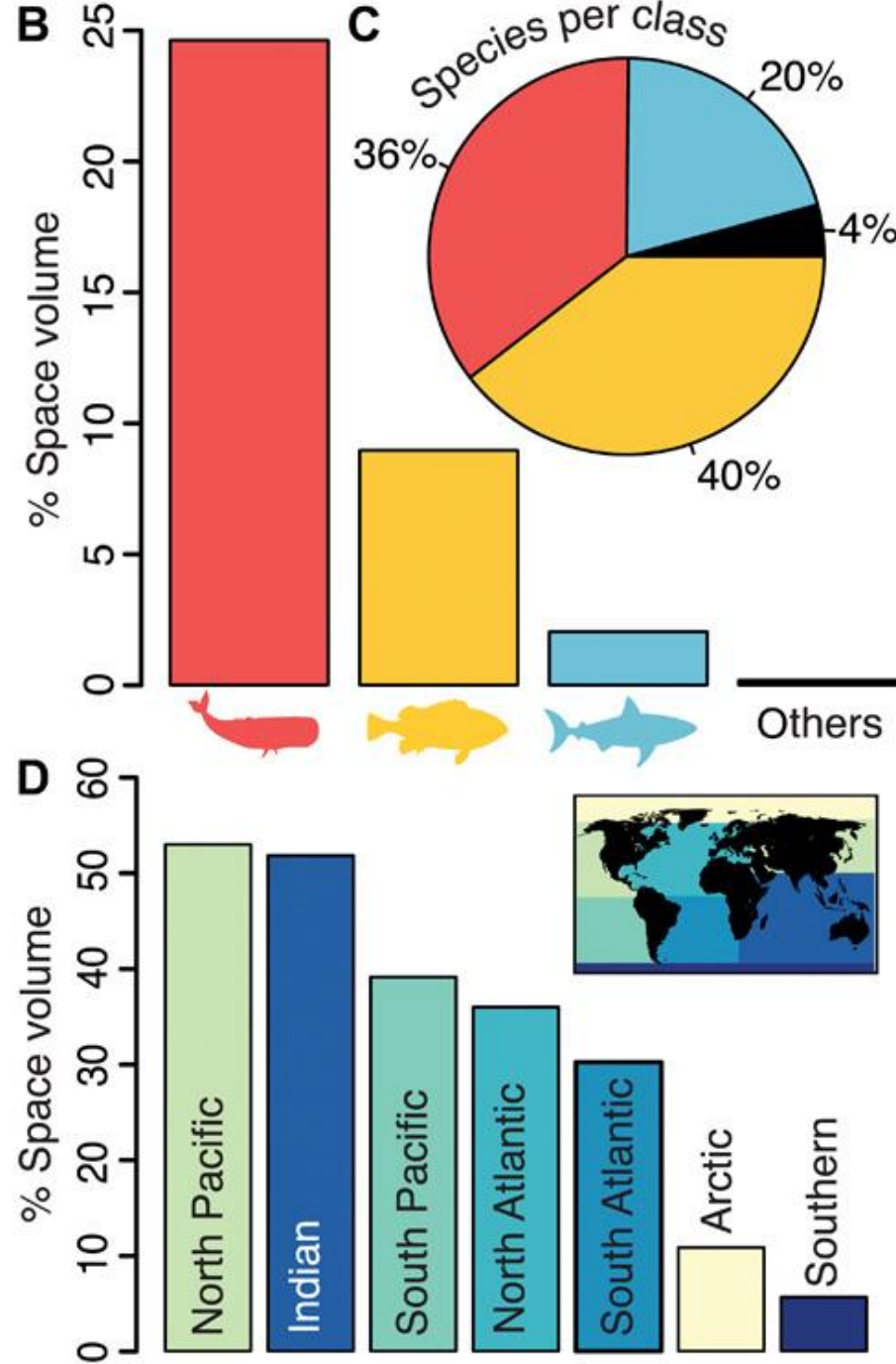
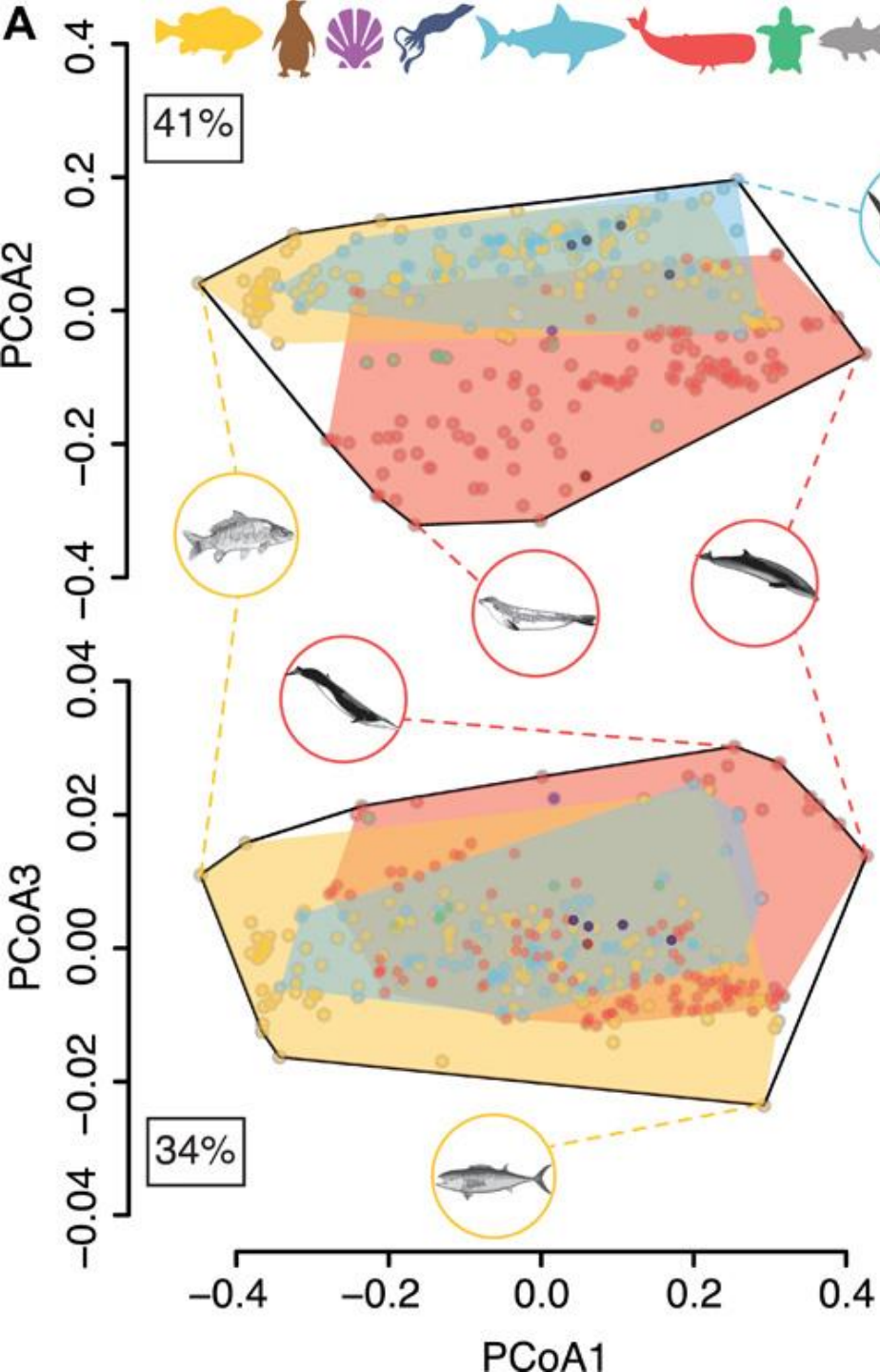
Low pH

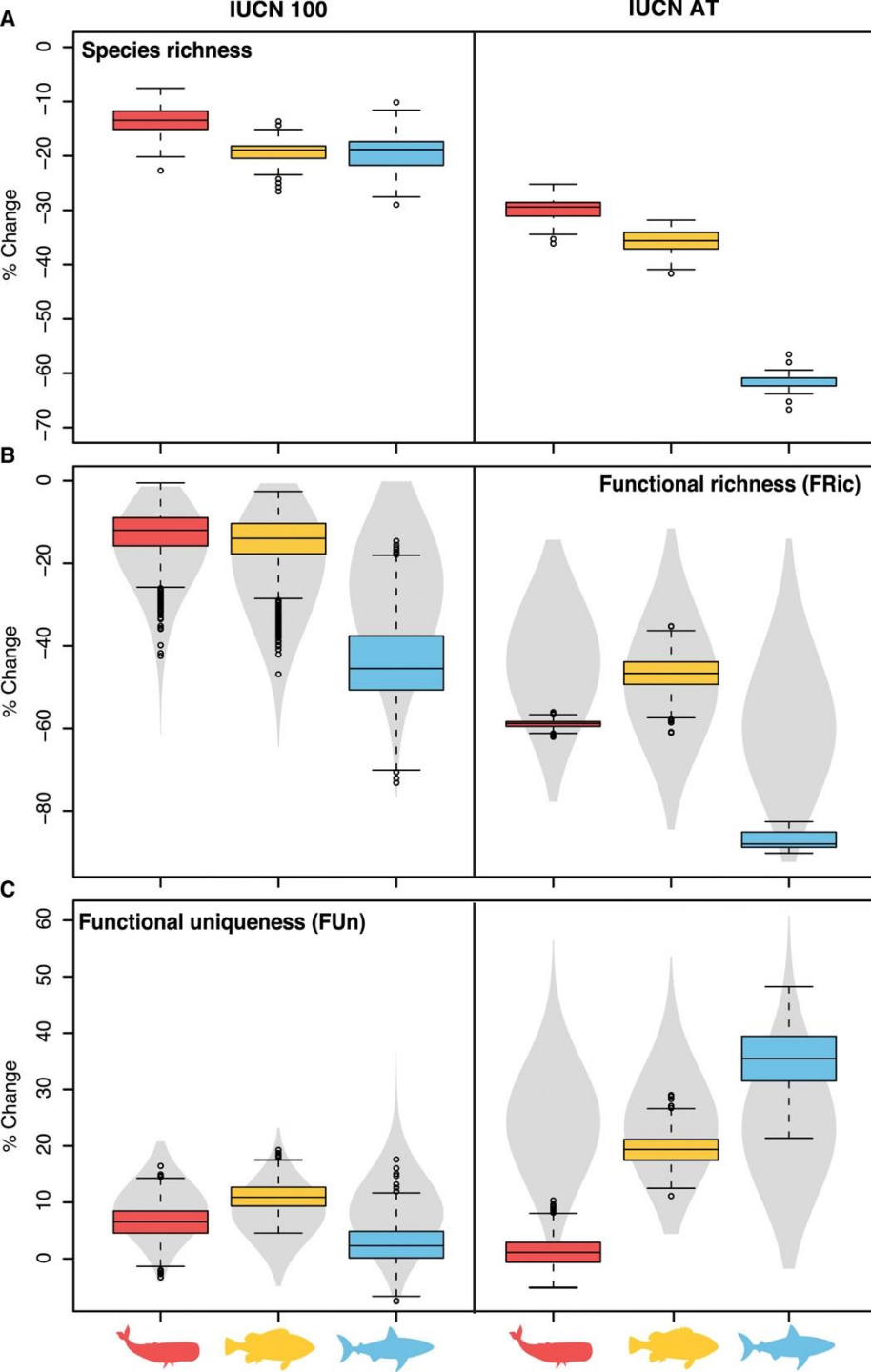


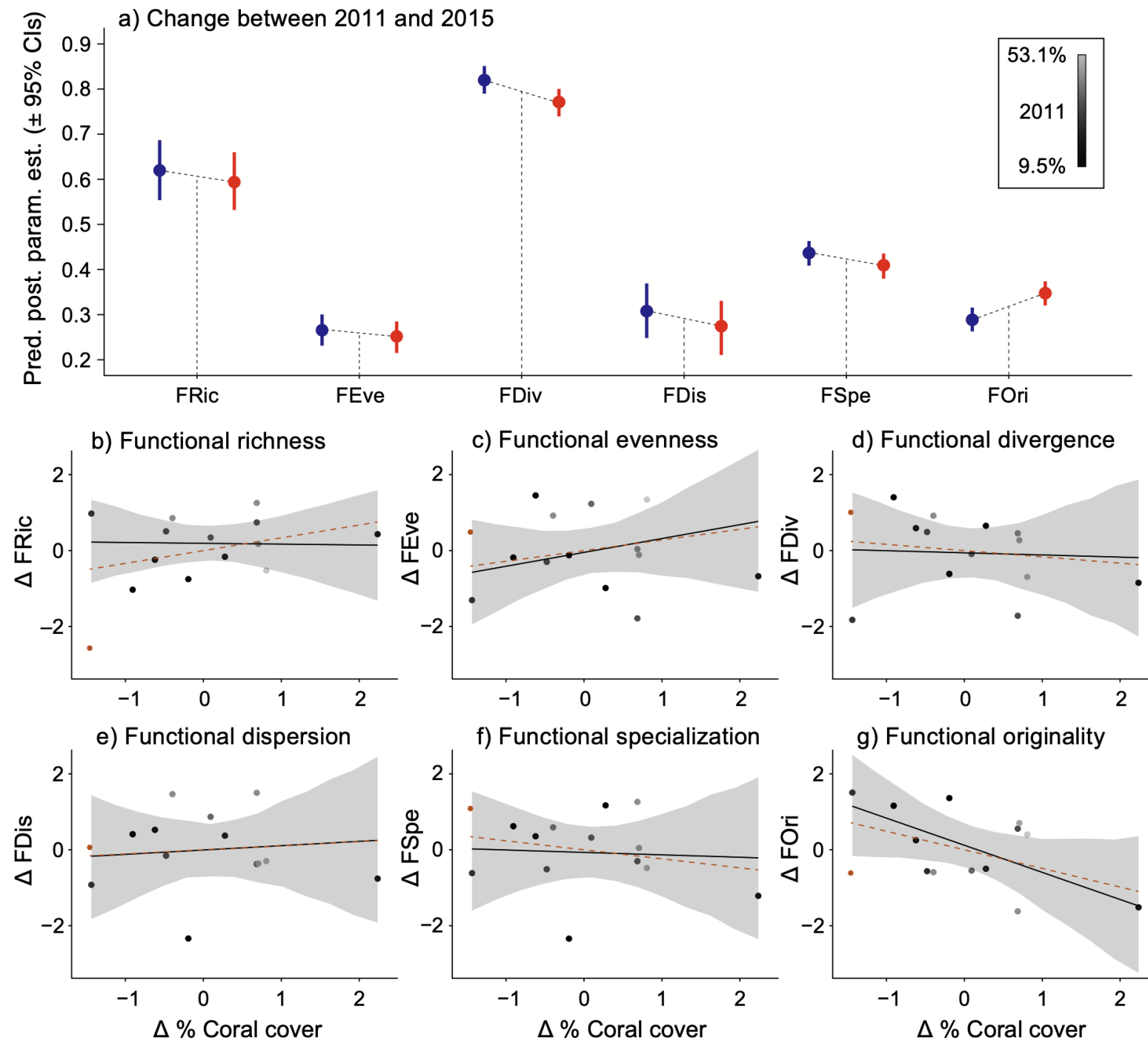
Extreme low pH



a**b**





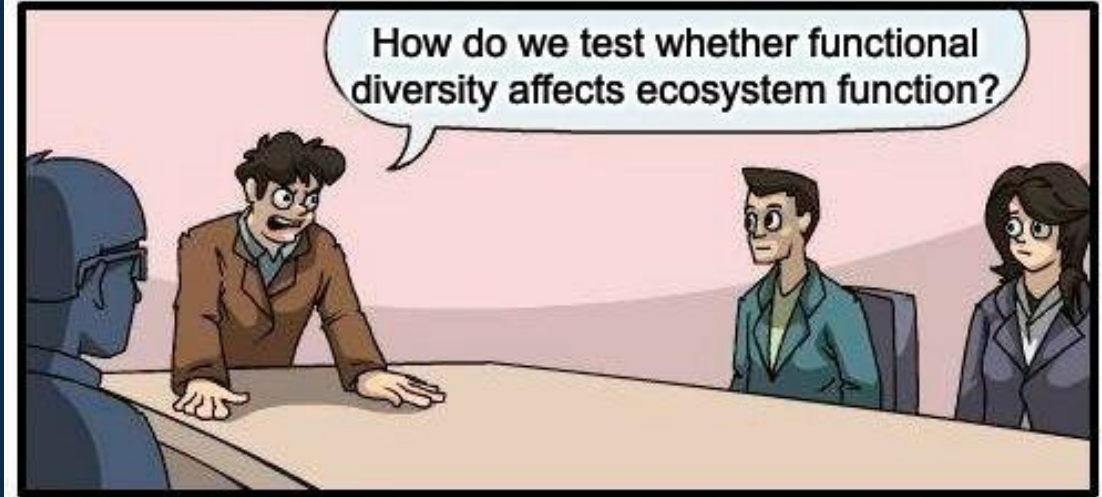
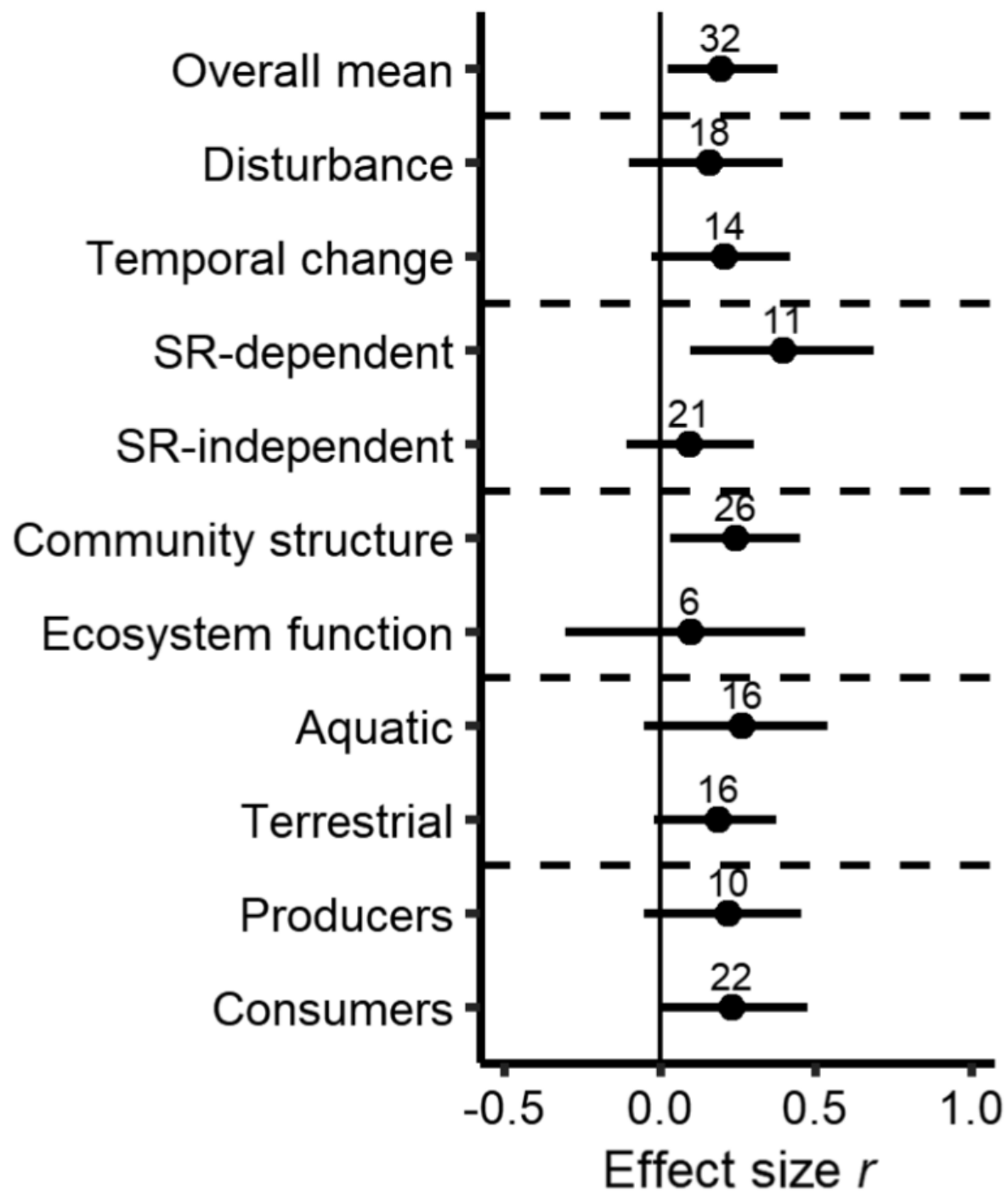


SYNTHESIS & INTEGRATION

Does functional redundancy affect ecological stability and resilience? A review and meta-analysis

CHRISTOPHER R. BIGGS ,† LAUREN A. YEAGER , DEREK G. BOLSER , CHRISTINA BONSELL,
ANGELINA M. DICHIERA, ZHENXIN HOU, SPENCER R. KEYSER, ALEXIS J. KHURSIGARA, KAIJUN LU,
ARLEY F. MUTH  BENJAMIN NEGRETE JR., , AND BRAD E. ERISMAN 

Marine Science Institute, The University of Texas at Austin, 750 Channel View Drive, Port Aransas, Texas 78373 USA





The diversity-stability relationship

WHATS STABILITY

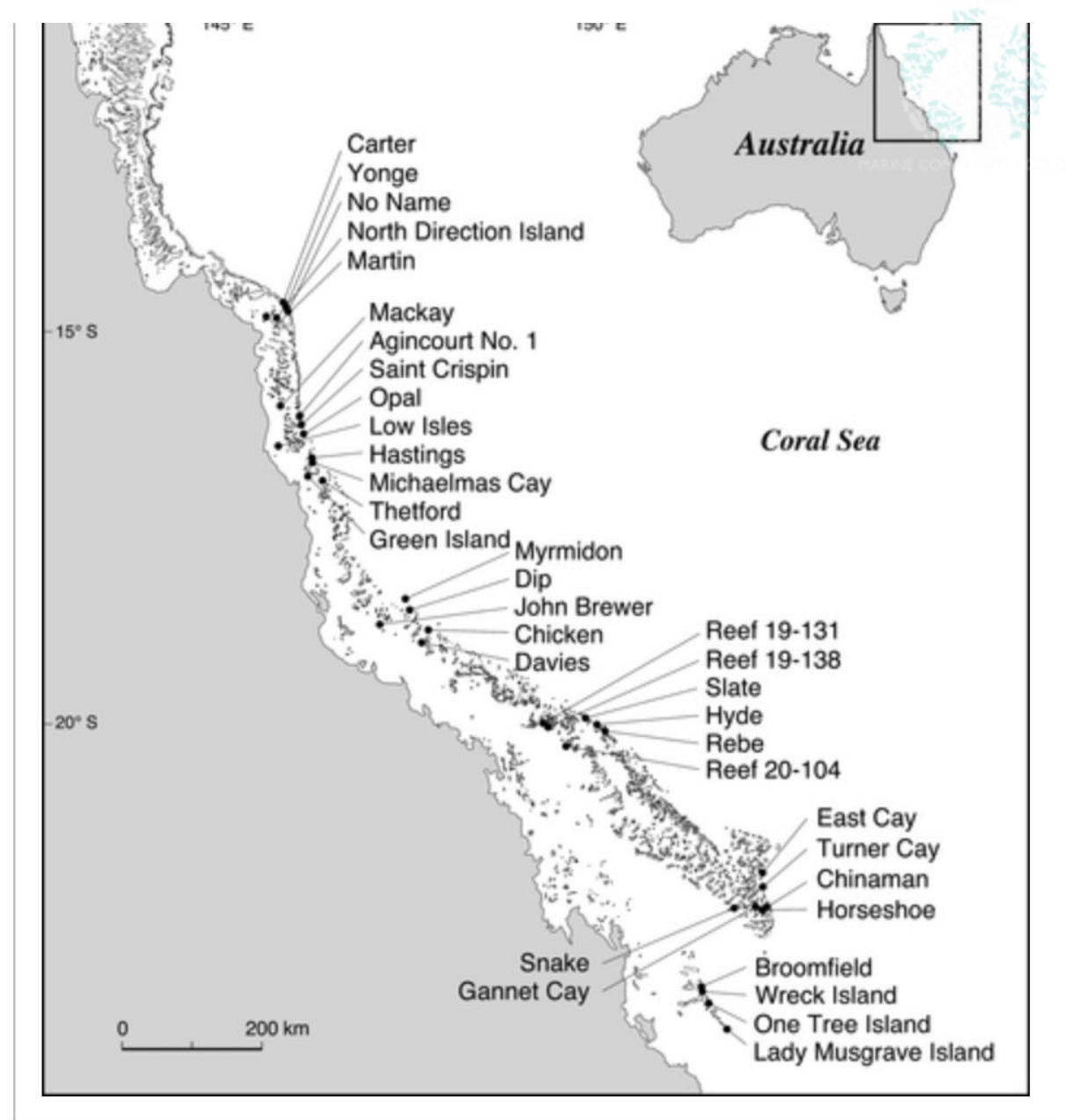


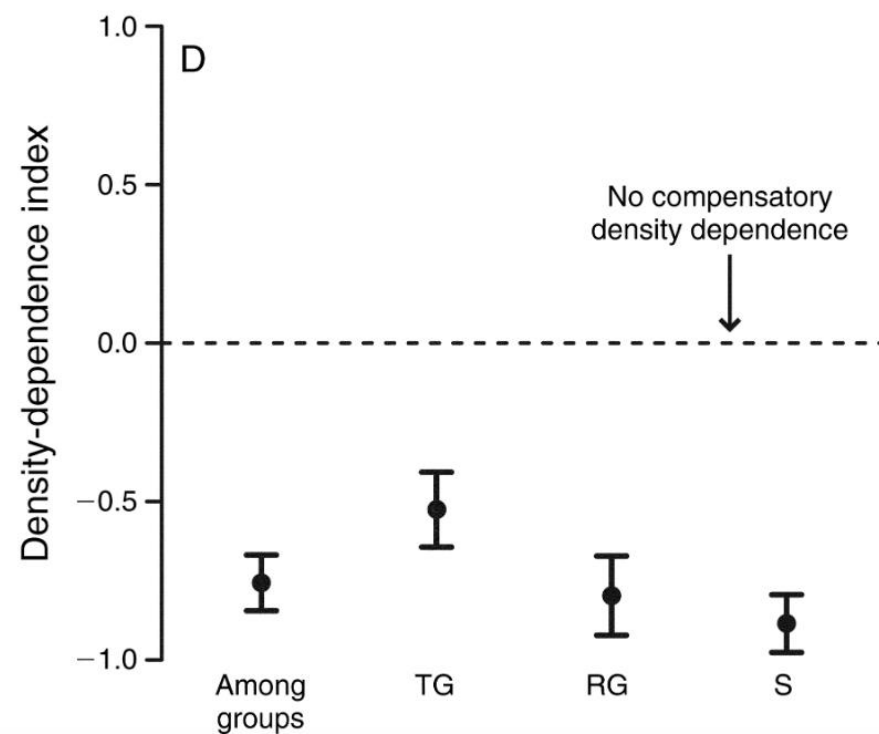
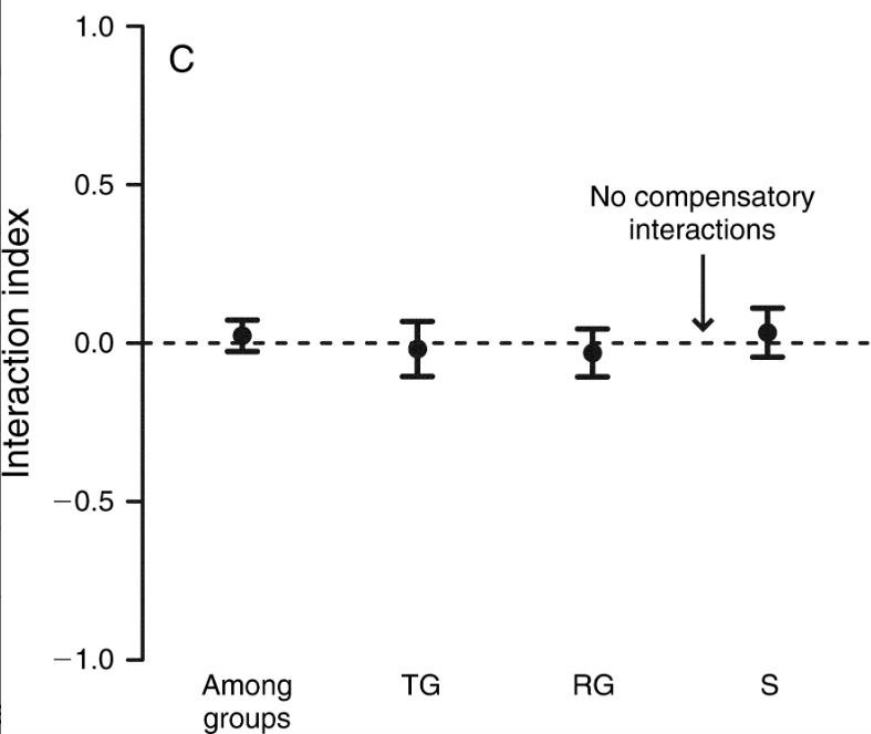
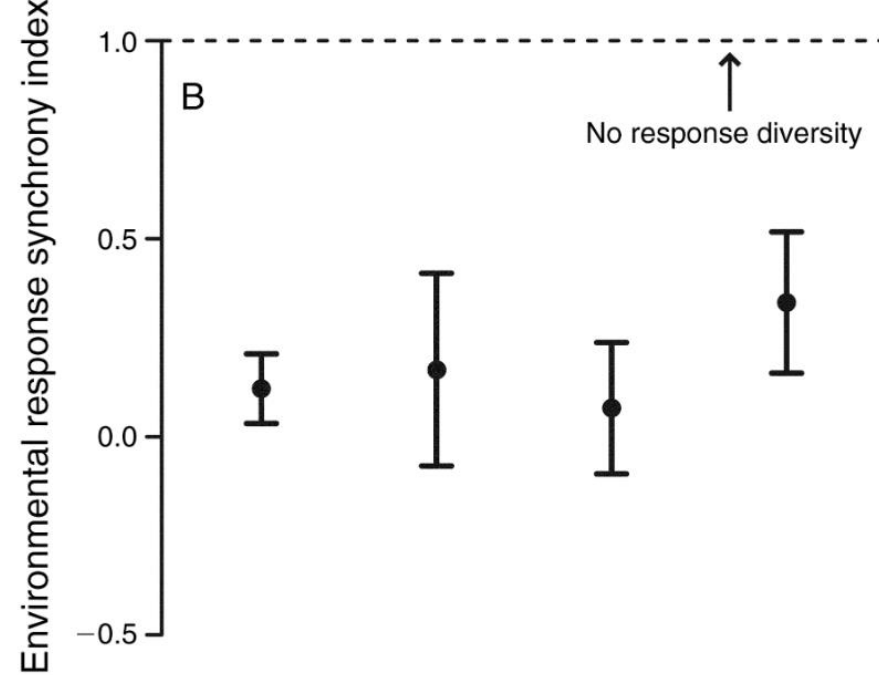
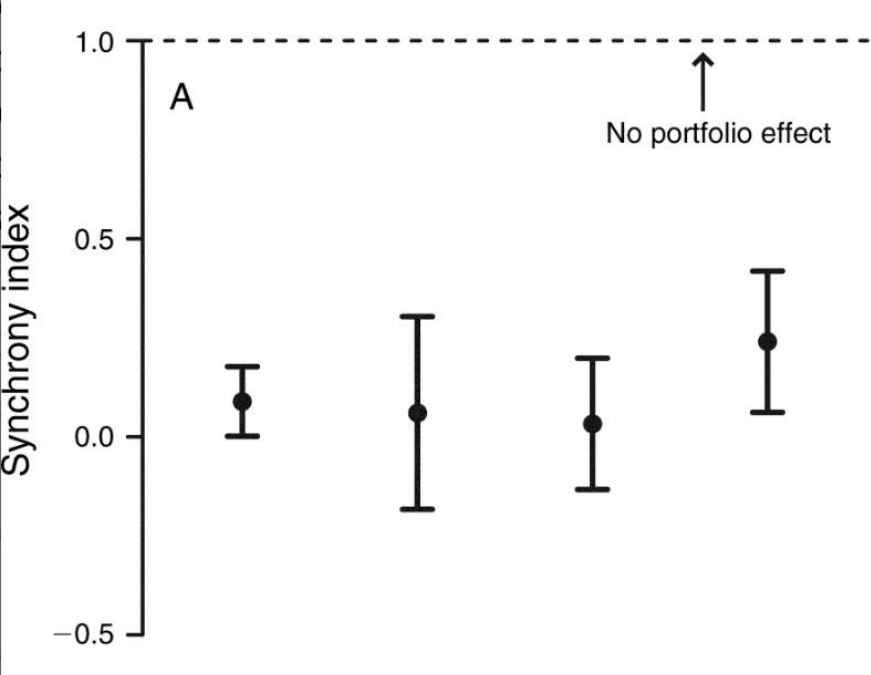
PRECIOUS

The Portfolio Effect

As long as species' responses to disturbances aren't synchronized, higher diversity will buffer ecological stability

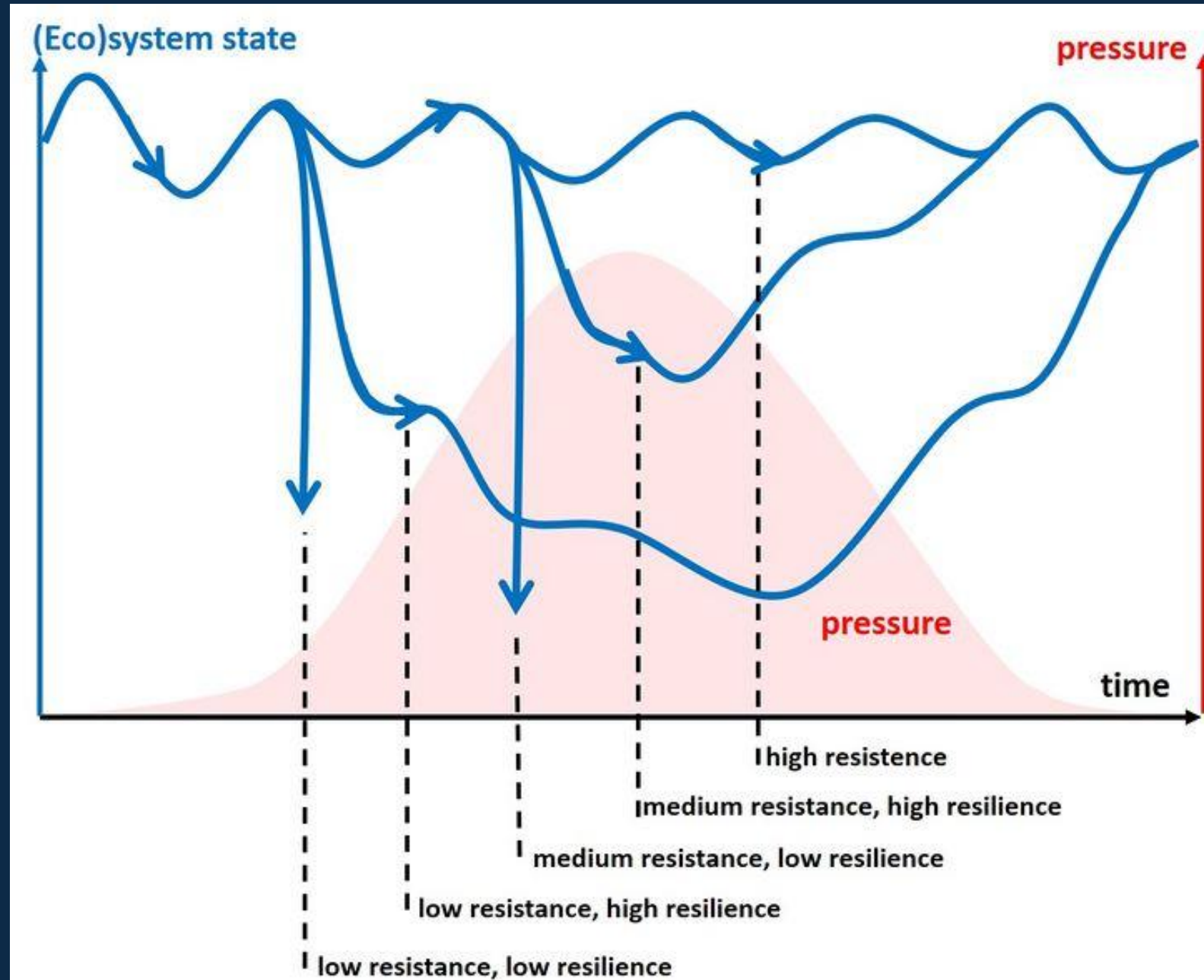








Resistance vs. resilience in ecosystem states



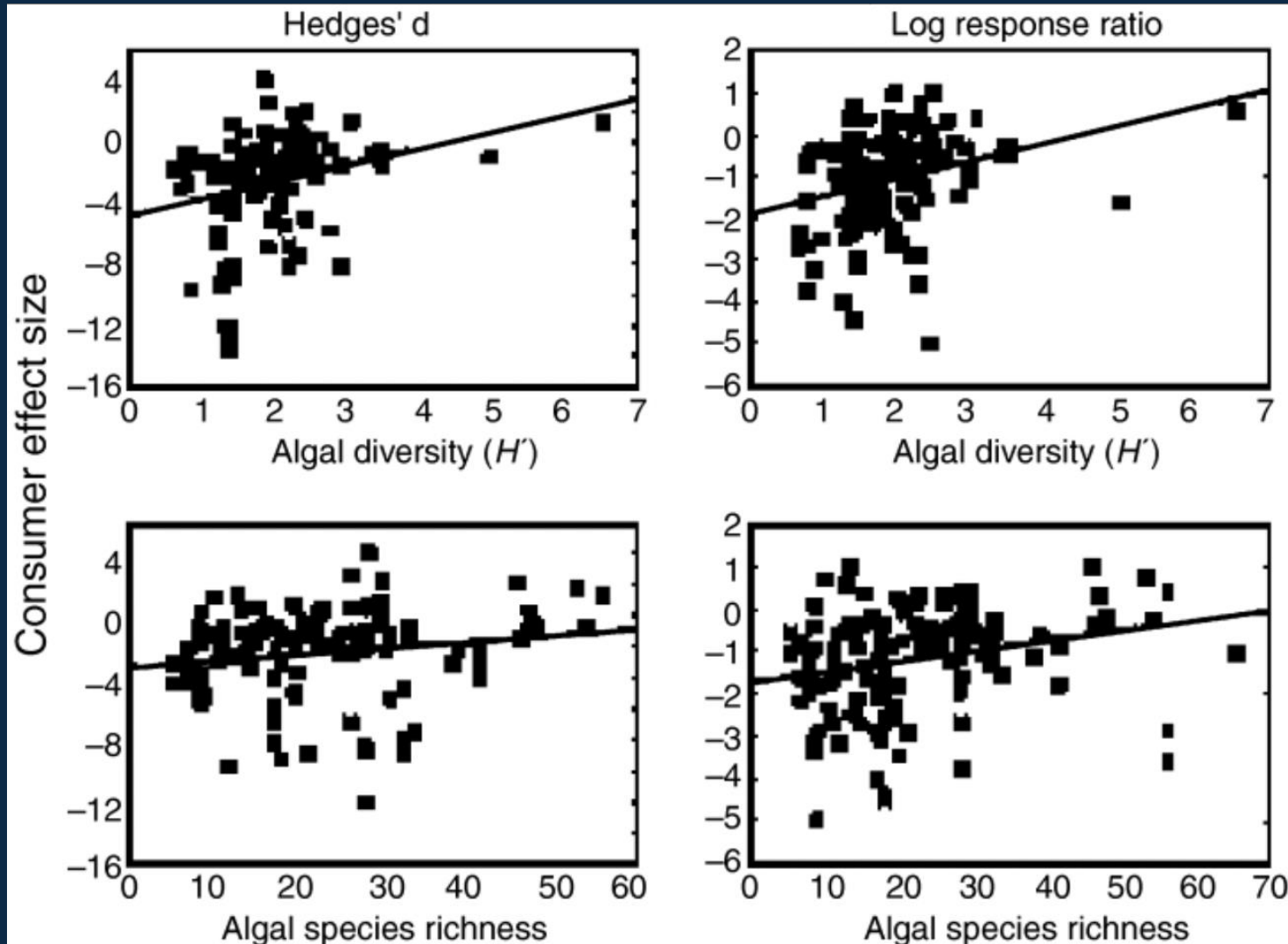
WHAT'S ECOSYSTEM STATE



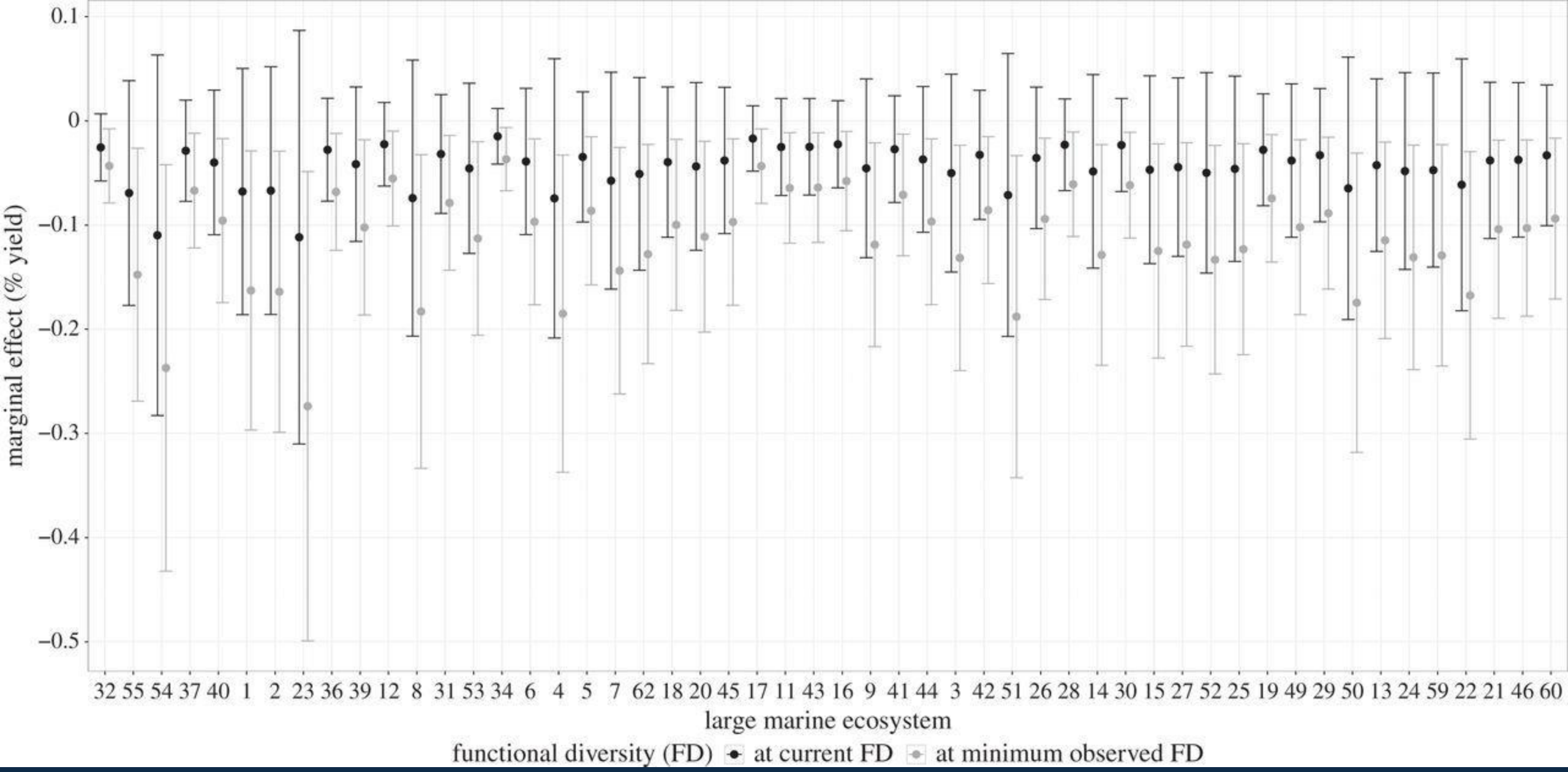
PRECIOUS?



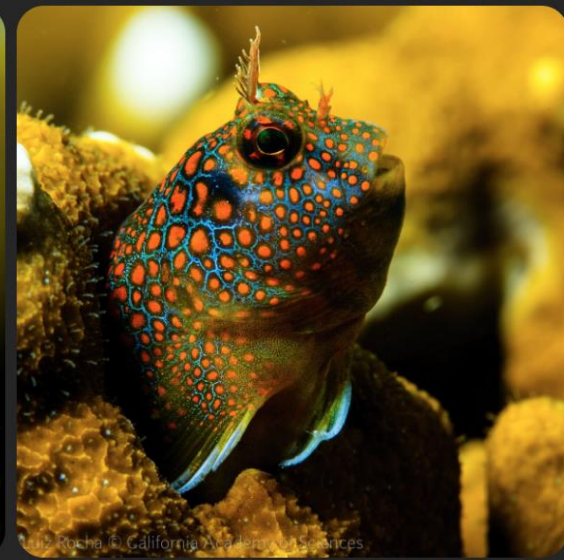
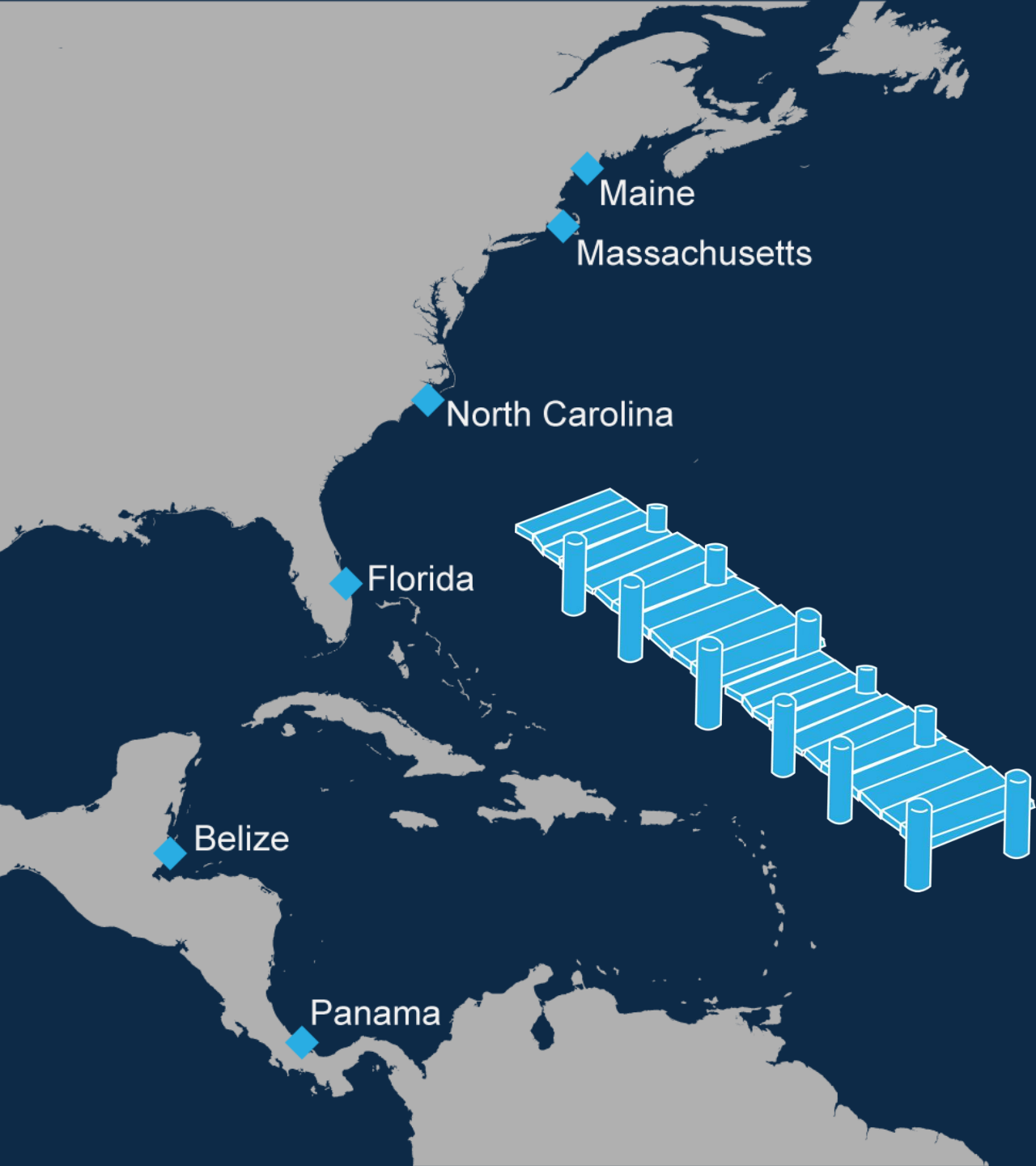
Algae diversity buffers grazer effects















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RESEARCH ARTICLE

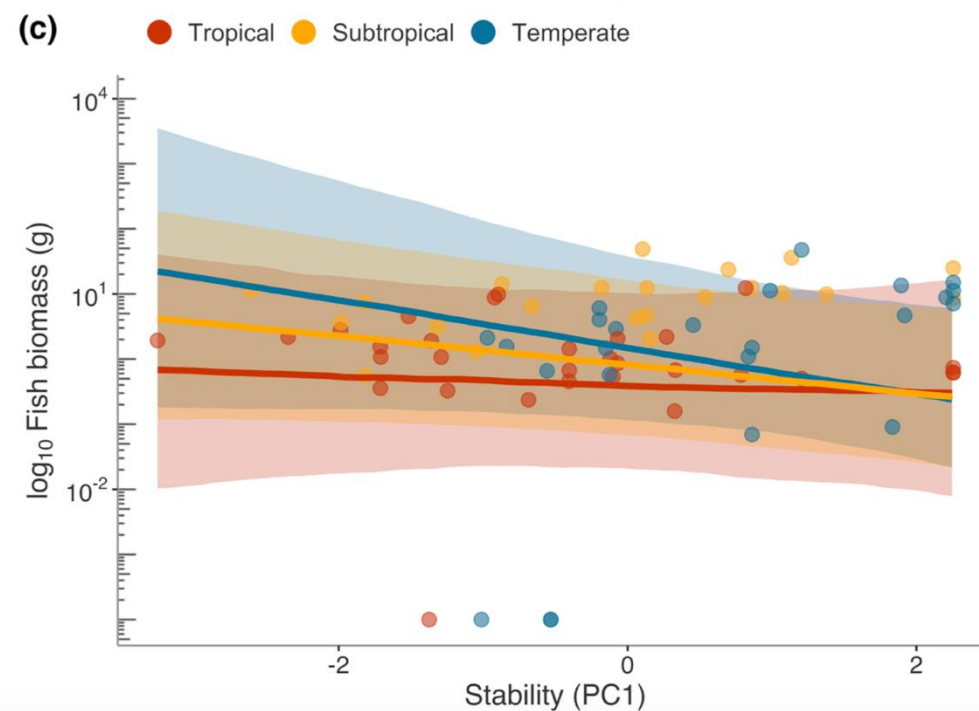
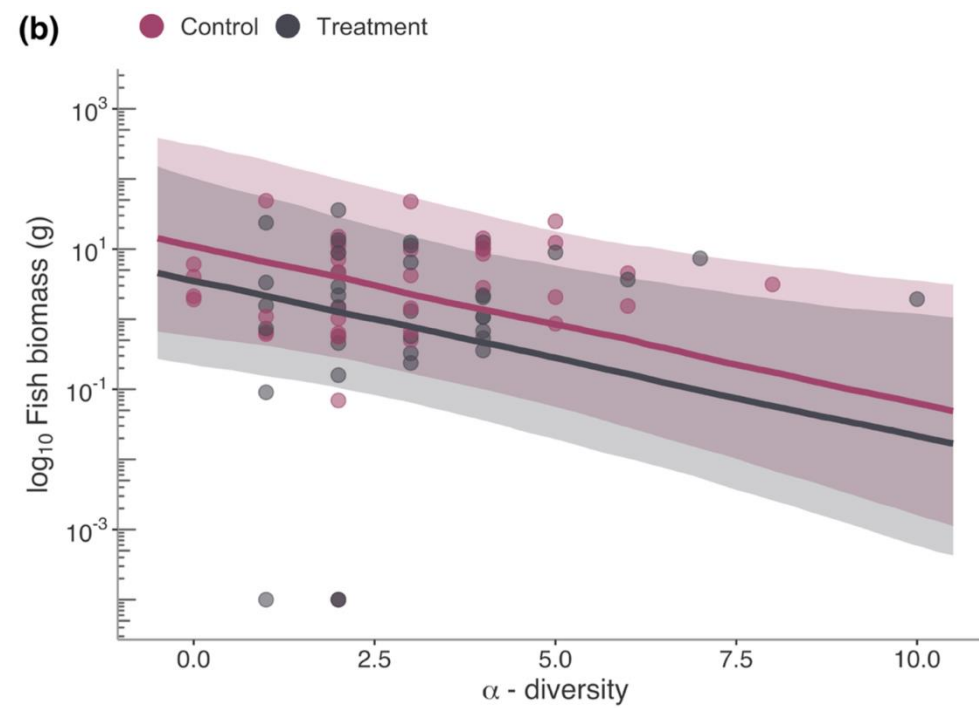
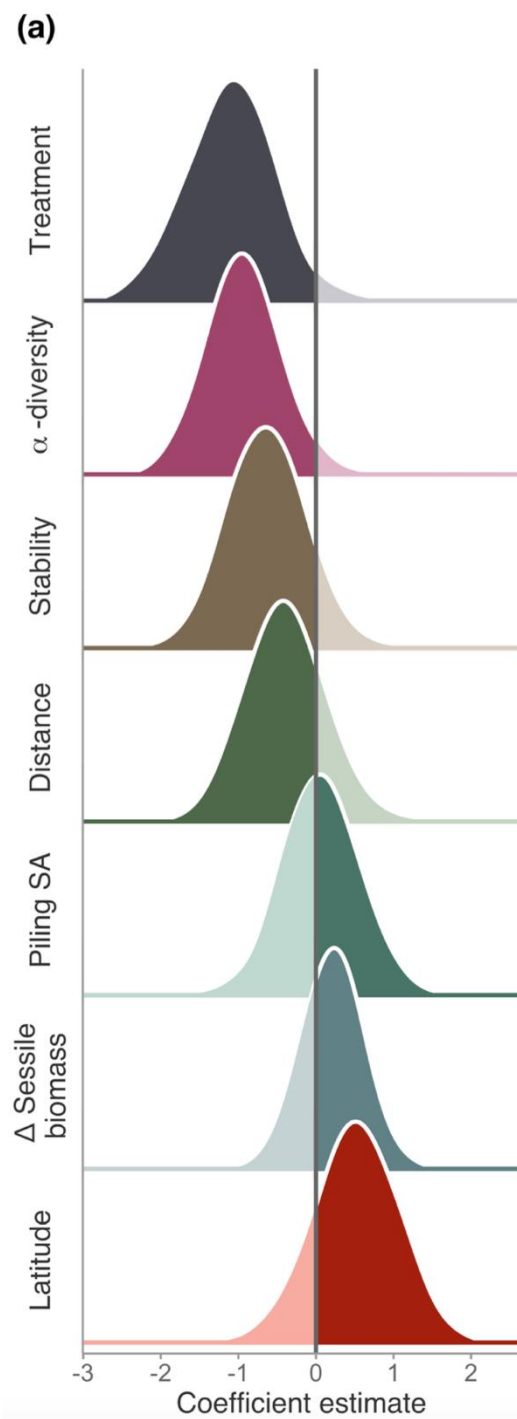
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Examining the diversity, stability and functioning of marine fish communities across a latitudinal gradient

Helen F. Yan^{1,2,3,4}  | Jordan M. Casey^{5,6} | Nancy Knowlton⁶ | J. Emmett Duffy⁷ |
Simon J. Brandl^{1,5,7}







NEXT LECTURE

Mon, April 21nd

Thursday: coding lab